

STUDIES ON DRY ROOT ROT DISEASE OF CHICKPEA
[*Cicer arietinum* L.] CAUSED BY *Rhizoctonia*
***bataticola* (Taub.) Butler.**

VEERENDRA KUMAR, K.V.

DEPARTMENT OF PLANT PATHOLOGY
COLLEGE OF AGRICULTURE, RAICHUR – 584 101
UNIVERSITY OF AGRICULTURAL SCIENCES
DHARWAD – 580 005

JANUARY, 2004

STUDIES ON DRY ROOT ROT DISEASE OF CHICKPEA

[*Cicer arietinum* L.] CAUSED BY *Rhizoctonia*

***bataticola* (Taub.) Butler.**

Thesis submitted to the
University of Agricultural Sciences, Dharwad
in partial fulfillment of the requirement for the

Degree of

MASTER OF SCIENCE (AGRICULTURE)

IN

PLANT PATHOLOGY

BY

VEERENDRA KUMAR, K.V.

**DEPARTMENT OF PLANT PATHOLOGY
COLLEGE OF AGRICULTURE, RAICHUR – 584 101
UNIVERSITY OF AGRICULTURAL SCIENCES
DHARWAD – 580 005**

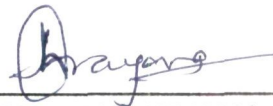
JANUARY, 2004

DEPARTMENT OF PLANT PATHOLOGY
COLLEGE OF AGRICULTURE, RAICHUR
UNIVERSITY OF AGRICULTURAL SCIENCES, DHARWAD

C E R T I F I C A T E

This is to certify that the thesis entitled "**STUDIES ON DRY ROOT ROT DISEASE OF CHICKPEA [*Cicer arietinum* L.] CAUSED BY *Rhizoctonia bataticola* (Taub.) Butler**" submitted by **Mr. VEERENDRA KUMAR, K.V.**, in partial fulfillment of the requirement for the degree of **MASTER OF SCIENCE (Agriculture)** in **PLANT PATHOLOGY** of the College of Agriculture, Raichur, University of Agricultural Sciences, Dharwad is a record of research work done by him during the period of his study in this University under my guidance and supervision and the thesis has not previously formed the basis for the award of any degree, diploma, associateship, fellowship or other similar titles.

RAICHUR.
JANUARY, 2004



(Y.D. NARAYANA)
Major Advisor

Approved by:

Chairman:



(Y.D. NARAYANA)

Members: 1.

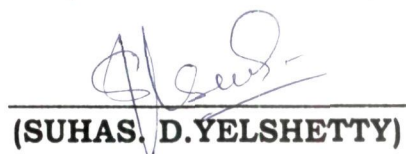


(M.K. NAIK)

2.

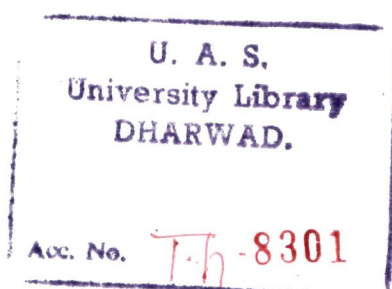
(P.S. DHARMARAJ)

3.



(SUHAS. D.YELSHETTY)

Released on — — — 1 4 MAY 2005



ACKNOWLEDGEMENT

With Regardful Memories....

It is a matter of pleasure and euphoria to glance back and recall the path one travels during the days of hard work and pre-perseverance and it is still great at this juncture to recall the faces and spirit in the form of parents, teachers, friends, near and dear ones. I would consider this work nothing more than incomplete without attending to the task of acknowledging the overwhelming help I received during this endeavor of mine.

*I consider myself fortunate to have worked under the guidance of the knowledge hungry, excellence pursuing, highly curious, ever co-operative and helpful personality, **Dr.Y.D. Narayana**, Associate Professor, Department of Plant Pathology, Agricultural Research Station, Gulbarga and the chairman of my advisory committee, he is a source of inspiration to me throughout the period of my investigation. I am grateful to him for his abysmal guidance, constant fomenting, punctilious and impeccable advice. Above all, his affectionate way of dealing with the thing throughout the course of my studies, which helped me to consummate the research work and with a grand success. I take this opportunity to express my heartfelt gratitude towards him. I had really a great pleasure and privilege, to be associated with him during this course of study.*

*I avail this opportunity to express my deep sense of reverence and gratitude to the members of my advisory committee namely **Dr. M.K. Naik**, Professor and Head, Department of Plant Pathology, College of Agriculture, Raichur for their constant help and suggestions during the investigation, sensible criticism in ameliorating this manuscript and valuable counsel during the period of study, with out which it would not have been possible to complete the thesis work.*

*I owe my thanks to **Dr. P.S. Dharmaraj**, Senior Scientist ARS, Gulbarga, **Dr. Suhas. D.Yelshetty**, Senior Scientist, Department of*

Entomology, ARS, Gulbarga, for their fruitful and constant help, valuable suggestions, sensible criticism, timely help during the study which enable the best of my ability in bringing this thesis.

I can not forget helpful and honest personalities, **Dr V.I. Benagi**, Extension Co-ordinator, KVK, Gulbarga and **Dr. V.B. Nargund**, Professor, Department of Plant Pathology, Agriculture College, Bheemarayanagudi, **Mr. M.B. Patil** and **Dr. Arun R. sataraddi**, Assistant Professors, Department of Plant Pathology, Agriculture College, Raichur.

Mr. R.K. Mesta, and **Mr. Gururaj Sunkad**, Junior Pathologists. RARS, Raichur, **Dr. M.N. Srinivas**, Professor of Agricultural Microbiology, **Mr. Mahadeva Swamy**, Assistant Professor of Agricultural Microbiology, College of Agriculture, Raichur, for their kind help at times.

It is my privilage to express my gratitude to **Dr. C.V. Patil** former DI (Agri.) and **Dr. B.Y. Kullaiswamy**, Director of Instruction (Agri.), College of Agriculture, Raichur, for their moral support during the course of study.

I felt the inadequacy of diction to express my sense of gratitude, heartfelt respect to my father Sri. **C.K. Veerabhadraiah** and mother Smt. **Varalakshmi**, Brothers **Sidda Raju, B.S, Satish Kumar** and **Shiva Kumar, K.V**, Sisters **Renuka, Bhagya** and my relatives for their needy inspiration with me. Without whose affection and moral support, I would not have come up to this level

Specially to my nearest and dearest chinnu, **Prapulla kumari** for her boundless love, needy inspiration, unshakable confidence, recurring encouragement with me, she is a source of inspiration to me throughout the period of my education, her help was evident at every stage of tension, anxiety and achievement, I dedicate this piece of work for their love and sacrifice.

My sincere thanks to **Mr. Lokesha, N.M.** for his moral support during my study and in completion of my research work. stay at Raichur, was

purposeful and meaningful, because of ocean of new friends, who are not legion in number, but more than a small group, to mention a few, Aswath, Padu, Krishnegowda, Venkatesh Murthy, Achari, Krishna Murthy, Kengegowda, Hanumanthappa, Ajith, Rafi, Veeru, Divyananda, Prasanna, Hitendra, Basava, Gogi and my junior friends, Vijay Kumar L, Chandramohan, Shivanna, Raju, Rudresh, Mithun, Praveen, Nataraj, Shivu, Goni, Nayak, Nemagowda and my senior friends Pramod, Prasad, Chidananda Swamy, Girish and Venkataramanappa, of course a long chain I remember always the warmth inspiration and encouragement.

I shall never forget the enormous and timely help rendered by Mr. Rehaman, Mr. Erannagowda, Smt. Yallamma, Moulanna and Narasimulu for their kind help, support in laboratory and field works.

It is my privilege to express gratitude to Department of Plant Pathology and College of Agriculture, Raichur and UAS Dharwad, for providing me an opportunity to complete my post -graduation.

I never forget the timely help by the staff of KVK and ARS, Gulbarga, during my survey work and in supplying the germplasm lines in conducting my research work.

I should not forget my friends and well wishers who have helped me directly or indirectly during my course of research work.

Finally, I wish to thank to Mr. Ramakrishna Shirwal, for neat and quick typing of this. Any omission of this brief of acknowledgement does not mean lack of gratitude.

Raichur
January, 2004


(VEERENDRA KUMAR, K.V.)

CONTENTS

CHAPTER NO.	PARTICULARS	PAGE NO.
I	INTRODUCTION	1 - 4
II	REVIEW OF LITERATURE	5 - 19
III	MATERIAL AND METHODS	20 - 40
IV	EXPERIMENTAL RESULTS	41 - 74
V	DISCUSSION	75 - 90
VI	SUMMARY	91 - 94
VII	REFERENCES	95 - 111
	APPENDIX	

LIST OF TABLES

Table No.	Topic	Page No.
1.	List of fungicides used for evaluation against <i>R.bataticola</i>	35
2.	Bio-agents used for evaluation against <i>R.bataticola</i>	38
3.	Chickpea genotypes used for screening against <i>R.bataticola</i>	38
4.	Incidence of dry root rot of chickpea in Gulbarga, Bidar and Raichur districts during 2002-2003	42
5.	Growth phase of <i>R.bataticola</i> on Potato Dextrose Broth (PDB) at different incubation period	45
6.	Effect of different solid media on mycelial growth of <i>R.bataticola</i>	47
7.	Growth characteristics of <i>R.bataticola</i> on different solid media at various periods	48
8.	Number of sclerotia of <i>R.bataticola</i> on synthetic/non-synthetic solid media on different incubation period	50
9.	Effect of different liquid media on dry mycelial weight of <i>R.bataticola</i>	52
10a.	Effect of temperature on mycelial growth and sclerotial production by <i>R.bataticola</i>	54
10b.	Effect of dry mycelial weight of <i>R.bataticola</i> on PDB at different temperature	55
11.	Effect of light on growth and sclerotial production of <i>R.bataticola</i>	57

Contd...2.

Table No.	Topic	Page No.
12.	Effect of relative humidity on mycelial growth and sclerotial production by <i>R.bataticola</i>	58
13.	Survival of the <i>R.bataticola</i> in soil, plant debris and refrigerator over a time	60
14.	Effects of dry root rot disease on various growth parameters of chickpea in sick pot culture under green house condition	62
15.	Effects of dry root rot disease on various growth parameters of chickpea under field condition	65
16a.	Efficacy of systemic fungicides on inhibition of growth of <i>R.bataticola</i>	67
16b.	Evaluation of non-systemic fungicides at various concentrations on growth inhibition on <i>R.bataticola</i>	69
17.	<i>In-vitro</i> evaluation of bioagents on inhibition of growth of <i>R.bataticola</i> in dual culture technique	70
18.	Evaluation of chickpea genotypes to dry root rot disease caused by <i>R.bataticola</i> in sick pot and blotter paper technique	73
19.	Categorization of chickpea genotypes screened against dry root rot by sick pot and blotter paper technique	74

LIST OF FIGURES

Fig. No.	Title	Between Page
1.	Incidence of dry root rot of chickpea in Gulbarga, Bidar and Raichur districts during 2002-2003	42 – 43
2.	Growth phase of <i>R.bataticola</i> on potato dextrose broth (PDB) at different incubation period	45 – 46
3.	Mycelial growth of <i>R.bataticola</i> on different solid media after four days of incubation	47 – 48
4.	Effect of dry mycelial weight on <i>R.bataticola</i> on different liquid media after 12 days of incubation	52 – 53
5.	Effect of temperature on mycelial growth and sclerotial production of <i>R.bataticola</i>	54 – 55
6.	Effect of temperature on dry mycelial weight of <i>R.bataticola</i>	55 – 56
7.	Effect of light on growth and sclerotial production of <i>R.bataticola</i>	57 – 58
8.	Effect of relative humidity on mycelial growth and sclerotial production by <i>R.bataticola</i>	58 – 59
9.	Efficacy of systemic fungicides on growth inhibition of <i>R.bataticola</i>	67 – 68
10.	Efficacy of non-systemic fungicides on growth inhibition of <i>R.bataticola</i>	69 – 70
11.	<i>In-vitro</i> inhibition of growth of <i>R.bataticola</i> by antagonists in dual culture technique	70 – 71

LIST OF PLATES

Plate No.	Title	Between page
1.	Growth of <i>R.bataticola</i> in Potato Dextrose Broth	32 – 33
2.	Mass multiplication of <i>R.bataticola</i> inoculum on sorghum grains	32 – 33
3.	Screening method used for identification of resistance	40 – 41
4.	Disease scoring scale used for identification of resistance	40 – 41
5.	Photograph showing drying of chickpea plants infected by <i>R.bataticola</i> in sick pot study	43 – 44
6.	Field dry root rot symptoms in comparison to healthy plants	43 – 44
7.	Chickpea plants infected by <i>R.bataticola</i> in Blotter paper study	45 – 46
8.	Microphotograph showing black sclerotia of <i>R.bataticola</i> at 10X	45 – 46
9.	Photograph showing pure culture of <i>R.bataticola</i> in slant	45 – 46
10.	Pure culture of <i>R.bataticola</i> obtained from infected chickpea plant in culture media	45 – 46
11.	Growth of <i>R.bataticola</i> on different solid media at 4 DAI	48 – 49
12.	Photograph showing early growth of <i>R.bataticola</i> in culture media	50 – 51

Contd...2.

Plate No.	Title	Between page
13.	Later growth showing sclerotial bodies initiation of <i>R.bataticola</i> on culture media	50 – 51
14.	Growth of <i>R.bataticola</i> on different liquid medium	52 – 53
15.	Growth of <i>R.bataticola</i> at different temperature levels	55 – 56
16.	Growth of <i>R.bataticola</i> in response to the varied light exposure period	58 – 59
17.	Photograph showing effect of relative humidity on growth of <i>R.bataticola</i>	58 – 59
18.	<i>In-vitro</i> inhibition of mycelial growth of <i>R.bataticola</i> by different systemic fungicide at different concentrations	70 – 71
19.	<i>In-vitro</i> inhibition of mycelial growth of <i>R.bataticola</i> by different non-systemic fungicides at different concentrations	70 – 71
20.	Effect of bio-agents on growth inhibition of <i>R.bataticola</i>	70 – 71

Introduction

I. INTRODUCTION

Chickpea (*Cicer arietinum* L.) is the third most important pulse crops after dry beans (*Phaseolus vulgaris* L.) and dry peas (*Pisum sativum* L.) and 15th among grain crops in the world. Chickpea being a self-pollinated diploid ($2n = 16$) is believed to be originated 7000 years ago in south eastern Turkey. Chickpea is predominantly used as dhal, for preparation of variety of snacks foods, sweets, condiments and fresh chickpea as vegetables because of its high protein digestibility and is richer in phosphorous and calcium. The germinated seeds of chickpea are rich in proteins and used as medicine to cure scurvy. Further Malic and oxalic acids collected from green leaves are prescribed for intestinal disorders. Soaked grain and husk are fed to cattle and horses as concentrate and roughage respectively (Anon., 1997). The chickpea crop has the ability to grow better with low inputs under harsh edaphic and arid environment than other crops. It is an important component in the cropping system of subsistence farming since it has the ability to derive 70 per cent of its nitrogen requirement through symbiotic nitrogen fixation.

In India, chickpea ranks second in area and third in production and can be grown in diverse soils and agro climatic conditions under conserved moisture. India is the largest producer of chickpea in the world covering 80 per cent area and exporter of chickpea to European countries. Rajasthan, Madhya Pradesh, Uttar pradesh, Haryana and Karnataka are

the major chickpea growing states accounting 80 per cent of total acreage and 84 per cent of total production with a productivity of 704 kg per ha with per capita consumption of 4.8 kg per year. Among the two types of chickpea cultivated, desi occupies 85 per cent area and 15 per cent of the area occupied by Kabuli with the introduction of high yielding varieties and irrigation in command areas, the acreage is nearing to high value crops like wheat, sunflower and mustard. Scope for increasing acreage is limited, to increase the production by avoiding major losses to diseases and pests (Masood Ali and Shivkumar, 2001).

In Karnataka, chickpea is grown in an area of 0.48 million hectares with a production of 0.29 million tonnes with the productivity of 606 kg/ha in the year 2001-2002 (Mishra, 2003). Among the several constraints affecting the productivity of chickpea 12-15 per cent loss in yields are due to wilt and dry root rot diseases (Masood Ali and Shivkumar, 2001).

About 50 pathogens have so far been reported on chickpea. Few of them are of great economic importance. These include Ascochyta blight (*Ascochyta rabiei* (Pass.) Lab.), Fusarium wilt (*Fusarium oxysporum* Schlecht emend. snyder and Hans. f. sp. *ciceri* (Padwick) Snyder & Hans). Dry root rot (*Rhizoctonia bataticola* (Taub.) Butler), Black root rot (*F. solani* (Mort.) Sacc.), Stem rot (*Sclerotinia sclerotiorum* (Lib.) de Bary), Collar rot (*Sclerotium rolfsii* Sacc.), Foot rot (*Operculella padwickii*

Kheswalla). *Botrytis* grey mold (*Botrytis cinerea* Pers. Ex. Fr.), rust (*Uromyces ciceris – arietini* (Grogg.) Jacz & Beyer) and stunt (pea leaf – roll virus). Among these, *Rhizoctonia bataticola* (Taub.) Butler, which causes dry root rot, is becoming severe in most of the chickpea growing regions of India (Nene *et al.*, 1981).

The dry root rot disease of chickpea caused by *R. bataticola* has wide range of hosts and is one of the most destructive plant pathogens in the tropics and subtropics (Dhingra and Sinclair, 1978). Disease development is favoured by dry soil conditions especially at flowering stage (whenever the crop is exposed to > 30° C (Singh and Mehrotra, 1982) in the presence of inoculum (Satish Lodha, 1998). This is a serious problem in late October to mid November, declines in December and January and increases again in February and March (Tripathi and Sharma, 1983).

Dry root rot of chickpea caused by *Rhizoctonia bataticola* appeared in chickpea during 1983 in CCS, Haryana Agricultural University, Bawal, India (Tripathi and Sharma, 1983). Hence, the disease is common in chickpea crop and becoming very severe in recent years because of its wide range of hosts. Whatever information available has been reviewed about the disease. However, very limited work has been done regarding incidence, cultural and physiological characters of the fungus, perpetuation of pathogen over time, effect of disease on growth parameters, integrated approach components like chemicals, bioagents

and resistant sources for effective management of this important soil borne disease, dry root rot disease caused by *R. bataticola*.

Keeping these points in background, the following investigations were undertaken in the present study:

1. Monitoring for incidence of dry root rot disease in chickpea growing areas of Gulbarga, Raichur and Bidar districts.
2. Isolation, cultural and physiological requirements of the pathogen.
3. Survival of the organism in root debris and soil over time.
4. Effect of disease on chickpea growth parameters.
5. *In-vitro* evaluation of efficacy of bio-agents and fungicides
6. Evaluation of chickpea genotypes for identifying resistance sources.

Review of Literature

II. REVIEW OF LITERATURE

Chickpea dry root rot disease caused by *Rhizoctonia bataticola* (Taub.) Butler. is one of the important diseases in Karnataka. *R. bataticola* is a well-known polyphagous, ubiquitous, omnivorous and non-target pathogen. These attributes have been made to this fungus since it is an economically important pathogen and *Macrophomina phaseolina* (Tassi) Goid has been reported to be the pycnidial stage of the fungus of *R. bataticola*. The literature reviewed here are partly pertaining to symptomatology, cultural characteristics, effect on growth, efficacy of fungicides and sources of resistance.

2.1 THE PATHOGEN AND SYMPTOMATOLOGY

Macrophomina phaseolina is an important pathogen on several crops in many parts of the world with a very wide host range (Reichert and Hellinger, 1947; Waller, 1976). The fungus was first observed as a cause of charcoal rot in sweet potato and was named as *Sclerotium bataticola* by Taubenhaus (1913). In 1925, Butler named the fungus as *Rhizoctonia bataticola* based on its sclerotial morphology and Ashby (1927) proposed the binomial *Macrophomina phaseoli* for the pycnidial stage of *R. bataticola*.

In 1947, Goidanich proposed the binomial *M. Phaseolina* which is currently, applied to both the micro sclerotial and pycnidial anamorph.

However, the micro sclerotial phase is the one predominantly observed and the named as *R. bataticola* and is a synonym.

Rhizoctonia bataticola, is the causal organism of dry root rot of chickpea, and was first observed in India by Haware and Nene (1978) in Hyderabad, Andhra Pradesh, India. Later, Tripathi and Sharma (1983) also reported the appearance of *R. bataticola* in CCS Haryana Agricultural University, Bawal, India.

Haware *et al.* (1990) reported that the disease development is influenced by dry soil conditions especially at flowering stage. The infected plant suddenly dries up in the field. The leaves and stems of affected plants are usually straw coloured. The taproot is dark and quite brittle in dry soil. The dark black sclerotia was observed in host tissues. Similarly, Sing *et al.* (1990) also reported that roots inoculated with *R. bataticola* showed disintegration of cortical tissues, while mycelium and sclerotial bodies plugging the xylem were seen in longitudinal sections. The sudden collapse of infected plants at the pod filling stage may be due to this plugging of the xylem vessels.

2.2 MODE OF INFECTION

Matana Srihuffagum *et al.* (1997) studied the systemic development of *M. phaseolina* from infected root part of blackgram seedlings with soil incorporated with 2 per cent inoculum of *M. phaseolina*

and soil moisture was kept at 56 per cent. The study revealed that the pathogen from infected roots invaded the stem tissues systemically and intercellularly at the rate of 4.4 cm a day. Its mycelia was located mostly in the xylem, phloem and pith but rarely in the cortex. The diseased plant was poor in vigour as a result of reduced soil moisture but no seed was found infected because the pathogen invaded the stem tissues and plants died abruptly within one to two days. Reciprocally, the abrupt death of the entire diseased plants terminated the growth and activities of the pathogen and subsequently initiated the formation of sclerotia in infected tissue.

2.3 SURVEY

A periodic survey conducted by Sharma *et al.* (1983) in Gwalior, reported that the incidence of dry root rot in all the five major chickpea producing districts of Madhya Pradesh causing yield loss ranging from 3.58 – 20.63 per cent. Later, Pandey *et al.* (1990) conducted the survey of chickpea diseases with reference to dry root rot in Allahabadh region reported that, the chickpea were infected by *R. bataticola*, *Fusarium oxysporum* f. sp. *ciceri*, and eighty three per cent of villages showed less than 25 percent disease intensity and remaining ones recorded 26 – 50 per cent disease intensity.

2.4 CULTURAL AND MORPHOLOGICAL STUDIES

2.4.1 Growth on solid Media

Vasudeva (1937) reported that *R. bataticola* grew equally well on Richard's agar, cotton root extract, synthetic agar, Brown's agar, but the sclerotia formation was favoured by cotton root extract and synthetic agar and was poor on Brown's agar.

Ghosh and Sen. (1973) reported that, Richard's agar supported good growth and sclerotia formation of *R. bataticola* as compared to PDA, Czapeck's Dox and Sabouraud's agar. On the contrary, Rose bengal agar was found to be the best media for isolation of *M. phaseolina*, propagules from infested soil (Shekar, 1983).

Waseer *et al.* (1990) isolated *M. phaseolina* from soybean and reported that, the pathogen grew profusely on PDA at 35° C, little growth occurred at either 20° C or 40° C. Later Sahi *et al.* (1992) observed that PDA supported the maximum colony diameter (9 cm) of *M. phaseolina*. Richard's agar supported maximum sclerotial production of the pathogen.

2.4.2 Growth in different liquid media

Shanmugam and Govinda Swamy (1973) reported that the maximum growth of the fungus in terms of dry mycelial weight in Richard's broth out of five media (Asthana and Hawker's, Brown's, Czapeck's – Dox, Leonian and Richard's) tested.

2.4.3 Growth phase in liquid media

Rhizoctonia sp. follows the classical growth pattern in liquid medium, having an initial period of accelerating growth followed by a phase of rapid growth and then decreases in growth due to autolysis (Israel and Ali, 1964).

In PDB, *R. bataticola* isolates varied with respect to attainment of maximum growth on potato dextrose broth. The isolates from bean, cowpea and safflower showed maximum growth on sixth day, cotton isolate on ninth day and sesamum and sunflower isolate on twelfth day (Anilkumar and Sastry, 1980). Byadgi and Hegde (1985) reported the significant variation in dry mycelial weight of six isolates of *M. phaseolina*., Bean isolate produced maximum growth with mean mycelial weight of 190 mg followed by bengal gram, cowpea, sorghum and soybean at 12 days after incubation.

2.4.4 Morphology of sclerotia

Ghosh and Sen (1973) recorded sclerotial diameter of justica, soybean, Bengal gram, jute, kochia, groundnut and pigeonpea isolates, which were 64.6 – 114.0 μm , 45.2 – 81.6 μm , 98.1 – 119.2 μm , 40.2 – 85.0 μm , 56.6 – 80.1 μm . 110 - 114 μm and 89.1 – 116.3 μm respectively.

Smits and Noguera (1988) used scanning electron microscope (SEM) and light microscopy and observed the sclerotial formation, which

began with branching and intertwining of adjacent hyphal filaments and an increase in size of associated cells and compaction of the sclerotial mats. The pycnidia state began with the merging of hyphal filaments towards a common point, followed by the development of ringed primordia and finally pycnidia. Later Dhar and Sarbhoy (1993) isolated the pod and seed isolate of soybean causing ashy stem blight and reported the smallest known sclerotia in the world (32 x 24 μm).

2.5 PHYSIOLOGICAL STUDIES

2.5.1 Effect of light

Vishwa Dhar and Sarbhoy (1989) opined that light had no effect on spore germination of *M. phaseolina* but germ tube elongation responded more to prolonged darkness.

Further, Vishwa Dhar *et al.* (1991) observed that, the exposure of *M. phaseolina* culture to alternate light/darkness for 12 hour each enhanced the sporulation. On the contrary, an increased darkness period had a detrimental effect on sporulation.

Muthukrishnan *et al.* (1995) noticed that, continuous light or light alternated with dark favoured the growth of *M. phaseolina in-vitro* on Czapek's Dox liquid medium.

2.5.2 Effect of Temperature

Singh and Mehrotra (1980) observed that, the incubation temperature of 35° C stimulated the mycelial growth of *R. bataticola*.

Vishwa Dhar and Sarbhoy (1989) observed that, the optimum temperature for germination and germ tube elongation of sclerotia was 20 to 30° C and maximum and minimum temperatures were 40° C and 5° C, respectively. The soilborne organisms such as *Fusarium oxysporum*, *Pythium aphanidermatum*, *Rhizoctonia bataticola* and *Rhizoctonia solani* were killed when exposed to temperature in the range of 42 – 63° C for various periods as reported by Anilkumar and Lakra (1990).

Sahi *et al.* (1992) noticed that, the maximum colony diameter (9 cm) and sclerotia formation of *M. phaseolina* was recorded at 30° C. Similarly, the maximum temperature of 35° C was ideal for growth of the pathogen, which also favoured maximum disease development (Amrit Sandhu *et al.*, 1999)

2.6 SURVIVAL OF THE FUNGUS

In the absence of host plants, *Rhizoctonia* survives either by active or inactive processes. In the inactive phase, it forms sclerotia, which can survive for a number of years (Byadgi, 1979).

Chowdary (1944) and Pitt (1964) reported sclerotial viability for 2½ years. While, Bhargava (1965) reported sclerotial viability for three years. Later Cook *et al.* (1973) and Dhingra and Sinclair (1975) reported that sclerotia of *M. phaseolina* remained viable for 12 to 18 months. Similarly, Watanabe (1973) observed that half-life of sclerotia in soil was 0.65 to 3.59 years. However, *M. phaseolina* depends largely on the sclerotial inoculum found either freely in soil or embedded in host tissue (Smith, 1969; Cook *et al.*, 1973 and Meyer *et al.*, 1973). Later Suriachandraselvam and Seetharaman (2000) reported the fungus *Rhizoctonia bataticola* survived for 31 months in the infected plant debris stored in ambient conditions.

2.7 EVALUATION OF MINIMUM INOCULUM FOR INFECTION

Garrett (1956) defined inoculum potential as the energy of growth of a parasite available for infection of a host, at the surface of the organ to be infected. The numbers of fungal propagules in the soil are important for causing root diseases.

Byadgi (1979) obtained considerable infection in bean seedlings with four per cent inoculum of *R. bataticola* and the infection increased with increase in inoculum levels. Maximum infection in cotton seedlings was obtained with eight per cent inoculum of *R. bataticola* (Ramamurthy, 1982). Similarly Pandey *et al.* (1999) reported that the competitive

saprophytic colonization of *Rhizoctonia bataticola* and *Rhizoctonia solani* of two facultative parasites appreciably increased as the concentrations of inoculum increased in the soil.

2.8 EVALUATION OF FUNGICIDES

Use of fungicides is an alternate method of managing the soil borne diseases in absence of resistant cultivars or when there is an out break of disease.

Anilkumar and Sastry (1980) studied the *in-vitro* efficacy of fungicides like captan, ceresan wet and dithane M-45 by poison food technique and they noticed fungus was less sensitive to captan but more sensitive to Dithane M-45. Similarly, Patil (1980) found that, Bavistin and Thiram were highly effective in inhibiting the growth of *M. phaseolina* of sorghum followed by brassicaol, agallol and captan.

Reddy *et al.* (1991) reported that, in an *in-vitro* study carbendazim was found to be most effective against *R. bataticola* followed by carbendazim + Thiram and captan. Later Singh and Kaiser (1994) reported that Carbendazim at 1000 ppm completely inhibited the growth of *M. phaseolina*.

Devi and Singh (1997) observed that fungicides such as carbendazim, thiophanate methyl and mancozeb were found effective *in-*

vitro conditions at 0.2 per cent concentration. Later Ebenezar and Wesely (2000) reported that among the six fungicides studied under *in-vitro* condition for their inhibitory effect on the growth of *M. phaseolina*. Carbendazim completely inhibited the growth of *M. phaseolina* at 0.1 per cent followed by copper oxychloride and thiram. Lokesh (2002) reported that in an *in-vitro* study Benomyl was found to be most effective against *M. phaseolina* followed by carbendazim and thiophanate methyl.

French bean seeds treated with benomyl and carbendazim @ 2 g/kg seed recorded the highest yield of French bean by reducing (39.4%) the incidence of ashy stem blight caused by *M. phaseolina* (Sharma, 1988).

Seed treatment with captan @ 3 g/kg seed reduced the root rot incidence of soybean by 20.2 per cent (Mohamed *et al.*, 1990). Similarly, Ahmed *et al.* (1992) noticed that, maize seeds treated with thiophanate methyl @ 2 g per kg seed was the best, followed by thiabendazole in reducing the intensity of maize stalk rot caused by *M. phaseolina* by 25.6 and 18.1 per cent and increased grain yield by 45.6 and 28.0 per cent respectively.

Black gram seeds treated with carbendazim (0.2%) effectively reduced the seedling mortality from 63.9 and 81.2 per cent to 8.9 and 11.2 per cent of two fungal isolates followed by thiophanate methyl and mancozeb (Devi and Singh, 1997).

Dwivedi and Dubey (1987) observed that soil application of Bavistin completely inhibited the survival of *M. phaseolina* @ 250 mg/g soil after 30 days, while such an effect was recorded with 1000 mg/g soil of Dithane M-45 and PCNB after 30 and 40 days incubation respectively.

Carbendazim and carboxin @ 0.2 per cent used as seed treatment, soil drench and seed treatment along with soil drench significantly reduced the incidence of dry root rot of chickpea, with a corresponding increase in phenolics and carbohydrates in the plants, which were followed by captan and thiram (0.2%). Seed treatment along with soil drenching was most effective in reducing the disease incidence followed by drenching and seed treatment alone (Rajinder Singh *et al.*, 1998). Soil drenching with Bavistin (0.2%) effectively reduced the charcoal rot of maize (*M. phaseolina*), which was followed by Bavistin and Topsin - M @ 0.2 per cent soil drenching (Savita Pareek *et al.*, 1998).

2.8.1 BIOLOGICAL CONTROL

Surprisingly, nature has achieved outstandingly successful biological control, which man has started perceiving rather dimly. The urgent need for increased crop production to feed the world's teeming millions will force for quick results perhaps away from biological control. Yet, he has finally began to realize that lasting success cannot be achieved by poisoning his environment and is increasingly turning to "natural control" by restoring a biological balance favorable to his crops Sanford (1926).

TH - 8301
University of Agricultural Sciences
University Library
Dharwad-5

Sanford (1926) was the first to realize that soil microorganisms themselves exert a natural biological control on root diseases. Then onwards there has been a dramatic increase in research efforts and several books (Baker and Cook, 1974; Huffaker and Messenger, 1976; Burges, 1981; Cook and Baker, 1983) and review articles (Naik and Sen, 1985 and Jayashree *et al.*, 2000) have come up, covering the use of specific microbes, their mechanisms and commercial evaluation. Use of antagonistic microorganisms against *Macrophomina* root rot has been well documented (Mukhopadyay, 1987; Raghuchander *et al.*, 1995; Jayashree *et al.*, 2000 and Rangeshwaran and Prasad, 2000) in several crops.

Trichoderma spp. represents the most thoroughly studied fungi that have shown antagonistic activities towards soilborne plant pathogens. The bio-control potential of *Trichoderma* spp. has been indicated in a number of reports (Jeyarajan *et al.*, 1991; Kehri and Chandra, 1991).

The mutant M₁ of *Trichoderma viridae* showed the highest *in-vitro* inhibition of mycelial growth (54.4%) and sclerotial germination (75.8%) of *M. phaseolina* followed by the mutant M₃ (Dinakaran and Marimuthu, 1988). The interaction between *T. harzianum* and *Rhizoctonia solani* was studied by both scanning (SEM) and transmission (TEM) electron microscopy. Macroscopic observations of fungal growth in dual cultures revealed that the pathogen growth inhibition occurred soon after contact with the antagonist. Scanning electron microscopy (SEM) investigations demonstrated that coiling of the

antagonist (*T. harzianum*) around *R. solani* was an early event preceding hyphal damage. The contact between the two fungi was mediated by a fine extra cellular matrix originating from cells of *R. solani*. Disorganization of the cell wall structure of *R. solani* appeared to be an early event that promoted internal osmotic imbalances that inturn triggered intracellular disorders, such as retraction of the plasma membrane and cytoplasm aggregation (Benhamou and Chet, 1993).

Nautiyal (1997) obtained 478 bacteria from chickpea rhizosphere competent *P. fluorescens* against *R. bataticola* and NBRI-7303 strain were found to inhibit major soil borne pathogens like *Fusarium oxysporum* f. sp. *ciceri*, *R. bataticola* and *Pythium* spp. Later Karunanithi *et al.* (2000) reported that *P. fluorescens* produced an antibiotic compound called pyrrolnitrin which inhibited the mycelial growth of *M. phaseolina* in dual culture by producing an inhibition zone of 12 mm.

Rangeshwaran and Prasad (2000) reported that chickpea seeds treated with talc based formulation of *P. fluorescens* (PDBCAB-29 and PDBCAB-30) effectively reduced the root rot incidence and recorded 100 per cent survival at 60 days after sowing, when compared to control (11%), Similar trend was noticed against *R. solani*. Similarly, Pradeep Kumar *et al.* (2000) observed that pigeonpea seeds treated with *T. viridae* completely eliminated seed borne infection of *R. bataticola* and *R. solani* with significant increase in seedling emergence, root and shoot length and

fresh weight of seedling when compared to *Chaetomium globosum* treated and untreated seeds as a control.

2.9 SCREENING FOR DISEASE RESISTANCE

2.9.1 Screening technique

Various techniques were employed by different investigations for inducing root rot symptoms in plants. The techniques included mixing of three weeks old wheat kernal culture at the rate of 50 g to sterile soil for each four pots (White, 1927); pouring a heavy mycelial fragment suspension to the plants grown in pots containing sterilized soil (Haymaker, 1928); dipping of injured roots of transplants in the mycelial suspension (Wellman, 1939); inoculating the soil with fungus culture and transplanting the seedlings (Schroeder *et al.*, 1953); sick plot and blotter paper technique (Nene *et al.*, 1981).

2.9.1.1 Resistant sources

Nene *et al.* (1989) reported that out of 225 *Cicer arietinum* genotypes evaluated for broad based resistance to *F. oxysporum f. sp. ciceris* and *R. bataticola*. The genotypes ICC-2862, ICC-9023, ICC-9032, ICC-10803, ICC-11550 and ICC-1551 were resistant to wilt and root rots at 11 to 12 of 18 locations tested.

Jayant *et al.* (1993) screened the chickpea genotypes for their resistance to *R. bataticola* at Jabalpur and found that the genotypes

BGM-416, BG-416, ICC-1376 and ICC-11314 showed resistant. Resistant cultivars had a greater number of lateral roots during early growth phases. Similarly, Gupta (1995) evaluated 300 diverse chickpea lines and compared for resistance to *Fusarium oxysporum f. sp. ciceris* and *R. bataticola* during 1990-92 in a multiple disease sick plot. Lines ICCV-89402, ICCV-90201, ICCV-90254, BG-209, H-88-2, GF-88426 and RSG-180 suffered less than 10 per cent incidence of both diseases and were considered as resistant.

Singh *et al.* (1996) reported field screening of chickpea cultivars against *F. oxysporum f. sp. ciceris* and dry root rot (*R. bataticola*) during 1990-92. Out of 1100 chickpea genotypes, 184 genotypes were found resistance to both the diseases. Genotypes like H83-125, G-186012, GC-186143, H90-22, H91-35 and H91-37 were identified as resistant to wilt and root rot disease.

Gangwar *et al.* (2002) reported that among thirty-five chickpea cultivars screened for resistance to dry root rot caused by *R. bataticola* in a field experiment at IIPR Kanpur, IICC-2644, IICC-10384, IICC-112244, IICC-11332, ICM-81002, ICCL-81010, ICC-12263 and ICC-12441 were resistant. Where as GCP-9504, G-96020, G-96105, G-96313 and GL-91059 were moderately resistant to dry root rot.

Material and Methods

III. MATERIAL AND METHODS

The present investigations on dry root rot of chickpea caused by *Rhizoctonia bataticola* were carried out during 2001-2003 in the laboratory of the Department of Plant Pathology, Agriculture College, Raichur. The details of the material used and methodology followed during the course of the present investigation are described under this chapter.

3.1 GENERAL PROCEDURE

3.1.1 Glassware and Cleaning

Corning and Borasil glasswares were used for all the experiments carried out. Before use, the glasswares were kept in the cleaning solution for a day containing 60 g of potassium dichromate [$K_2Cr_2O_7$] and 60 ml of concentrated sulphuric acid [H_2SO_4] in one litre of water. They were washed with detergent [vim] powder followed by washing in running tap water and then rinsed in distilled water before use.

3.1.2 Sterilization

Sterilization of glassware and media were achieved by autoclaving at 1.1kg/cm^2 (121.6°C) for 20 minutes. Soil mixture were sterilized by autoclaving at 1.1kg/cm^2 for one hour for three consecutive days. The seeds and plant tissues were surface sterilized in 1:1000 mercuric chloride solutions. All cultural studies of the pathogen were

carried out under aseptic conditions under laminar flow. The tips of inoculation needle and forceps were flame sterilized.

3.2 STUDY MATERIAL

3.2.1 Monitoring for Incidence of Dry Root Rot Disease of Chickpea during 2002

An Intensive roving survey was conducted during December 2002 to know the incidence (%) and severity of dry root rot on chickpea in the farmers field at Raichur, Gulbarga and Bidar districts. In each district a minimum of five villages and in each village ten fields were examined. Then the incidence was calculated and expressed in percent by using the formula mentioned below.

$$\text{Per cent disease incidence} = \frac{\text{Number of plants infected}}{\text{Total number of plants examined}} \times 100$$

3.2.2 Collection and isolation of the fungus

Dry root rot affected chickpea plants were collected from Agriculture Research Station (ARS) Gulbarga. The isolation of the fungus was done by following standard tissue isolation method under aseptic condition. The infected tissues of the root (dark coloured Sclerotial bodies) were cut in to small bits of size 1-2 mm and surface sterilized in 1:1000 mercuric chloride solution for a minute and then washed repeatedly thrice in sterile distilled water to remove the traces of mercuric chloride before

transferring them to sterile potato dextrose agar [PDA] slants under aseptic conditions and incubated at $30 \pm 1^\circ\text{C}$ for growth of the fungus.

3.2.3 Hyphal tip Culture

Ten ml of clean-filtered two per cent water agar solution was poured in to sterile petriplates and allowed to solidify. A small bit of the fungal culture from the periphery of agar slants was picked and placed on solidified water agar in Petriplate under aseptic conditions. After 16-18 hours of incubation at $30 \pm 1^\circ\text{C}$, the plates were examined and newly growing hyphal tip was located and marked on the lower surface of the petriplate. The portion of the agar directly below the mark was cut and transferred to fresh PDA slants in such a way that the hyphal tip bearing surface was in contact with PDA surface and incubated at $30 \pm 1^\circ\text{C}$.

3.2.4 Maintenance of Culture

The fungus was sub cultured on PDA slants and allowed to grow at $30 \pm 1^\circ\text{C}$ for ten days. Such slants were preserved in refrigerator at 4°C and maintained by subculturing once in 20 - 25 days interval. Such a culture was used through out the study.

3.3 PATHOGENICITY TEST

“Blotter paper technique” was employed for proving the pathogenicity of *Rhizoctonia bataticola* on chickpea (Nene *et al.* 1981).

The fungus was cultured on PDB. Twenty ml of PDB was poured into the 100 ml conical flasks and sterilized. The flasks were then inoculated and incubated for 10 days. The mycelial mat from the flask was removed and macerated in a waring blender along with distilled water for a minute. The inoculum was later collected in a beaker. In the mean time, the chickpea seedlings were raised in polythene bags containing sterilized sand mixed soil. The two weeks old seedlings were uprooted and the roots were immersed in sterile water in order to remove the soil particles, the seedlings were then immersed completely in the inoculum in a beaker for a minute. The seedlings were then placed side by side on a blotter paper (45 cm x 25 cm with one fold), in such a way that only the cotyledons and roots are covered and the green tops of seedlings remain out side and then blotter paper was folded. The folded blotter paper were then placed in trays, and kept in an incubator at 35° C for 8 days. The blotters were moistened every day. At the end of the incubation period the seedlings were examined for the extent of root damage. The fungus was then re- isolated on PDA slants, incubated for 4 days and compared with initially obtained culture.

3.3.1 Identification of the fungus

The morphology, cultural characters, formation of sclerotia, and branching of mycelium were the principal characters considered for identification of pure culture of *Rhizoctonia bataticola*. These characters

were studied and compared with the description given by Ashby (1927) and Goidanich (1947) and identified as *Rhizoctonia bataticola* (Taub.) Butler.

3.4 Cultural characters of *Rhizoctonia bataticola*

3.4.1 Cultural characters on different solid media

The cultural characters of *Rhizoctonia bataticola* were studied on the following five different solid media.

1. Potato dextrose agar (PDA)
2. Czapeck's agar
3. Malt extract agar
4. Oat meal agar
5. Rose Bengal agar

The composition and preparation of the above mentioned synthetic and non-synthetic/semi synthetic media were obtained from Ainsworth and Bisby's "Dictionary of the fungi " by Hawksworth *et al.* (1983). The composition and preparation of the media are given below.

1. Potato dextrose agar (PDA)

In most of the experimental studies, PDA medium was used. The composition of PDA is as follows,

Peeled potato	200.00 gm
Dextrose	20.00 gm
Agar - agar	20.00 gm
Distilled water	1000 ml (volume to make up)

Two hundred grams of peeled potato was cut into small bits and boiled in distilled water and extract was collected by filtering through muslin cloth. Dextrose 20 gram and agar agar powder 20 gram each were dissolved in potato extract and the final volume was made up to 1000 ml with distilled water. Later it was sterilized at 1.1 kg/cm² pressure for 15 minutes and preserved for future use.

2. Czapeck's agar

Sucrose	30.00 gm
Sodium nitrate (NaNO ₃)	2.00 gm
Potassium dihydrogen phosphate (KH ₂ PO ₄)	1.00 gm
Magnesium sulphate (MgSO ₄ 7H ₂ O)	0.50 gm
Ferric chloride (FeCl ₃ 6H ₂ O)	0.01 gm
Potassium chloride (Kcl)	0.50 gm
Agar - agar	20.00 gm
Distilled water	1000 ml (volume to make up)

Agar - agar was melted in 500 ml distilled water. The other ingredients were dissolved in remaining 500 ml of distilled water. The two

solutions were mixed thoroughly and the volume was made up to one litre. The medium was sterilized at 1.1 kg/cm² pressure for 15 minutes.

3. Malt extract agar

Malt extract	20.00 gm
Agar - agar	20.00 gm
Distilled water	1000 ml (volume to make up)

All the ingredients were dissolved in 400 ml distilled water and agar was dissolved separately in 500 ml of distilled water and mixed with the above solution and the volume was made up to one litre. The medium was sterilized at 1.1-kg/cm² pressure for 15 minutes.

4. Oat meal agar

Oat flakes	40.00 gm
Agar - agar	20.00 gm
Distilled water	1000 ml (volume to make up)

The oat flakes were boiled in 500 ml of distilled water for half an hour and filtered through a muslin cloth. Agar was melted in 500 ml of distilled water separately. Both the solutions were mixed thoroughly. Then the volume was made up to one litre and sterilized.

5. Rose bengal agar

Rose Bengal	0.03 gm
Magnesium sulphate (MgSO ₄)	0.5 gm
Potassium dihydrogen phosphate (KH ₂ PO ₄)	1.00 gm
Peptone	5.00 gm
Glucose	10.00 gm
Agar - agar	20.00 gm
Distilled water	1000 ml (volume to make up)

All the ingredients were dissolved in 400 ml of distilled water and agar was dissolved separately in 500 ml of distilled water and mixed with the above solution and the volume was made up to one litre. The medium was sterilized at 1.1 kg/cm² pressures for 15 minutes.

3.4.2 Growth of *R. bataticola* in different liquid media.

The composition and preparation of different liquid media used were same as that of solid media, but with out adding agar. 20 ml of each medium was added to each of the 250 ml flasks. Each medium was replicated thrice. After sterilization, the flasks were inoculated with 5mm disc of *R. bataticola* from 10 days old well grown culture and incubated at $30 \pm 1^{\circ}$ C. After the incubation period the cultures were filtered through previously weighed Whatman No. 42 filter paper discs of 9.0 cm diameter, which were dried to a constant weight at 60^oc in an electric oven prior to filtration. The mycelial mat on the filter paper was thoroughly washed with

sterile distilled water to get rid of the salts likely to be associated with mycelium. The filter paper along mycelial mats were dried to a constant weight at 60° C and then cooled in dessicator and weighed immediately on an electronic balance. Results were analyzed statistically.

3.4.3 Growth on solid media

Twenty ml each of solid media was poured into 90 mm diameter Petriplates. Such medium was replicated thrice. Five mm culture discs of the fungus was taken from the periphery of the 10 days old colony grown on PDA and placed at the center of the plates. Such plates were incubated at $30 \pm 1^{\circ}$ C for four days. Colony diameter was recorded for each plate. Observations on the various cultural characters *viz.* type of margin, colour and surface elevation and number of sclerotial bodies formed per microscopic field (10X) at three spots in a Petriplate and size of the sclerotial bodies were recorded by using standard stage and ocular micrometer. The results were analyzed statistically.

3.4.4 Growth phase of *R.bataticola* on liquid media (PDB)

Twenty ml of potato dextrose broth (PDB) was poured into each of the 150 ml conical flasks and these flasks were sterilized at 1.1 kg/cm² for 20 minutes. The inoculum discs of 5 mm diameter of 10-day-old culture were transformed aseptically into the PDB flasks separately. Each treatment was replicated thrice. The flasks were incubated at $30 \pm 1^{\circ}$ C.

Three flasks were harvested separately at a time, starting from fourth day onwards up to twenty second day by leaving a gap of 48 hours between the two successive harvests. After the incubation period, the dry mycelial weights were recorded as described earlier.

3.5 PHYSIOLOGICAL STUDIES ON *R. bataticola*

3.5.1 Effect of temperature levels on growth of *R. bataticola*.

The growth of *R. bataticola* was tested at 15, 20, 25, 30, 35 and 40° C. Twenty ml potato dextrose broth was dispensed and sterilized in 150 ml flasks. Each flask was inoculated and incubated for 12 days in incubators adjusted to required temperature levels mentioned above. Each treatment was replicated thrice. After the incubation period, the dry mycelial weights were recorded as described earlier.

3.5.2 Effect of temperature levels on growth and sporulation of *R. bataticola*.

The growth of *Rhizactonia bataticola* was tested at 15, 20, 25, 30, 35 and 40° C. Twenty ml of PDA was poured into the petriplate aseptically and allowed to solidify. Then the plates were inoculated with 0.5 cm mycelial discs of *R. bataticola* on to the center of the petri plates. Each treatment was replicated thrice. The plates were incubated for 6 days in incubators adjusted to required temperature levels mentioned above. After the

incubation period, the mycelial growth (mm) and number of sclerotia per microscopic field (10X) was recorded.

3.5.3 Effect of light on growth and sporulation of *R. bataticola*

Ten ml of sterilized PDA was poured into sterilized Petri plates aseptically and allowed to solidify. Then the plates were incubated with 0.5 cm mycelial discs from the 10 days old culture of the fungus. The incubated plates were exposed to continuous light and continuous darkness for 4 days, alternate light and dark condition of 12 hours each for 4 days. After 4 days of exposure mycelial growth and number of sclerotia per microscopic field (10X) were recorded.

3.5.4 Effect of Relative humidity on growth and sporulation of *R. bataticola*

To test the optimum relative humidity required for the growth and sporulation of the fungus, five relative humidity levels *viz.* 65, 75, 85, 95 and 100 per cent were maintained by following method described by Booth (1971). From 10 days old culture 0.5 cm discs were inoculated on potato dextrose agar plates and incubated at different relative humidity levels for 6 days in desicator. Five mm discs were removed from each humidity level after incubation and number of sclerotia per microscopic field (10X) was recorded.

Different humidity levels were maintained under laboratory conditions using sodium chloride as described by Booth (1971).

Percent relative humidity at

<u>$30 \pm 1^{\circ}\text{C}$</u>	<u>NaCl (g)</u>
65	56
75	40
85	24
95	8
100	0

3.6 SURVIVAL STUDIES

Naturally infected chickpea plants were collected from Agriculture Research Station (ARS), Gulbarga during *rabi* 2002. The infected root pieces were cut into small pieces (2 to 3 cm) and buried in earthen pot containing sterilized soil, stored under laboratory conditions and in a refrigerator. The samples were retrieved at 30 days interval from stored infected root. The collected samples were surface sterilized and plated on PDA. After incubation period the observations on mycelial growth and sclerotia present per microscopic field was recorded. The sampling procedure continued for a period of 6 to 7 months to check the survivalability of the pathogen under different storage conditions.

3.7 MASS MULTIPLICATION OF INOCULUM OF *R. bataticola*

3.7.1 Mass multiplication of *R. bataticola* on PDB

The pathogen was mass multiplied on potato dextrose broth (Plate-1). One hundred ml of PDB was taken into 250 ml of (borosil) conical flask and sterilized at 1.1 kg/cm² for 20 minutes. The mycelium disc of 0.5 cm cut from the margin of 10 days old culture grown on petridish which was inoculated into PDB under aseptic condition. The flasks were then incubated at 30 ± 1° C for two weeks. The mycelial mats were collected after incubation period by filtering through Whatman No. 42 filter paper. The mycelial mat was washed with sterile water and used for further experiments.

3.7.2 Mass multiplication on sorghum grain

The fungus was multiplied on sorghum grain medium at 30 ± 1°C for three weeks and used as soil inoculum. The medium was prepared by soaking the surface sterilized sorghum grains for overnight with 2 per cent sucrose solution and water drained off. One hundred grams of sorghum grains were packed in autoclavable 500 ml conical flask, wrapped the conical flask with cotton plug and then autoclaved at 1.1 kg /cm² for half an hour. The sterilized sorghum grains were inoculated with *R. bataticola* under aseptic conditions and incubated for three weeks (Plate-2). The inoculum was added to the sterilized soil at the rate of 4 per cent for making the soil sick.



Plate 1. Growth of *R.bataticola* in Potato Dextrose Broth

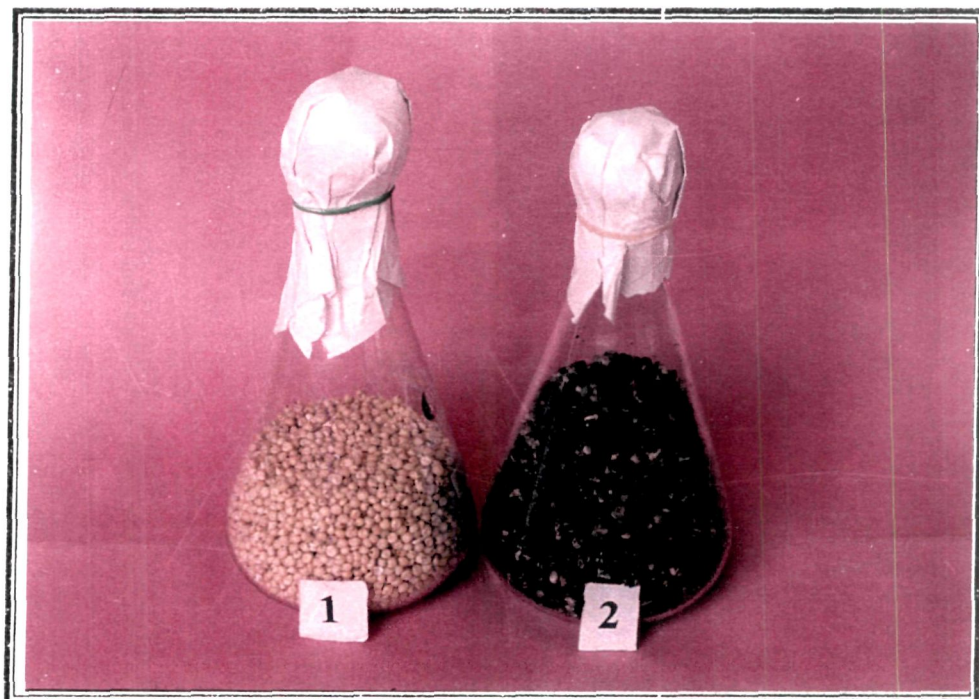


Plate 2. Mass multiplication of *R.bataticola* inoculum on sorghum grains

3.8 Evaluation of minimum inoculum level for infection

Finely powdered soil sieved through 2 mm sieve was sterilized at 15 lb for 30 minutes for three consecutive days and mixed with inoculum of the fungus grown on sterile sorghum grain mixture to give 1, 2, 3 and 4 per cent by weight of inoculum in the infested soil.

Number of earthen pots of size 20 cm x 17 cm were taken and washed with 5 per cent copper sulphate solution. 2 kg of soil were taken in each pot and these soils were mixed with 80, 60, 40 and 20 grams of inoculum were added to give a 4, 3, 2 and 1 per cent respectively. Uninoculated sterile soil served as a control. The healthy chickpea seeds of variety Annigeri-1 were surface sterilized with 1:1000 mercuric chloride solutions for 2 minutes washed thrice in tap water and were sown in the pots at the rate of 15 seeds per pot. The pots were watered to 50 per cent moisture holding capacity. The loss in soil moisture was made up by adding whenever required.

Observations were recorded 15 days after sowing by counting number of seedlings survived. Both pre and post emergence death of seedlings were used for calculating percent infection.

3.8.1 Effect of dry root rot disease on growth parameter

Effect of disease on chickpea growth parameters was studied based on standard inoculation technique. One, two, three and four percent

concentration of fungus culture was mixed to each kg of sterilized soil, then chickpea seeds were be sown in sick pots with suitable control without inoculation of fungus. Observations regarding days required for disease development, intensity of the disease and growth parameters like plant height, number of branches and biomass both in healthy and diseased plants were recorded both in flowering and pod formation stage under pot culture technique.

3.9 DISEASE MANAGEMENT

3.9.1 Evaluation of fungicides

3.9.1.1 *In vitro* evaluation

The efficacy of four Non-systemic and five systemic fungicides were tested against the pathogen. Under *in-vitro* conditions The non - systemic fungicides were tested at 1000,2000 and 3000ppm. Whereas systemic fungicides were evaluated at 250,500 and 1000ppm respectively. The list of fungicides used along with their chemical and trade names are presented in Table1.

3.9.1.2 Poison food technique

In the poison food technique, (Sharvelle, 1961) the fungus *R.bataticola* was grown on PDA medium in Petriplates for 10 days prior to setting up of the experiment. The fungicidal formulations were added to melted PDA medium to obtain the desired concentrations on the basis of

Table 1: List of fungicides used for evaluation against *R.bataticola*

Common name	Trade name	Chemical name
Systemic fungicides		
Carbendazim	Bavistin 50 WP	Methyl-2 benzimidazole carbamate
Benomyl	Benlate 50 WP	Methyl-N (1 butyl carbamyl)- 2 benzimidazole carbamate
Hexaconazole	Contaf 5 EC	(RS) 2- (2,4 – dichlorophenyl) –1 (1H-1, 2,4-triazol-1-yl) hexan-2-01
Thiophanate methyl	Tospin M-70 WP	1,2-bis (3 methoxy carbonyl-2) thiourea benzene
Carboxin	Vitavax 75 WP	2,3, dihydro-5 carboxy anilidac-6-methyl, 4, oxathin
Non-systemic fungicides		
Captan	Captan 50 WP	Trichloro methyl thio-4-cyclohexene 1-2, dicarboximide
Thiram	Thiridae 75 WP	Tetra methyl thiuram disulfide
Mancozeb	Dithane M-45 75 WP	A combination of manganese and zinc ethylene bis dithio carbamate
SAAF (Carbendazim 12% + Mancozeb 63%)	SAAF 75 WP	A combination product of Carbendazim and Mancozeb

active ingredients present in the chemical. Twenty ml of poisoned medium was poured in each sterilized petriplates. Control was maintained without addition of fungicides. Half cm mycelial disc taken from the periphery of ten days old culture was placed in the center of the petriplate on the solidified poisoned PDA and incubated at $30 \pm 1^{\circ} \text{C}$ till the full growth of the fungus. Three replications were maintained for each treatment. The diameter of the colony was measured in two directions and average were recorded. Percent inhibition of growth was calculated by using the formula given below by Vincent (1947)

$$I = \frac{C - T}{C} \times 100$$

Where,

I= Per cent inhibition.

C= Growth in control.

T= Growth in treatment.

3.10 EVALUATION OF BIO-AGENTS

Bioagents obtained from the Project Directorate of Biological Control (PDBC), Bangalore (Table 2) were evaluated for their efficacy under *in vitro* using dual culture technique against *R.bataticola*..

3.10.1 Dual culture test

The bio agents were evaluated for their efficacy through dual culture technique. The fungal bio agents (*T.harzianum*, *T.viride* and

T.virens) and the test fungus were inoculated simultaneously side by side on a single Petriplate containing solidified PDA medium. Three replications were maintained for each bioagent with one control maintaining only pathogen and bioagent. They were incubated for six days. The diameter of the colony of both bio agents and the pathogen was measured. The percent inhibition of growth of the test pathogen was calculated as described earlier.

The efficacy of *P.fluorescens* was tested by streaking bacteria at one side of the Petriplate opposite to the test pathogen. The growth of the fungus was inhibited when it started growing towards the bacterial colony on PDA. The inhibition zone was measured from the edge of test fungal mycelium to the edge of bacterial colony after six days of incubation and expressed as percent inhibition over control.

3.11 Screening of chickpea genotypes

Twenty-five genotypes collected from ARS, Gulbarga (Table 3) were screened against dry root rot using following technique.

3.11.1 Sick pot technique

Chickpea seeds were surface sterilized with HgCl_2 (0.1%) for a minute and washed repeatedly thrice in sterile water to remove the traces of HgCl_2 . Fifteen seeds were sown in each pot containing sterilized black soil incorporating the fungal culture at the rate of 40 gm per kg soil. Each

Table 2: Bio- agents used for evaluation against *R.bataticola*.

Sl.No.	Bio- agents	Source
1.	<i>Trichoderma harzianum</i> (PDBCH-10)	PDBC, Bangalore
2.	<i>Trichoderma virens</i>	PDBC, Bangalore
3.	<i>Trichoderma viride</i> [PDBCH – 24]	PDBC, Bangalore
4.	<i>Pseudomonas fluorescens</i>	PDBC, Bangalore

Table 3: Chickpea genotypes used for screening against *R.bataticola*

Sl. No.	Genotypes	Sl. No.	Genotypes
1.	ICCV – 98933	14.	GPC - 14
2.	ICCV – 4958	15.	Annigeri – 1
3.	ICCV - 98904	16.	BG – 362
4.	ICCV – 98936	17.	GPC – 21
5.	ICCV – 94916-8	18.	GPC – 3
6.	ICCV – 98901	19.	GPC – 32
7.	ICCV – 98906	20.	GPC – 13
8.	ICCV – 98905	21.	GPC – 42
9.	ICCV – 98907	22.	GPC – 20
10.	ICCV – 94924-3	23.	MLT – D – 14
11.	PBG – 68	24.	AVT -5
12.	GPC – 12	25	GPC - 23

treatment was replicated thrice. Data on germination of seeds and root rot incidence was recorded.

3.11.2 Computation of disease index

The disease incidence was computed by taking in to consideration of pre- and post- emergence seedling death. Pre- emergence and post emergence seedling deaths were calculated by employing formula.

$$\text{Pre emergence Seedling death} = \frac{\text{No. of seedlings emerged in healthy soil} - \text{No. of seedlings emerged in sick soil}}{\text{No. of seedlings emerged in healthy soil}} \times 100$$

$$\text{Post emergence seedling death} = \frac{\text{No. of seedlings killed}}{\text{No. of seedlings emerged}} \times 100$$

In sick pot technique the genotypes were evaluated based on percent incidence using disease scale (Nene *et al*, 1981).

Reaction	Rating (Percent incidence)
Immune	0
Resistant	1 – 10
Moderately resistant	11 – 25
Susceptible	26 – 50
Highly susceptible	51 - 100

3.11.3 Blotter paper technique

Blotter paper technique (Plate-3) developed by Nene *et al* (1981) was used to screen different chickpea genotypes. The fungus was cultured on PDB. The flasks were then inoculated and incubated for 10 days. The procedure for inoculation was already described previously under chapter 3.3.

Further, the genotypes were evaluated based on percent disease index using a 1-9 scale (Plate-4) as described by Nene *et al* (1981).

Grade	Reaction	Symptoms	Percent disease
1 - 2	Immune	No infection on roots	0
3 - 4	Resistant	Very few small lesions on roots	10% or less mortality
5 - 6	Moderately resistant	Lesions on roots clear, but small. New roots are free from infection	11 - 20%
7 - 8	Susceptible	Lesions on roots many. New roots generally free from lesions	20 -50%
9	Highly susceptible	Roots infected and completely discolored	51% or more mortality

The percent disease index was calculated by using formula given by Wheeler (1969).

$$\text{Per cent disease Index (PDI)} = \frac{\text{Sum of numerical ratings}}{\text{Total number of Observations}} \times \frac{100}{\text{Maximum disease Grade}}$$



Plate 3. Screening method used for identification of resistance



Plate 4. Disease scoring scale used for identification of resistance

Experimental Results

IV. EXPERIMENTAL RESULTS

The present investigations on dry root rot (*R.bataticola*) disease of chickpea with reference to the survey for the disease incidence, isolation, cultural and physiological studies, survival of the fungus, effect of disease on chickpea growth parameters, *in-vitro* evaluation of fungicides, biocontrol agents, screening of chickpea genotypes and their studies were carried out at the Department of Plant Pathology, College of Agriculture, Raichur. The results of the investigations presented here under.

4.1. SURVEY FOR INCIDENCE OF CHICKPEA DRY ROOT ROT DISEASE

An intensive roving survey was conducted in different taluks of Gulbarga, Bidar and Raichur districts in 2002, where the chickpea is predominantly grown during *rabi* season, to assess the incidence of dry root rot of chickpea. The data pertaining to survey are given in Table 4; Fig .1

The results revealed that, the highest severity of dry root rot incidence was observed in Gulbarga district (9.97%) followed by Raichur (5.16%) and Bidar district (4.28%). In Gulbarga district, the maximum dry root rot incidence was recorded in Chittapur (12.21%) and the least incidence of 7.09 per cent was recorded in Sedam taluk. However, in Bidar district the maximum and minimum dry root rot incidence was

Table 4. Incidence of dry root rot of chickpea in Gulbarga, Bidar and Raichur districts during 2002-03.

District	Place	No. of fields	Percent Dry Root Rot Incidence	
			Range	Average
Gulbarga	ARS, Farm	10	11-12	11.21
	Chittapur	18	6-13	12.21
	Sedam	23	5-8	7.09
	Aland	20	6-9	8.19
	Afzalpur	20	10-15	11.19
	Mean			9.97
Bidar	Humnabad	20	4-5	4.31
	Nirana	15	3-4	3.75
	Hanikuni	10	4-6	5.11
	Manakhalli	10	4-7	5.03
	Hudugi	10	3-4	3.21
	Mean			4.28
Raichur	Deodurga	15	4-5	6.11
	Mandalagi	10	4-8	6.75
	Jalahalli	10	3-6	4.63
	Devadakallu	10	5-7	5.13
	RARS Farm	05	3-4	3.21
	Mean			5.16
Range				6.47

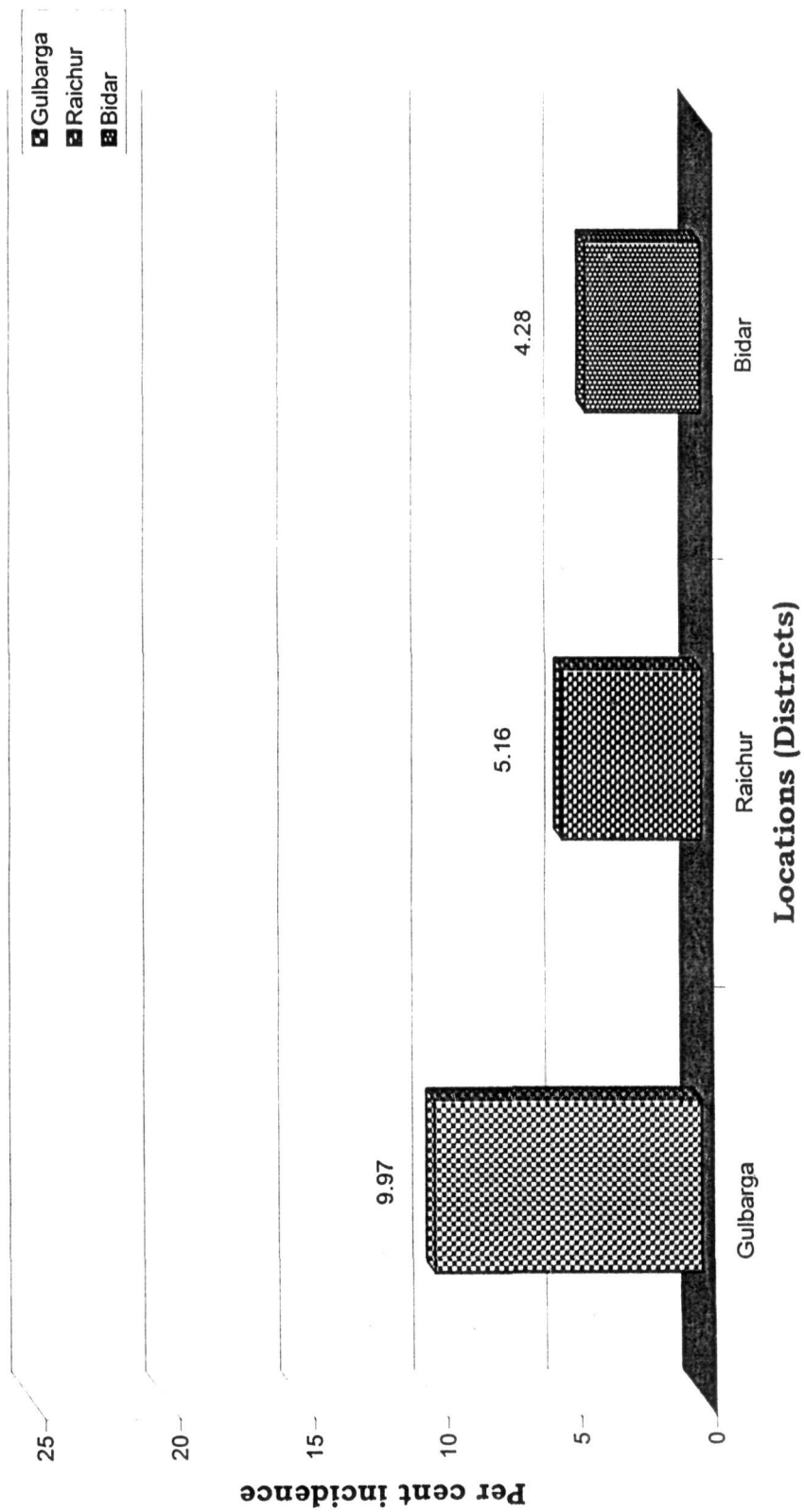


Fig.1. Incidence of dry root rot of chickpea in Gulbarga, Bidar and Raichur districts during 2002-2003

observed in villages of Hanikuni (5.11%) and Hudugi (3.21%), respectively. Whereas in Raichur district the dry root rot incidence was maximum (6.75%) in Mandalagi village and minimum (3.21%) was recorded in Regional Research Station Farm, Raichur. The average dry root rot incidence of chickpea was 6.47 per cent with range of 3.21 to 12.21 per cent.

4.2 SYMPTOMATOLOGY

Chickpea plants infected by dry root rot were characterized by drooping and drying of leaves (plate-5). The roots system of infected plant showed dark and extensive rotting, destroyed lateral roots, roots become brittle and shredding of bark was also seen. Drooping of petioles and leaflets at top and plants appeared straw coloured. The diseased plants could be very easily pulled out of the soil and formations of black coloured sclerotial bodies on root and stem portion of plant were observed (Plate-6).

4.3 PROVING THE PATHOGENCITY

Blotter paper technique was employed for proving the pathogenicity. In Blotter paper technique, the leaves started drying on the infected plant and they remain attached to the plant, root were rotten and formation of sclerotial bodies on the root portions were seen 12 days after inoculation.. Reisolation from diseased plants consistently yielded the *R. bataticola*. The uninoculation plant appeared healthy without any root rots (Plate-7).



Plate 5. Photograph showing drying of chickpea plants infected by *R.bataticola* in sick pot study



Plate 6. Field dry root rot symptoms in comparison to healthy plants

4.4 IDENTIFICATION OF THE FUNGUS

The morphological and cultural characters of *R. bataticola* were studied for identification of the fungus. The mycelium appeared white in the initial stages of growth, but soon turned to black colour as and when the sclerotia formation initiated (Plate-8). The germination of sclerotia and right angle branching of mycelium was noticed with a small constriction at the basal part of branched hyphae. The fungus developed in to black colour on PDA four days after inoculation.

4.5 CULTURAL STUDIES

4.5.1 Maintenance of culture of *R. bataticola*

The culture of *R. bataticola* on chickpea were isolated and their pure culture was maintained on PDA (Plate 9 & 10). Further, the results are presented here under. The culture of *R. bataticola* was further identified by sending the culture to India Type Culture Collection (ITCC), IARI, New Delhi.

4.5.2 Growth at different incubation period

The results of the growth phase of *R. bataticola* after incubation (Table 5, Fig 2) over a period of time (4 – 22 days) showed a significant difference between incubation periods and dry mycelial weight. The maximum dry mycelial weight was observed at 16 days after incubation (251.16 mg). This was significantly superior over other treatments. Dry

Table 5. Growth phase of *R. bataticola* on Potato dextrose broth (PDB) at different incubation periods.

Incubation period (Days)	Dry mycelial weight (mg)*
4	58.03
6	113.80
8	137.00
10	164.66
12	192.06
14	204.83
16	251.16
18	190.00
20	179.26
22	142.73
S.Em $\pm$	1.00
CD at 1%	4.069

* Mean of three replication

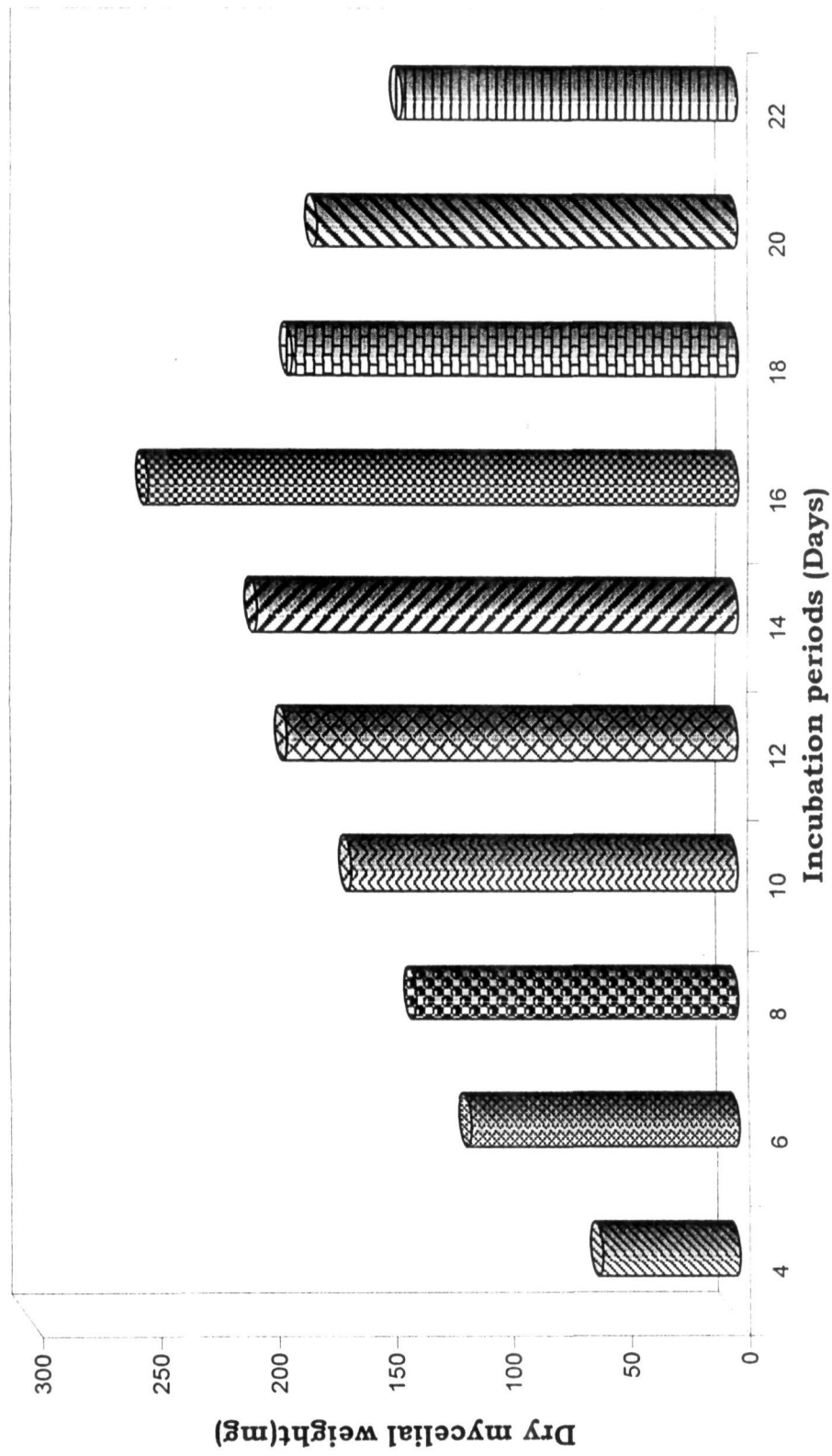


Fig. 2. Growth phase of *R. bataticola* on potato dextrose broth (PDB) at different incubation periods.



Plate 7. Chickpea plants infected by *R.bataticola* in Blotter paper study

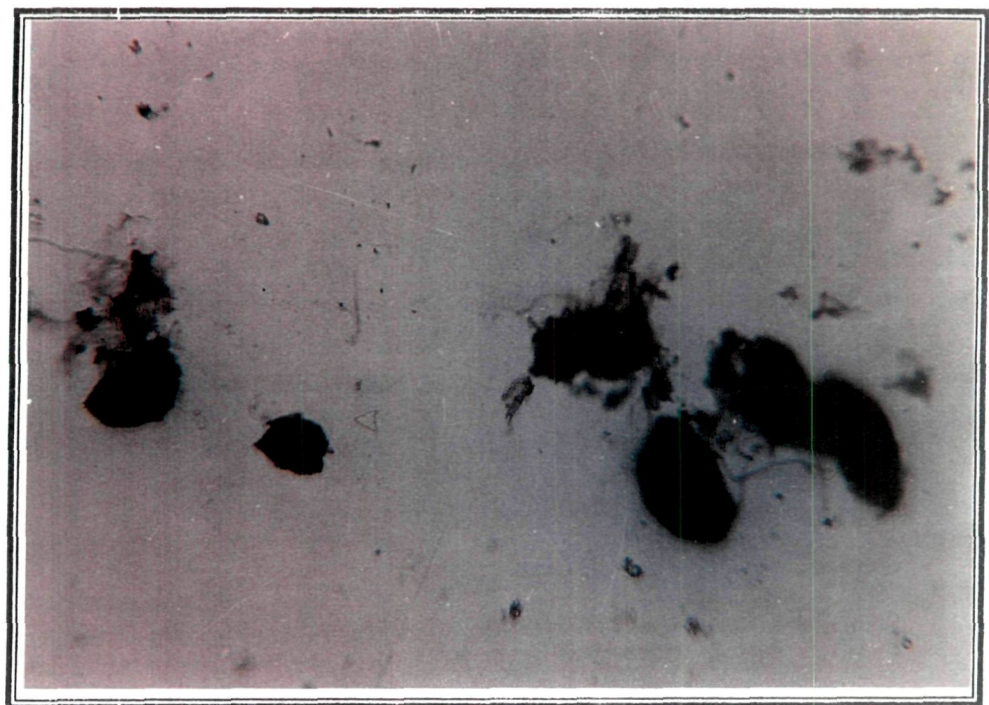


Plate 8. Microphotograph showing black sclerotia of *R.bataticola* at 10X

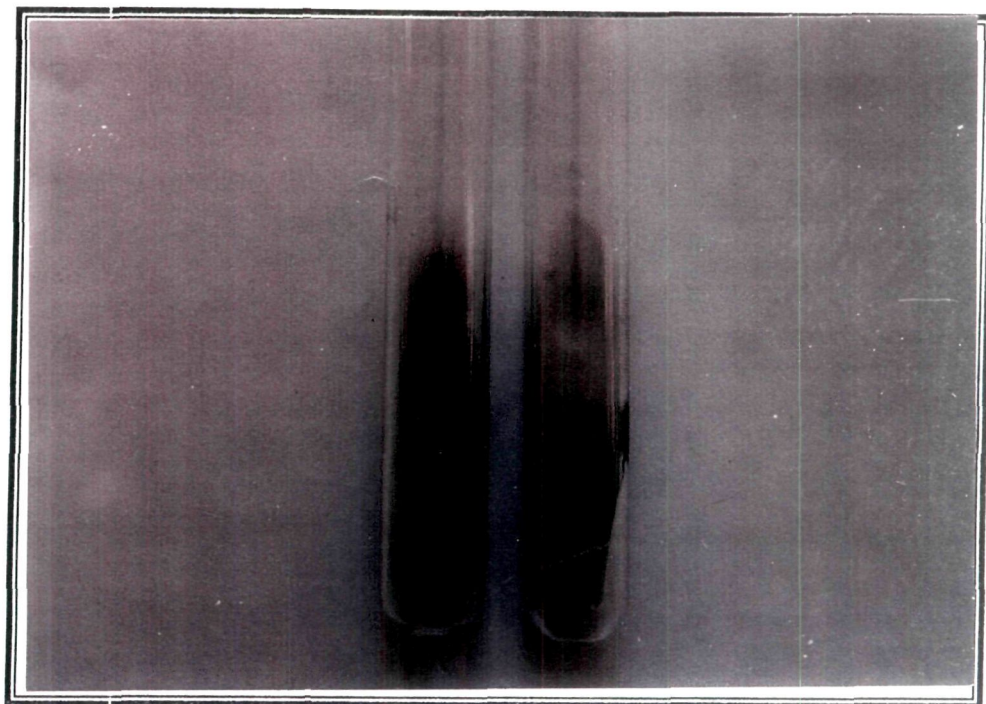


Plate 9. Photograph showing pure culture of *R. bataticola* in slant

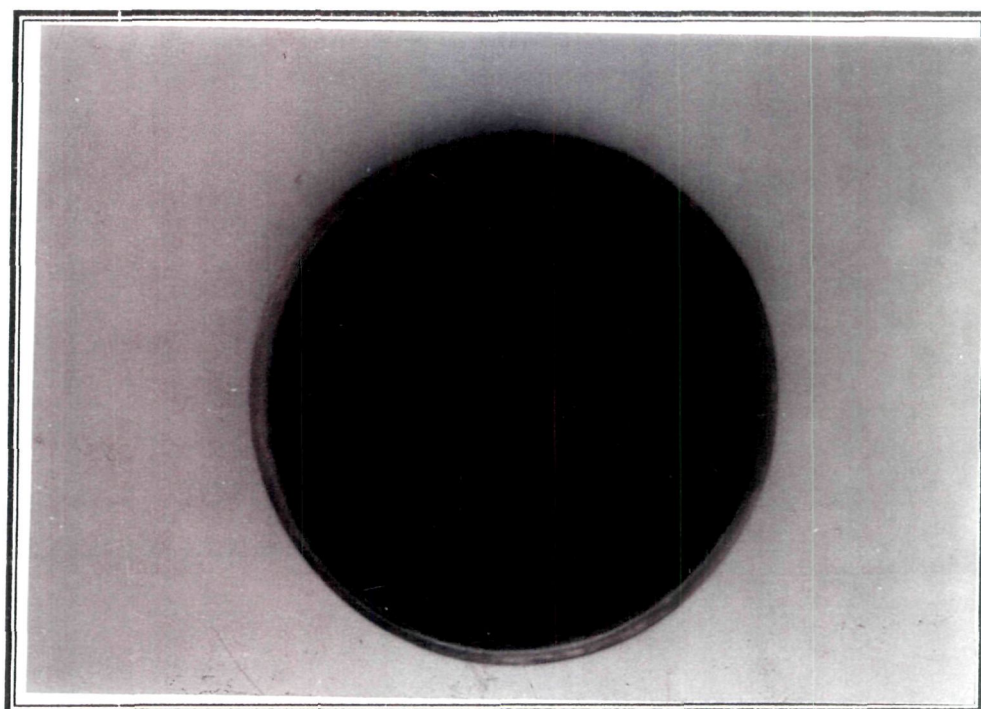


Plate 10. Pure culture of *R. bataticola* obtained from infected chickpea plant in culture media

mycelial weight of *R. bataticola* increased up to 16 days and least mycelial dry weight was observed at 4 days after incubation period (58.03 mg). After 16 days it was gradually decreased.

4.5.3 Growth, sclerotial size and sclerotial production of *R. bataticola* on different solid media.

The culture media on growth, sclerotial size and sclerotial production of *R. bataticola* was carried out on five different solid media and the results are presented here under Table 6 and Fig.3. The radial growth of *R. bataticola* reached a maximum of 90.00 mm by the end of fourth day on potato dextrose agar and Czapeck's agar, which was significantly superior over the other treatments. This was followed by malt extract agar with 78.13 mm radial growth, and the least growth of 68.50 mm was recorded on oatmeal agar (Plate-11).

Among five different solid media tested for sclerotial size variation, it was found that, the maximum sclerotial size was observed in potato dextrose agar (7.68 x 6.8 μ m) followed by Rose bengal agar (7.22 x 5.62 μ m). The least sclerotial size was observed in oatmeal agar (4.8 x 4.03 μ m).

Observations on colony colour, type of growth, type of margin and sclerotial production on different solid media were recorded on second, fourth, fifth and sixth day after inoculation and the result are presented in Table 7.

Table 6. Effect of different solid media on mycelial growth of *R. bataticola*

Media	Radial growth (mm)*	Sclerotial size (µm)
Potato Dextrose Agar	90.00	7.68 x 6.8
Czapeck's agar	90.00	6.68 x 5.70
Malt extract agar	78.13	5.7 x 4.8
Rose Bengal agar	72.83	7.22 x 5.62
Oatmeal agar	68.50	4.8 x 4.03
S.Em ±	0.234	
CD at 1%	1.110	

* Mean of three replication and the radial growth observation recorded four days after inoculation

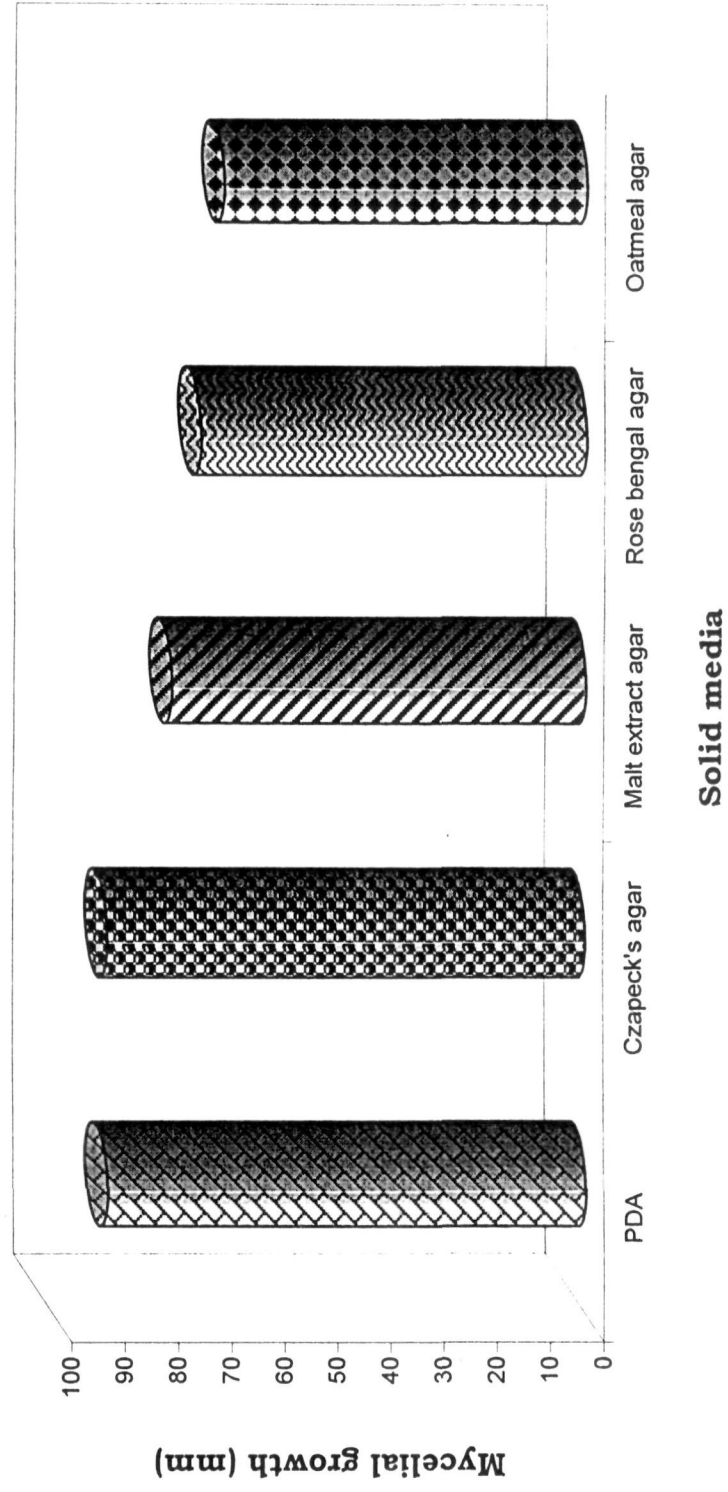


Fig. 3. Mycelial growth of *R. baraticola* on different solid media after four days of incubation.

Table 7. Growth characteristics of *R. bataticola* on different solid media at various periods.

Media	Characters	Cultural characters on different days.			
		2 nd	4 th	5 th	6 th
PDA	Colony colour	White	Black	Black	Black
	Type of growth	Flat growth	Flat growth	Flat growth	Flat growth
	Type of margin	Uniform	Uniform	Uniform	Uniform
	Sclerotial production	*	**	***	***
MA	Colony colour	Light black	Black	Black	Black
	Type of growth	Raised growth	Flat growth	Flat growth	Flat growth
	Type of margin	Uneven	Uneven	Uneven	Uneven
	Sclerotial production	*	**	**	***
CA	Colony colour	White	Black	Black	Black
	Type of growth	Flat growth	Flat growth	Flat growth	Flat growth
	Type of margin	Uniform	Uniform	Uniform	Uniform
	Sclerotial production	*	**	***	***
OMA	Colony colour	White	Black	Black	Black
	Type of growth	Flat growth	Flat growth	Flat growth	Flat growth
	Type of margin	Uniform	Uniform	Uniform	Uniform
	Sclerotial production	*	*	*	**
RA	Colony colour	Light black	Light Black	Light black	Light black
	Type of growth	Raised growth	Flat growth	Flat growth	Flat growth
	Type of margin	Uneven	Uneven	Uneven	Uneven
	Sclerotial production	*	*	**	**

PDA = Potato Dextrose Agar, MA = Malt Extract Agar, CA= Czapeck's Agar,
RA = Rose Bengal Agar OMA = Oat Meal Agar

* Scanty Sclerotia Production, ** Medium Sclerotial Production,
*** Heavy Sclerotia Production

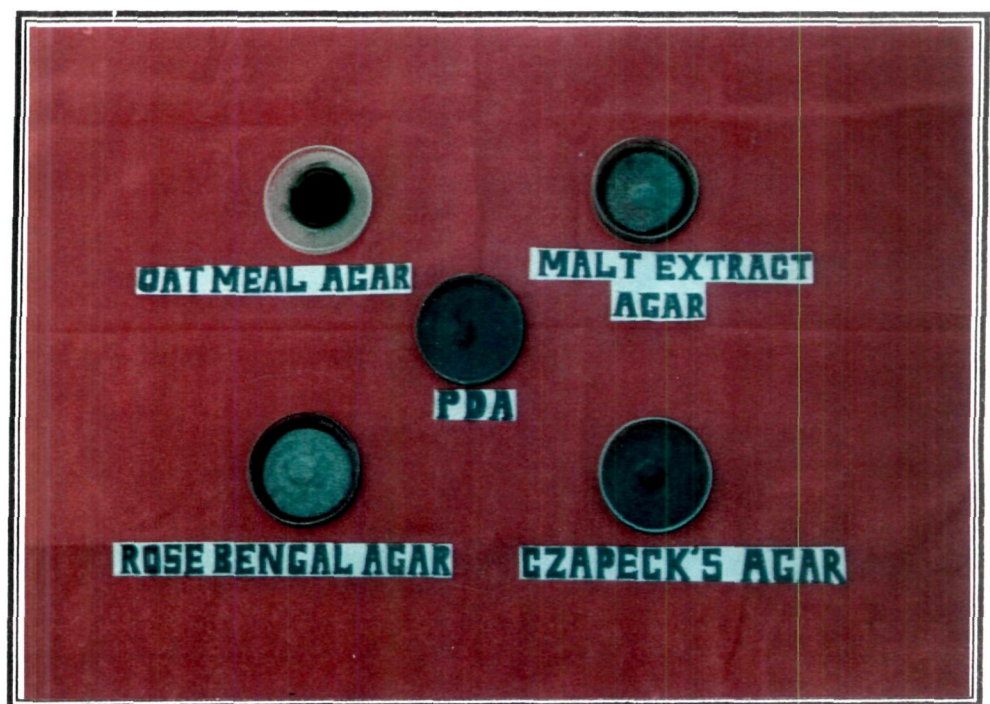


Plate 11. Growth of *R. bataticola* on different solid media at 4 DAI

The colony colour, type of growth, type of margin and sclerotial production on different solid media, the media were observed. It was noticed that the mycelium was white colour with flat growth with uniform margin in PDA, Czapeck's agar and oatmeal agar, except in maltose extract agar and Rose bengal agar. However, in maltose extract agar and Rose bengal agar, the mycelium showed light black colour with slightly raised growth with uneven margin.

Sclerotial production:

The results in sclerotial production of *R. bataticola* on synthetic/non-synthetic/semi-synthetic solid media are presented in Table 8. The results revealed that, all the five media of the fungus showed difference in their sclerotial production. Among the five media tested, sclerotial production started on second day onwards on all the media tested. The maximum sclerotial production on PDA with mean sclerotial number of 68.2 per microscopic field (10 X), which was significantly superior to other media, followed by czapack's agar (55.8) and malt extract agar (53.6). The least sclerotial production was noticed on oatmeal agar (38.5).

Incubation period has been tested for sclerotia production on different solid media (Plate 12 & 13). All the five media started producing sclerotial after second day onwards. On fourth day, the maximum sclerotia production was noticed on PDA (69.6) which was followed by malt extract agar (63.3) and czapack's agar (62.3). Least sclerotia

Table 8. Number of sclerotia of *R. bataticola* on synthetic / non-synthetic solid media on different incubation periods.

Synthetic / Non-synthetic solid media	Number of sclerotia present at different days (10X)*						Mean
	2	4	6	8	10	12	
Potato dextrose agar	0.00	69.6	80.3	89.3	88.6	81.3	68.2
Czapeck's agar	0.00	62.3	65.3	68.6	69.6	69.0	55.8
Malt extract agar	0.00	63.3	68.0	71.3	60.0	59.0	53.6
Rose Bengal agar	0.00	57.0	60.0	62.0	68.3	70.3	52.9
Oatmeal agar	0.00	41.3	50.6	51.0	48.6	39.3	38.5
Mean	0.00	58.7	64.8	68.4	67.0	63.8	53.8
	Media		Days		Interaction		
S.Em ±	0.274		0.300		0.671		
CD at 1%	1.031		1.130		2.526		

* Mean of three replication

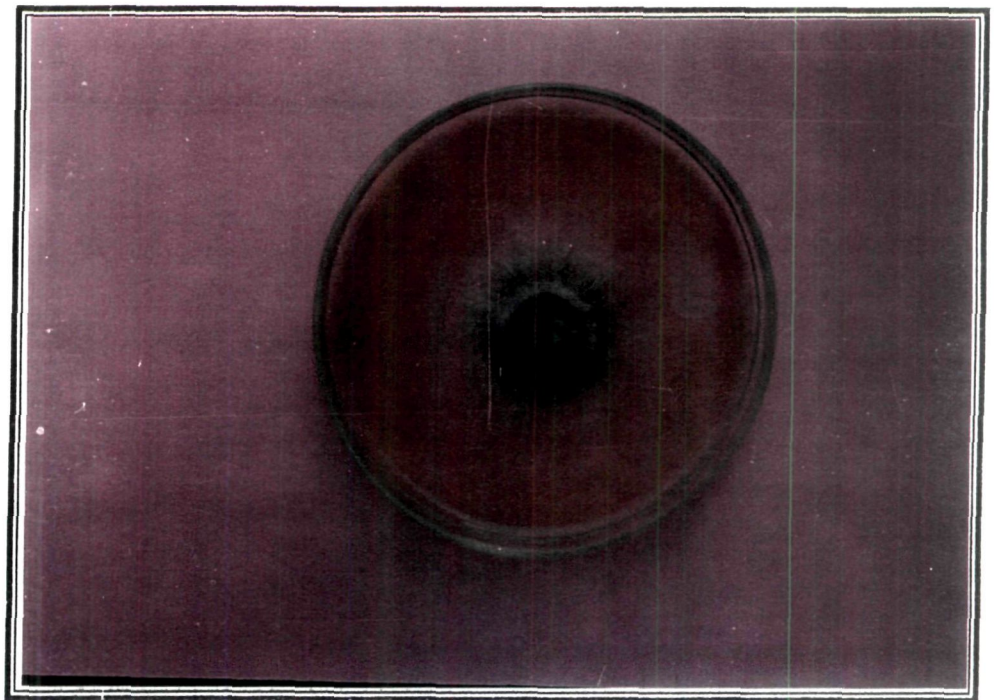


Plate 12. Photograph showing early growth of *R. bataticola* in culture media

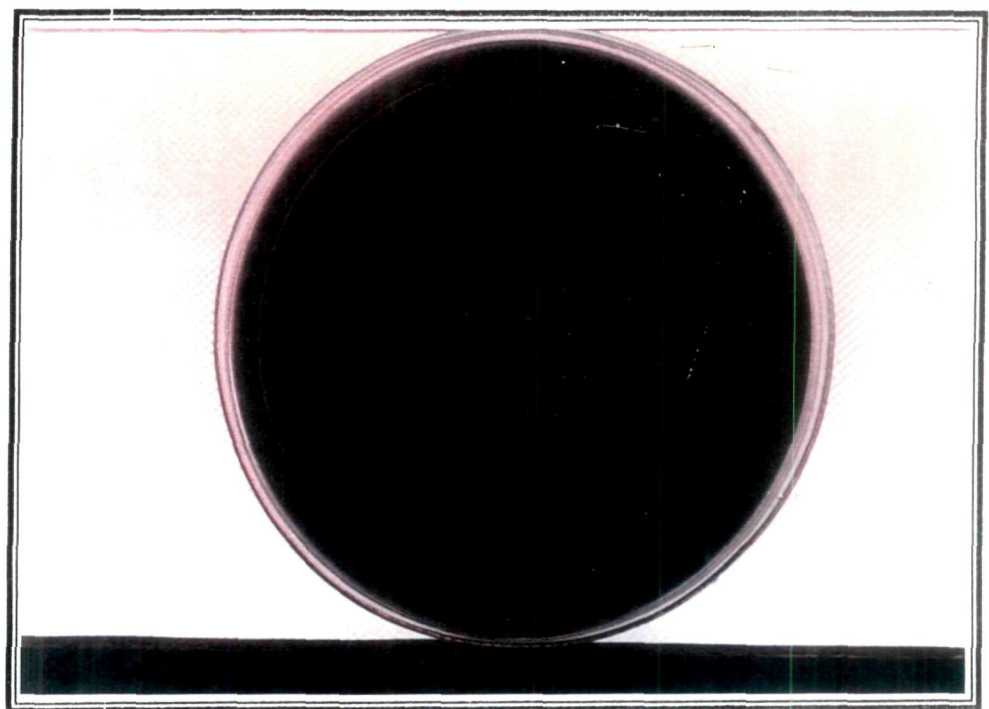


Plate 13. Later growth showing sclerotial bodies initiation of *R. bataticola* on culture media

production was noticed on fourth day on oatmeal agar (41.3). Whereas on sixth day, maximum sclerotial production was noticed on PDA (80.3 per microscopic field) followed by malt extract agar ((68.0) and czapeck's agar (6.53) and the least was on oatmeal agar (50.6). However, on eighth day, maximum sclerotia was noticed on PDA (89.3 per microscopic field (10 X) followed by malt extract agar 71.3 and cazpeck's agar 68.6. Least sclerotial production was noticed on oatmeal agar (51.0). Whereas on tenth day, the maximum sclerotial production of 88.6 was noticed on PDA which was followed by czapeck's agar (69.6) and Rose bengal agar (68.3). Least sclerotial production was noticed on oatmeal agar (48.6). On twelfth day, 81.3 per microscopic field were present on PDA followed by Rose bengal agar (70.3) and czapeck's agar (69.0). Least sclerotial production was noticed on twelfth day on oatmeal agar (39.3).

4.5.4 Growth on different liquid media

The results showed that among the five liquid media tested for mycelial growth (Table 9 and Fig. 4) the maximum dry mycelial weight was observed in potato dextrose broth (181.86 mg) which was significantly superior over other media tested, followed by Czapeck's media (159.42 mg). The least dry mycelial weight was observed in oatmeal media (88.47 mg) after 12 days of incubation (Plate 14).

Table 9. Effect of different liquid media on dry mycelial weight of *R. bataticola*

Media	Dry mycelial weight (mg)*
Potato dextrose broth	181.86
Czapeck's broth	159.42
Malt extract broth	153.52
Rose Bengal broth	119.51
Oatmeal broth	88.47
S.Em $\pm$	0.530
CD at 1%	2.517

* Mean of three replications and the observation was recorded 12 days after inoculation

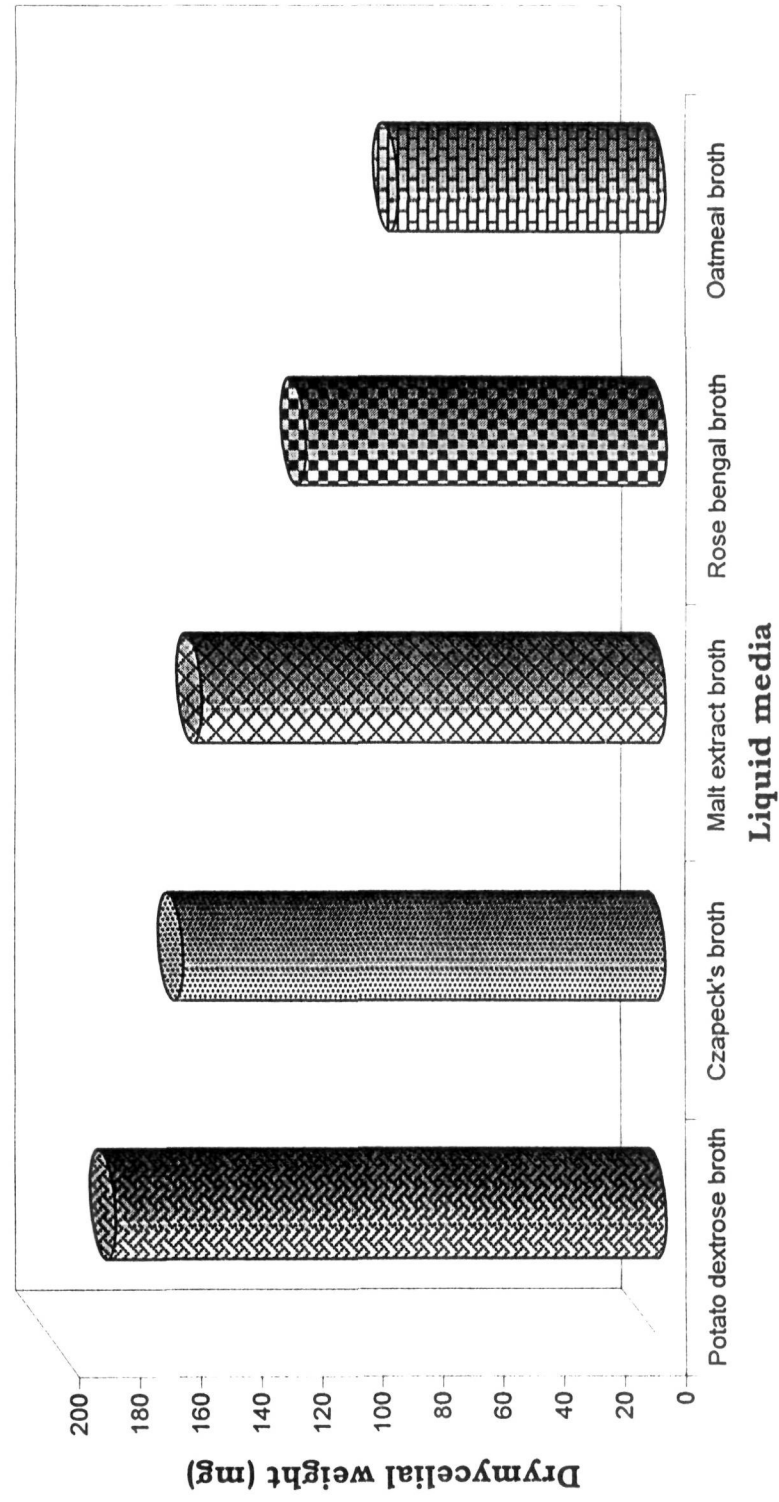


Fig. 4. Effect of dry mycelial weight on *R. bataticola* on different liquid media after 12 days of incubation.

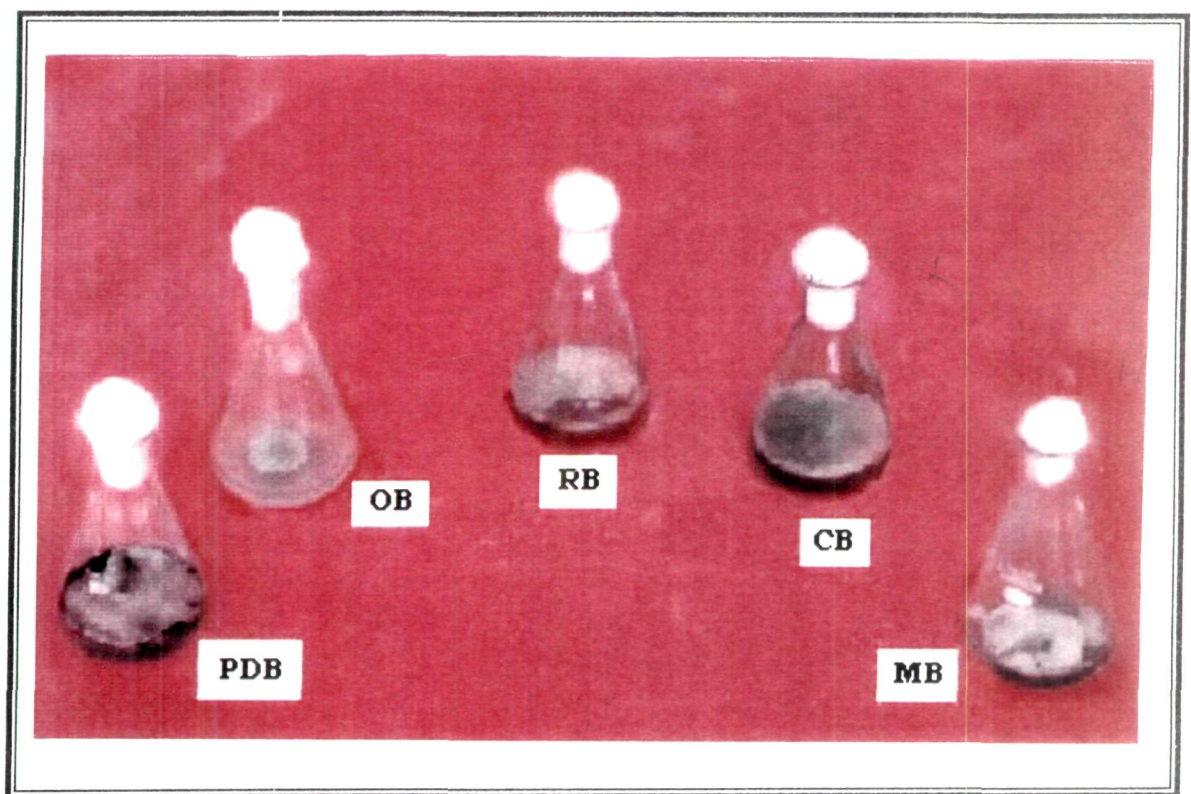


Plate 14. Growth of *R. bataticola* on different liquid medium

4.6 PHYSIOLOGICAL STUDIES

4.6.1 Effect of temperature on growth and sporulation of *R. bataticola*

The effect of six temperatures *i.e.*, 15, 20, 25, 30, 35 and 40° C was evaluated for growth and sporulation of *R. bataticola*. The results (Table 10a and Fig.5) showed that the maximum growth of the fungus was observed at both 30 and 35° C (90 mm) (Plate-15) followed by 40° C (87.3 mm). The least growth of the fungus was observed at 15° C (35.3 mm). Similarly, the maximum sclerotial production was observed at 35° C (87.0) and less sclerotial production was recorded at 15° C (45.3) followed by 20° C (54.3).

R. bataticola was grown on potato dextrose broth (PDB) at temperature of 15, 20, 25, 30, 35 and 40° C for 12 days to know the requirement of optimum temperature for its maximum growth. Average dry mycelial weights of *R. bataticola* after 12 days of incubation at different temperatures are presented in Table 10b and Fig. 6.

The fungus showed maximum growth of 196.70 mg at 35° C that was significantly superior over other treatments. Least dry mycelial weight was observed at 15° C (42.10 mg).

Table 10a. Effect of temperature on mycelial growth and sclerotial Production by *R. bataticola*.

Temperature (°C)	Mycelial growth (mm)*	No of sclerotia/ microscopic field (10X)*
15	35.3	45.3
20	45.6	54.3
25	60.6	68.6
30	90.0	81.3
35	90.0	87.0
40	87.3	79.0
S.Em ±	0.279	0.502
CD at 1%	1.248	2.249

* Mean of three replication and observation recorded six days after inoculation.

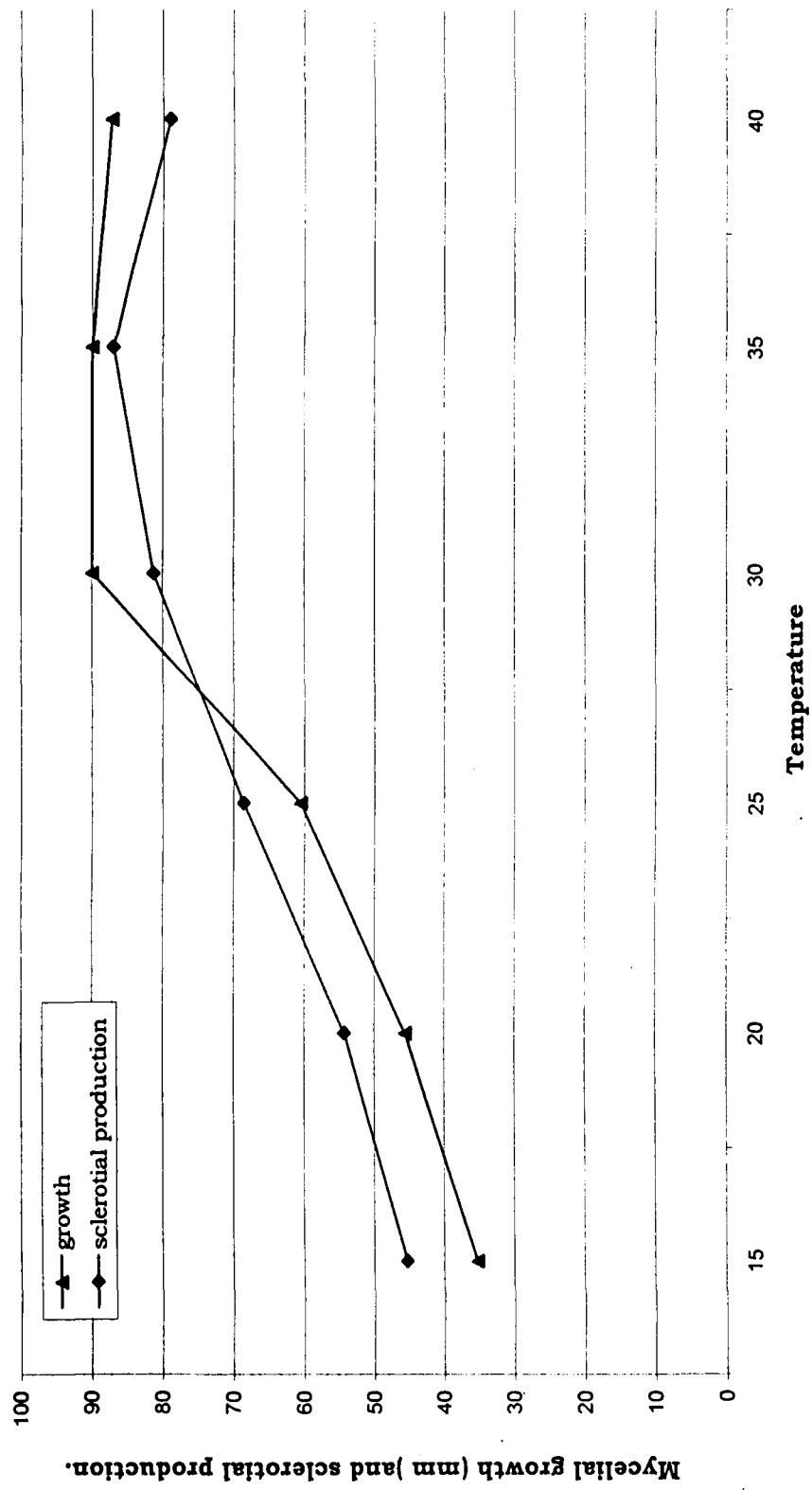


Fig .5. Effect of temperature on mycelial growth and sclerotial production of *R. bataticola*.

Table 10b. Effect of dry mycelial weight Of *R. bataticola* on PDB at different temperature.

Temperature (°C)	Dry mycelial weight (mg)*12 DAI
15	42.10
20	52.66
25	135.60
30	186.43
35	196.70
40	182.43
S.Em ±	0.622
CD at 1%	2.789

* Mean of three replication and the observation on dry mycelial weight recorded 12 days after inoculation.

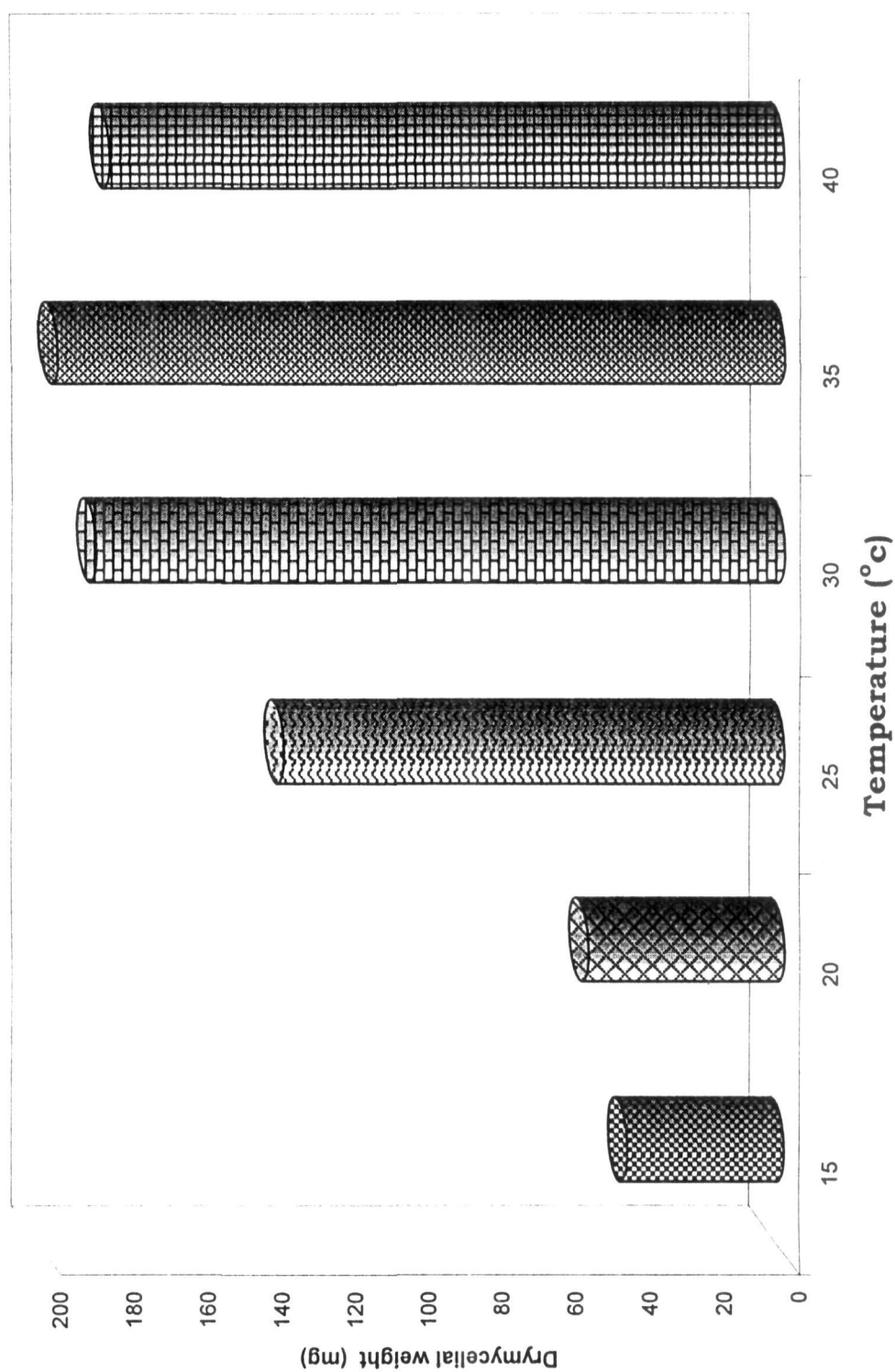


Fig.6. Effect of temperature on Dry mycelial weight of *R.bataticola*.

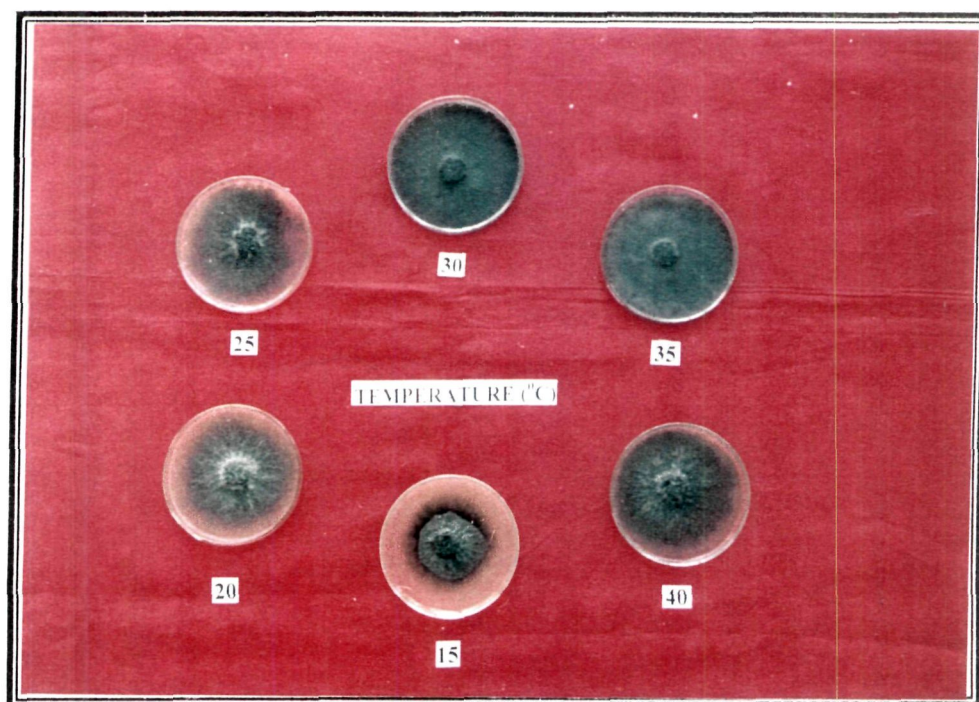


Plate 15. Growth of *R. bataticola* at different temperature levels

4.6.2 Effect of light on growth and sporulation of *R. bataticola*

The effect of alternate light (12 hour) and dark (12 hour), continuous light (24 hour) and complete darkness (24 hour) on growth and sporulation of *R. bataticola* was studied under laboratory conditions. The results (Table 11 and Fig.7) showed that the maximum growth and sclerotial production was observed in alternate light and darkness with 89.50 mm and 87.3 sclerotia per microscopic field (10 X) respectively followed by continuous light with 89.30 mm and 83.3 sclerotial per microscopic field respectively. The minimum growth and sclerotial production was observed in total darkness with 81.6 mm and 72.6 sclerotia per microscopic field (10 X) respectively (Plate-16).

4.6.3 Effect of relative humidity on growth and sporulation of *R. bataticola*

The growth and sporulation of *R. bataticola* was studied at different relative humidity (RH) levels i.e., 65, 75, 85, 95 and 100 per cent. The results (Table 12 and Fig.8) showed that the maximum growth and sclerotial production were observed at 75 percent RH (90.00 mm (Plate 17) and 87.6 sclerotia per microscopic field). The least growth and sclerotial production was observed at 65 percent RH (47.00 mm and 62.3 sclerotia per microscopic field).

Table 11. Effect of light on growth and sclerotial Production of *R. bataticola*.

Exposure to light and dark	Mycelial growth (mm)*	No of sclerotia/ microscopic field (10X)*
Continuous light	89.3	83.3
Continuous dark	81.6	72.6
Alternate with light and dark	89.5	87.3
S.Em $\pm$	0.184	0.304
CD at 1%	1.198	1.980

* Mean of three replication and observation recorded on mycelial growth four days after inoculation.

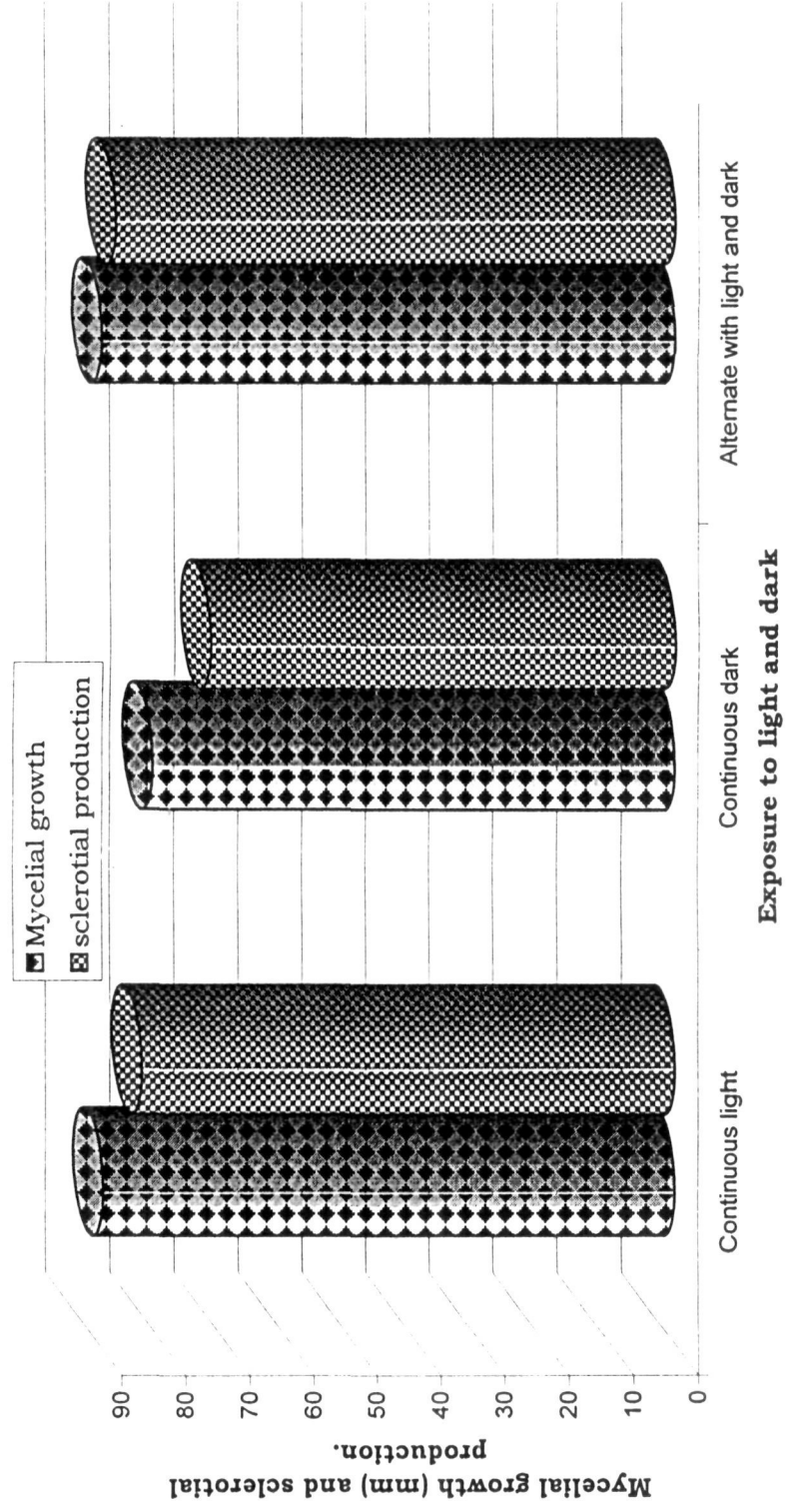


Fig. 7 . Effect of light on growth and sclerotial production of *R. bataticola*.

Table 12. Effect of relative humidity on mycelial growth and sclerotial production by *R. bataticola*.

Relative Humidity (%)	Mycelial growth (mm)*	No of sclerotia / microscopic field (10X)*
65	47.0	62.3
75	90.0	87.6
85	90.0	79.3
95	56.3	70.3
100	52.3	63.6
S.Em $\pm$	0.342	0.605
CD at 1%	1.622	2.873

* Mean of three replication and mycelial growth and sclerotial production as recorded six days after inoculation.

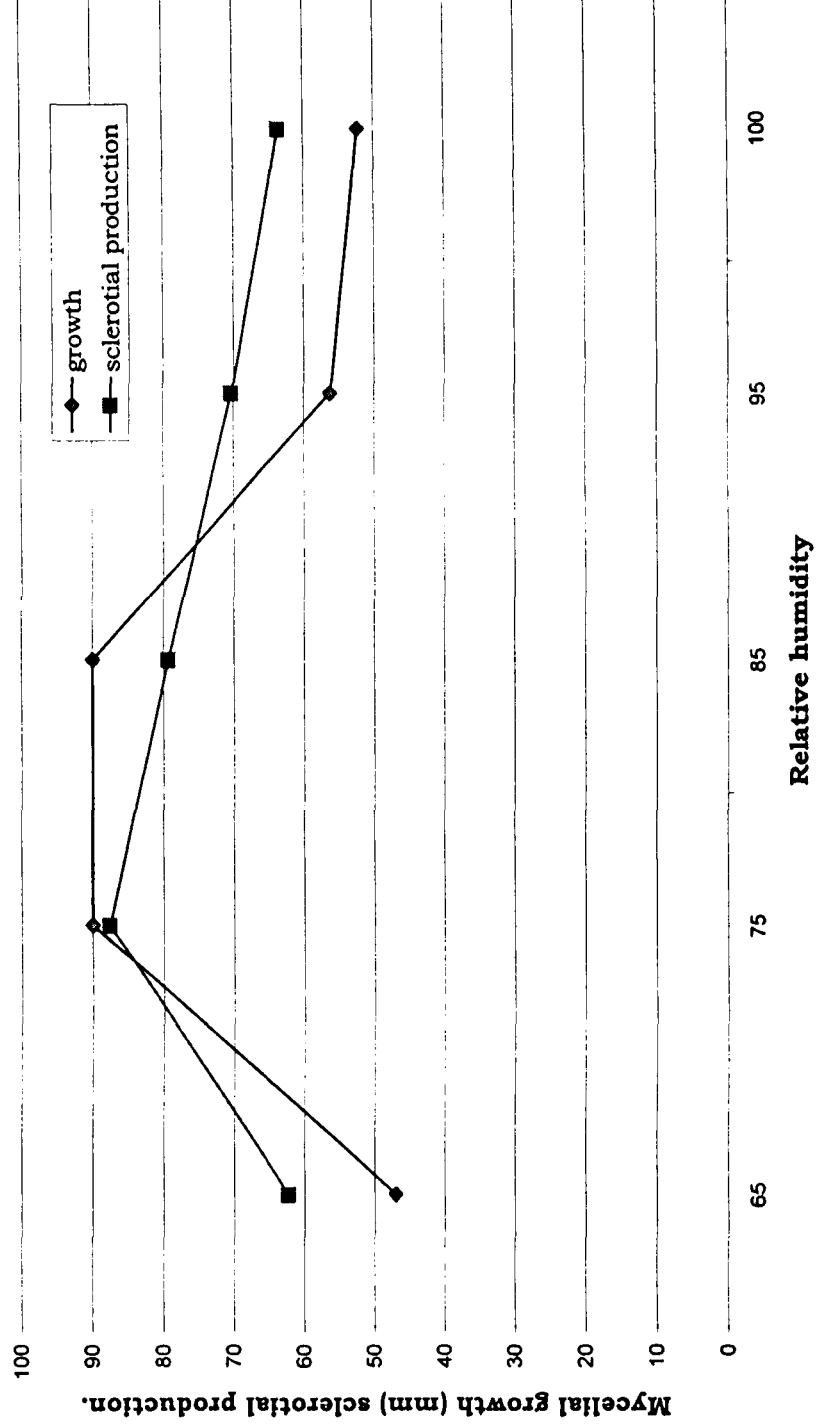


Fig .8 . Effect of relative humidity on mycelial growth and sclerotial production by *R. bataticola*.

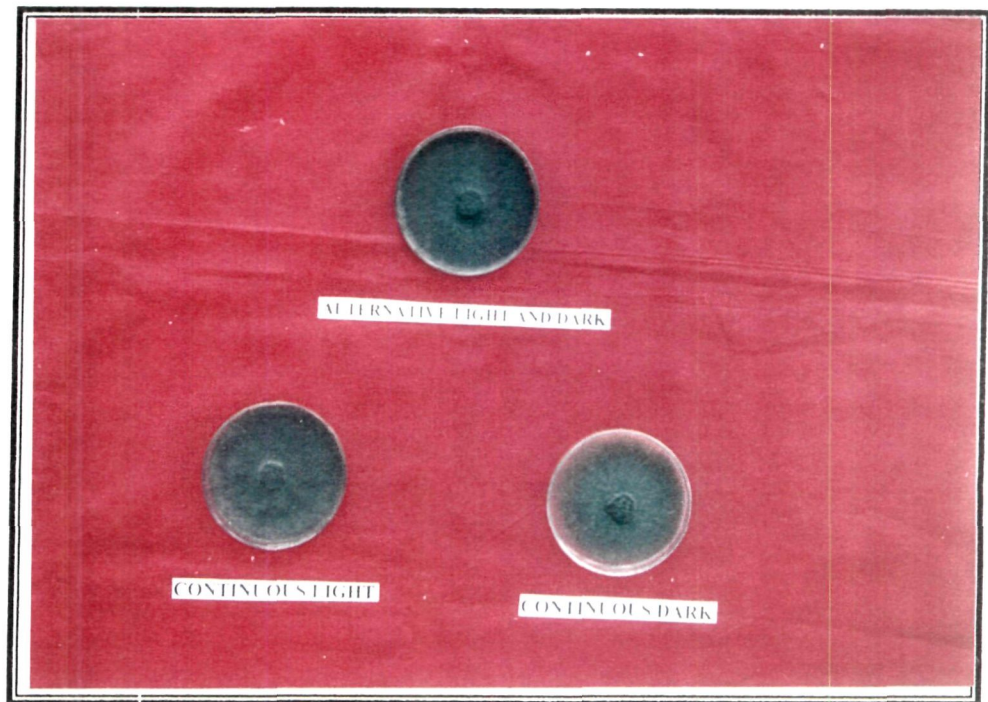


Plate 16. Growth of *R. bataticola* in response to the varied light exposure period

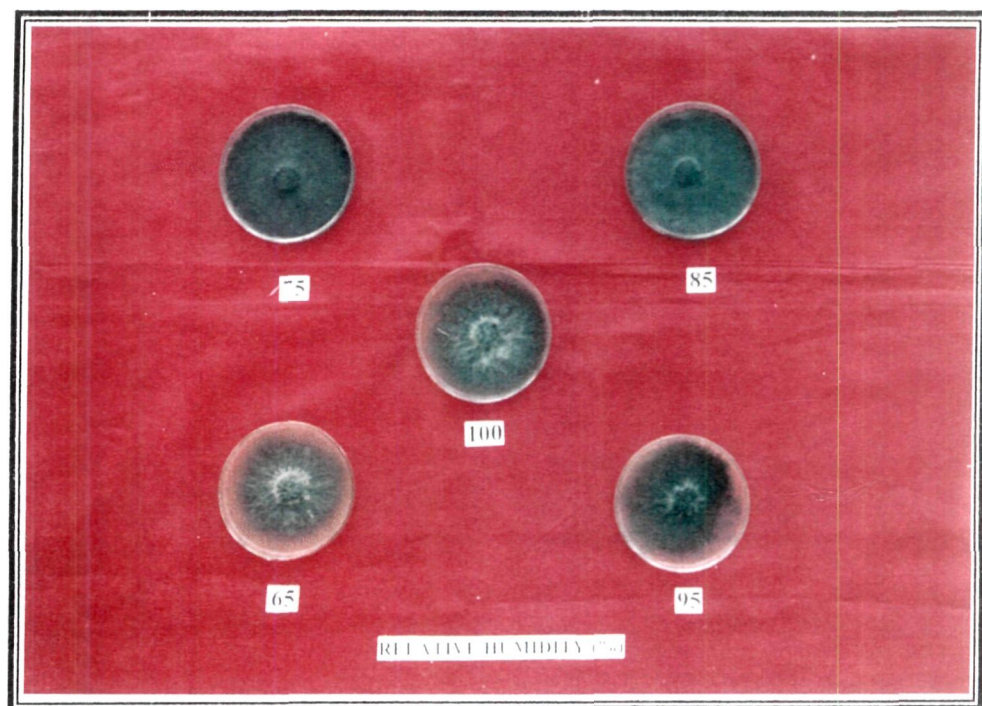


Plate 17. Photograph showing effect of relative humidity on growth of *R. bataticola*

4.7 SURVIVAL OF *R. bataticola* IN SOIL AND PLANT DEBRIS

Survival of *R. bataticola* in soil and plant debris was studied by burying the naturally infected chickpea samples in pots containing sterilized soil up to a depth of 2 cm, stored in polythene bags and exposed in refrigerator. The samples were retrieved from the soil at 30 days interval to know the survivalability of the pathogen over a period of time.

The results (Table 13) showed that the fungus survived in soil up to 210 days. The least growth was noticed in soil up to September 15th (71.81 mm). Whereas, in plant debris the fungus survived up to 210 days. The least growth was noticed in plant debris up to September 15th (82.00 mm). Whereas, in a refrigerator, *R. bataticola* could be recovered up to 90 days. Survival capacity of the fungus was found to be decreasing over a period of time in both soil and plant debris.

Similarly, sclerotial production of *R. bataticola* in soil and plant debris increased slightly up to two months. Whereas in refrigerator sclerotial production decreased from initial stage in comparison to soil and plant debris. The maximum sclerotial production in soil was noticed up to 60 days and decreased thereafter subsequently (April 15th)(84.10). The least sclerotial production (60.12) was noticed in 210 days (September 15th). Whereas in plant debris the maximum sclerotial production was (90.50) noticed up to 60 days and decreased subsequently (April 15th). Least (65.00) was noticed in 210 days (September 15th). Similarly, in

Table 13. Survival of the *R. bataticola* in soil, plant debris and refrigerator over a time.

Period (Months)	Days after storage	Mycelial growth (mm)*			Number of sclerotia per microscopic field (10X)*		
		Soil (2 cm)	Plant debris	Refrigerator (4° C)	Soil (2 cm)	Plant debris	Refrigerator (4° C)
February 15 th	0	90.00	90.00	90.00	80.36	88.00	51.76
March 15 th	30	90.00	90.00	90.00	82.30	89.20	47.20
April 15 th	60	90.00	90.00	76.66	84.10	90.50	38.73
May 15 th	90	90.00	90.00	54.76	77.40	84.50	27.83
June 15 th	120	90.00	90.00	00.00	70.50	74.90	00.00
July 15 th	150	88.00	90.00	00.00	68.20	72.16	00.00
August 15 th	180	77.20	86.66	00.00	67.00	69.10	00.00
September 15 th	210	71.81	82.00	00.00	60.12	65.00	00.00
	S.Em±	0.325	0.334	0.176	0.336	0.255	0.250
	CD at 1%	1.406	1.443	0.758	1.453	1.102	1.079

* Mean of five replication

refrigerator maximum sclerotial production was noticed in February 15th (51.76) followed by March 15th (47.20). After May 15th, there was no sclerotial production in refrigerator.

4.8 Evaluation of minimum inoculum level for infection

Mass inoculum of the fungus was produced on sorghum grains and it was added to sterile soil to give 1.2.3 and 4 per cent inoculum level by weight in infested soil. The surface sterilized chickpea seeds were sown. The effect on growth of chickpea seedlings was visually assayed and also measured in terms of percentage of seedlings showing symptoms. The data are presented in Table 14.

Among the four different inoculum levels tested, first symptom expression noticed in 35 days after sowing where highest inoculum (4%) was incorporated in soil and was followed by 3, 2 and 1 per cent inoculum where seedlings expressed symptoms after 40, 47 and 58 days respectively. The percent disease was also found to be highest (76.6%) at higher inoculum followed by 64.2, 43.7 and 28.5 at 3, 2 and 1% inoculum in the soil in comparison to healthy control. The effect of disease on growth parameters of chickpea at flowering stage indicated that the plant height (25.0 cm) was significantly more in lower percent of inoculum (1%) compared to 21.0, 18.0 and 16.0 cm plant in 2 per cent, 3 and 4 per cent inoculum respectively. The lowest plant height of 16.00 cm was noticed in

Table 14. Effects of dry root rot disease on various growth parameters of chickpea in sick pot culture under green house conditions.

Inoculum (%)	First symptom expression (Days)	Intensity of disease (%)	Flowering stage						Pod formation stage					
			Plant height (cm)	No. of Branch	Shoot length (cm)	Root length (cm)	Weight of plant (gm)	% Reduction	Plant height (cm)	No. of Branch	Shoot length (cm)	Root length (cm)	Weight of plant (gm)	% Reduction
1	58	28.5	25	1.8	17.6	7.4	1.52	81.8	28.3	2.0	20.8	7.50	1.65	88.0
2	47	43.7	21	1.2	15.4	5.6	0.91	89.1	24.8	1.2	19.1	5.70	1.28	90.0
3	40	64.2	18	1.2	13.6	4.4	0.82	90.2	21.4	1.2	16.1	5.30	0.90	93.3
4	35	76.6	16	1.0	12.3	3.7	0.64	92.3	19.0	1.0	14.8	4.20	0.74	94.5
Control	00	00.0	35	3.0	26.8	9.0	8.38	00.0	40.3	3.0	30.8	9.50	13.48	00.0
Mean reduction								70.7						73.16
S,Em ±	-	-	0.82		0.21	0.54	0.30		0.11		0.90	0.41	0.52	
CD at 1%	-	-	2.68		0.69	1.84	1.02		0.46		2.90	1.13	1.58	

* Mean of five replication

4 per cent inoculum in comparison to 35.8cm in control. Similarly less number of main branches (1.00) was noticed in higher percent inoculum and both at 2 and 3 per cent level, 1.20 branches were noticed as compared to control (3.00).

The shoot length of 12.3cm was recorded at 4 per cent inoculum followed by 3,2 and 1 per cent inoculum respectively in comparison to control where shoot length was 26.8 cm. Similarly with respect to root length that is significantly lower root length (3.70 cm) was noticed in higher inoculum as compared to lower inoculum as well as control. The total dry weight of plant is 0.64 g in 4 per cent inoculum followed by 1.52, 0.91 and 0.82 g were recorded by 1, 2 and 3 per cent inoculum respectively.

Similarly the effect of disease on growth parameters of chickpea at pod formation stage indicated that, the plant height was significantly more in lower per cent inoculum (1%) as (28.30 cm) followed by 2 (24.80 cm), 3 (21.40 cm) and 4 (19.00 cm) per cent as compared to control (40.30 cm). Less number of branches (1.0) were observed at the plant weight of higher per cent inoculum when compared to control (3.00).

The shoot length was less (14.80 cm) in 4 % inoculum and was followed by 3, 2 and 1 per cent inoculum. In control the shoot length was 30.80 cm. Similar was the trend with respect to root length where the lower root length (4.20 cm) was noticed in higher inoculum as compared to control (9.50 cm). The total weight of the plant was significantly higher

(13.48 g) in control when compared to plant weight of 0.74 g in 4 per cent inoculum.

Effect of disease on chickpea growth parameters under field conditions at different growth stages of the crop presented here under Table 15.

The results clearly indicated that dry root rot disease drastically reduced the chickpea growth parameters. Among the different stage, plant height of 25.4 cm was noticed in diseased plant when compared to healthy (36.8 cm), 34.4 cm in diseased and 57.8 cm in healthy and 39.3 cm in diseased and 59.8 cm in healthy in flowering, pod formation and harvesting stage respectively. There was marked effect of the disease on production of branches. The number of branches decreased in dry root rot infected plants as compared to healthy plant in all the stages of crop. The reduction in the production of branches in 2.4 in diseased and 5.4 in healthy in flowering stage, 3.4 in diseased and 6.1 in healthy in pod formation and 3.4 in diseased and 7.3 in healthy in harvesting stage. With respect to shoot length 19.00 cm in diseased and 26.8 cm in flowering stage, 25.6 in diseased and 47.8 cm in pod formation and 29.4 cm in diseased and 49.2 cm is healthy in harvesting stage. Similarly in root length 6.4 cm was observed in diseased when compared to healthy 10.0 cm in flowering stage. Where as in pod formation stage 8.8 cm and 10.0 cm both in diseased and healthy respectively. Where as in harvesting

Table 15. Effects of dry root rot disease on various growth parameters of chickpea under field conditions.

Stage of the crop		Plant height (cm)	No. of branches	Shoot length (cm)	Root length (cm)	Weight of the plant (gm)	% Reduction over healthy
Flowering	Disease	25.4	2.4	19.0	6.4	2.97	77.15-
	Healthy	36.8	5.4	26.8	10.0	13.0	
Pod formation	Diseased	34.4	3.4	25.6	8.8	8.36	65.10 -
	Healthy	57.8	6.1	47.8	10.0	23.96	
Harvesting	Diseased	39.3	3.4	29.4	9.9	12.80	53.28
	Healthy	59.8	7.3	49.2	10.6	27.40	

* Mean of five replication

stage 9.9 cm of root length was noticed in diseased plant when compared to healthy (10.6 cm). There was also significant reduction in total biomass of the plant in all the stages of infection. 2.97 g of plant weight was noticed in diseased plant in flowering stage compared to 13.0 g in healthy. Similarly in pod formation stage 8.36 g and 23.96 g of plant weight was noticed in both diseased and healthy plant respectively. Where as in harvesting stage 12.80 g was noticed in diseased plant when compared to healthy (27.40 g). The per cent bio mass of 77.15, 65.10, 53.28 at flowering, pod formation and at harvesting stage were recorded over healthy plants (00.0).

4.9 EVALUATION OF FUNGICIDES

4.9.1 *In vitro* evaluation by poison food technique

There was a significant difference between the fungicides and concentrations (Table 16a and Fig. 9). All the five systemic fungicides differed significantly with respect to inhibition of *R. bataticola*. Among these, hexaconazole was found to be superior in inhibiting the radial growth of *R. bataticola* (100.0%) (Plate 18) in all the concentrations tested followed by benomyl (99.9%) and carbendazim (96.19%). The least inhibition of growth was observed in carboxin treatment (52.2%).

Hexaconazole inhibited the radial growth by 100.00 per cent at both 250, 500 and 1000 ppm. Further, benomyl showed 100 per cent inhibition of radial growth of pathogen at 500 and 1000 ppm. However,

Table 16a. Efficacy of systemic fungicides on inhibition of growth of *R. bataticola*.

Systemic fungicides	Percent inhibition of mycelial growth			Mean
	Concentration (ppm)			
	250	500	1000	
Carbendazim	90.36 (71.88)	98.55 (82.97)	99.68 (86.76)	96.19 (80.54)*
Benomyl	99.75 (87.18)	100.00 (89.96)	100.00 (89.96)	99.90 (89.03)
Hexaconazole	100.00 (89.96)	100.00 (89.96)	100.00 (89.96)	100.00 (89.96)
Carboxin	32.50 (34.78)	50.30 (45.17)	73.90 (59.27)	52.20 (46.40)
Thiophanate methyl	34.00 (35.67)	50.20 (45.15)	81.70 (64.70)	55.30 (48.50)
Control	00.00 (00.00)	00.00 (00.00)	00.00 (0.00)	00.00 (00.00)
Mean	71.32 (57.61)	79.81 (63.29)	91.05 (72.54)	80.71 (63.94)

	Fungicides	Concentrations	F X C
S.Em ±	0.350	0.248	0.607
CD at 1%	1.351	0.955	2.340

* Figures in parenthesis are arc-sin transformed values

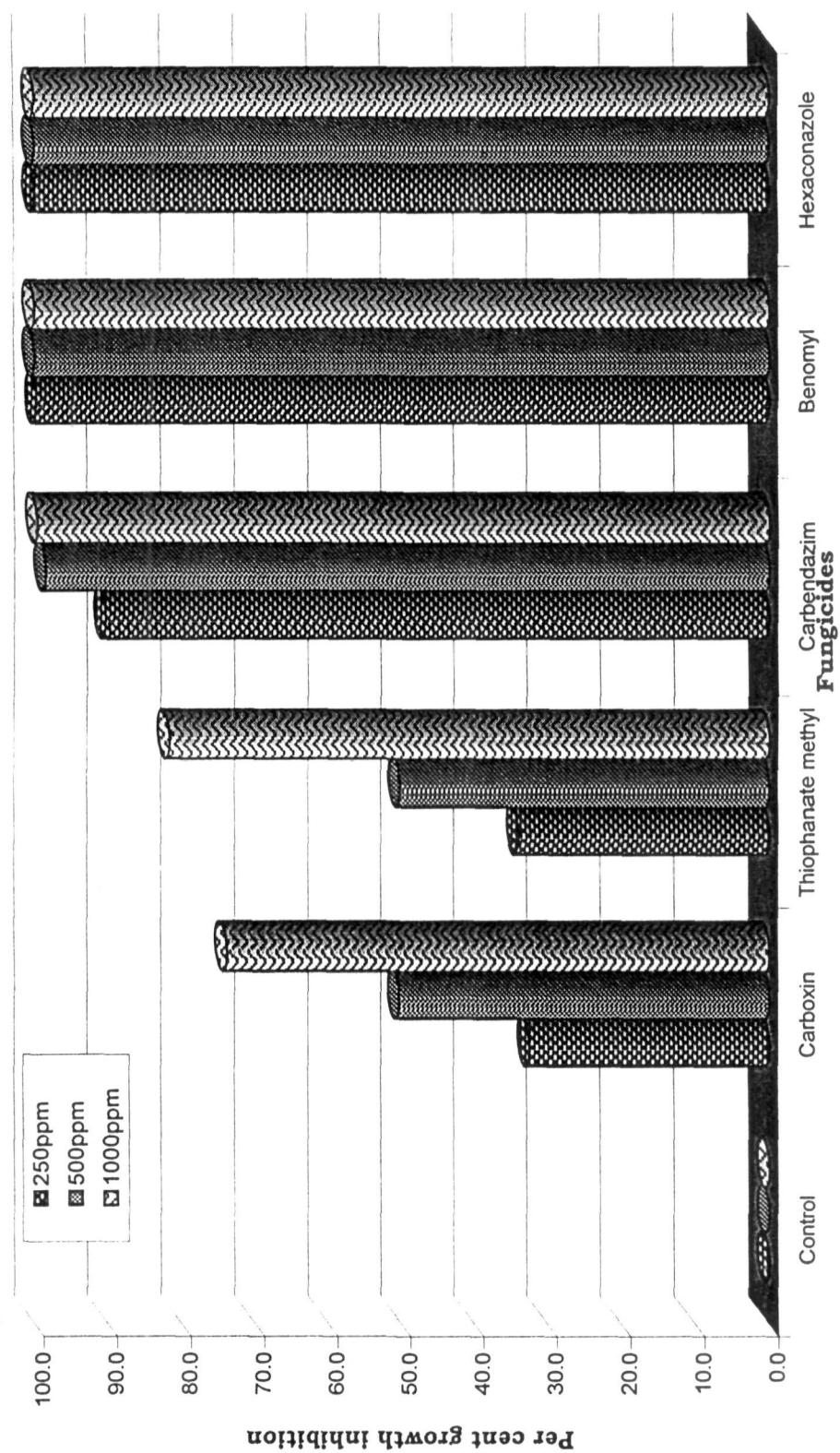


Fig.9 . Efficacy of systemic fungicides on growth inhibition of *R.bataticola*

carboxin recorded least inhibition of growth of *R. bataticola* (32.5 %) at 250-ppm concentration.

Among the non-systemic fungicides (Table 16b and Fig.10) tested, SAAF was found to be effective in inhibiting the radial growth (100.0%) of *R. bataticola* (Plate 19) followed by mancozeb (99.4%), thiram (87.4%), and the least inhibition was recorded with captan (59.7%).

At different concentrations, SAAF inhibited the radial growth of the pathogen by 100 per cent at 1000, 2000 and 3000 ppm. Further, mancozeb showed 100 per cent inhibition of radial growth at 2000 and 3000 ppm. However, captan recorded least inhibition of growth of *R. bataticola* (36.2%) at 1000 ppm concentration.

4.10 BIOLOGICAL CONTROL

Four bioagents were tested against *R. bataticola* in dual culture technique. Observations on *in vitro* inhibition of mycelial growth of *R. bataticola* by various bio-agents are presented in Table17.

The results in the Table 17 and Fig. 11. Showed that all the bioagents tested were found to be effective in inhibiting the mycelial growth of *R. bataticola*. Among the four antagonists, *Pseudomonas fluorescens* was found significantly superior in inhibiting (79.2%) the growth of *R. bataticola* (Plate-20). But remained on par with *Trichoderma*

Table 16b. Evaluation of non-systemic fungicides at various concentrations on growth inhibition of *R. bataticola*.

Non -Systemic fungicides	Percent inhibition of mycelial growth			Mean Per cent inhibition
	Concentration (ppm)			
	1000	2000	3000	
SAAF (Carbendazim 12%+ Mancozeb 63%)	100.00 (89.96)	100.00 (89.96)	100.00 (89.96)	100.00 (89.96)*
Thiram	71.80 (57.92)	88.00 (69.78)	97.00 (80.06)	87.40 (69.25)
Mancozeb	94.70 (76.80)	100.00 (89.96)	100.00 (89.96)	99.40 (85.57)
Captan	36.20 (36.99)	59.20 (50.30)	81.50 (64.51)	59.70 (50.60)
Control	00.00 (00.00)	00.00 (00.00)	00.00 (0.00)	00.00 (00.00)
Mean	75.67 (60.40)	86.80 (68.70)	94.62 (76.56)	86.62 (68.53)

	Fungicides	Concentrations	F X C
S.Em $\pm$	0.374	0.290	0.648
CD at 1%	1.462	1.132	2.532

* Figures in parenthesis are arc-sin transformed values

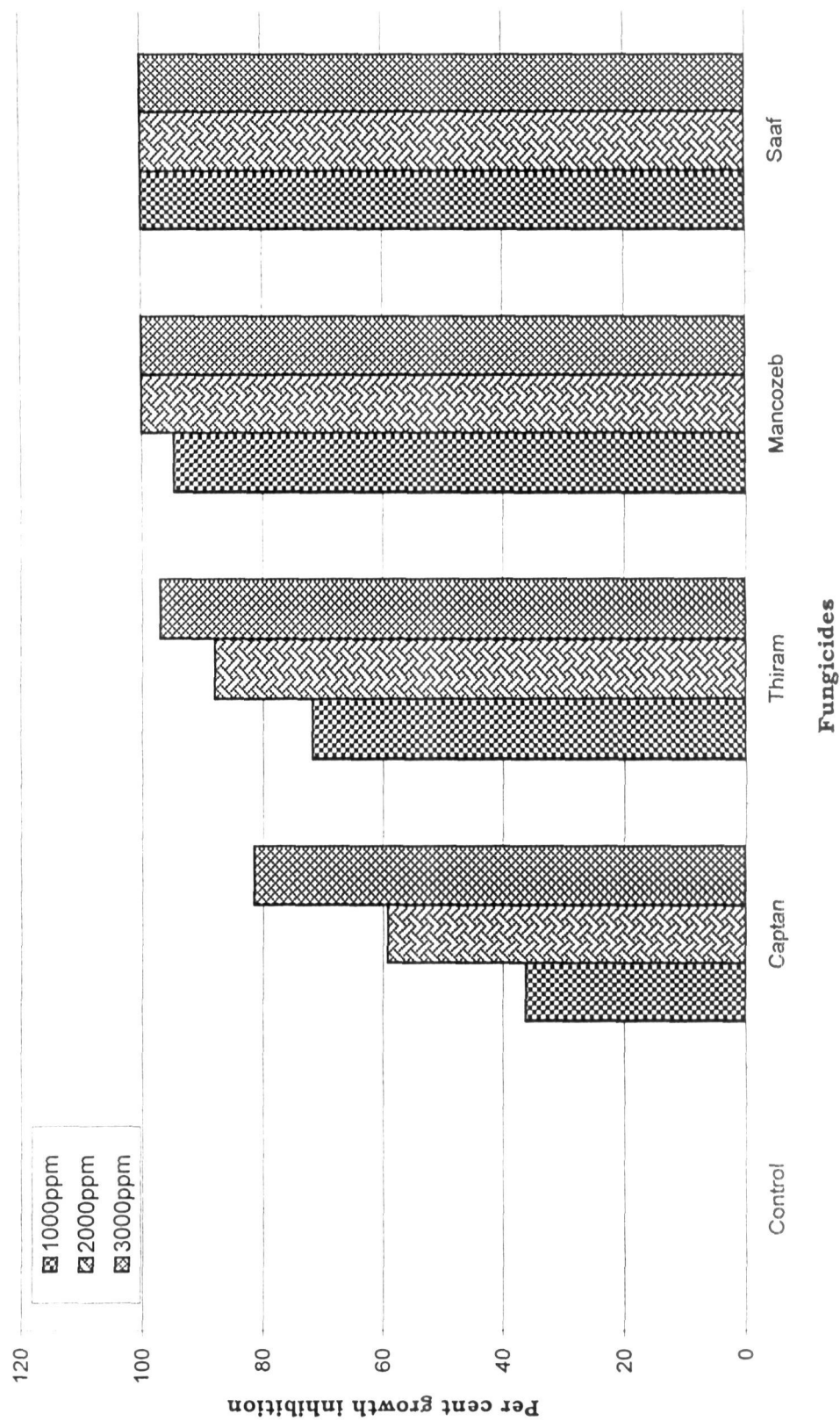


Fig. 10. Efficacy of non-systemic fungicides on growth inhibition of *R. bataticola*

Table 17. *In-vitro* evaluation of bioagents on inhibition of growth of *R. bataticola* in dual culture technique.

Bioagents	Percent inhibition over control
<i>Trichoderma harzianum</i>	72.90 (58.63)*
<i>Pseudomonas fluorescense</i>	79.20 (62.89)
<i>Trichoderma viride</i>	55.80 (48.37)
<i>Trichoderma virens</i>	57.30 (49.22)
Control	00.00 (00.00)
S.Em $\pm$	0.312
CD at 1%	1.480

* Figures in parenthesis are arc-sin transformed values

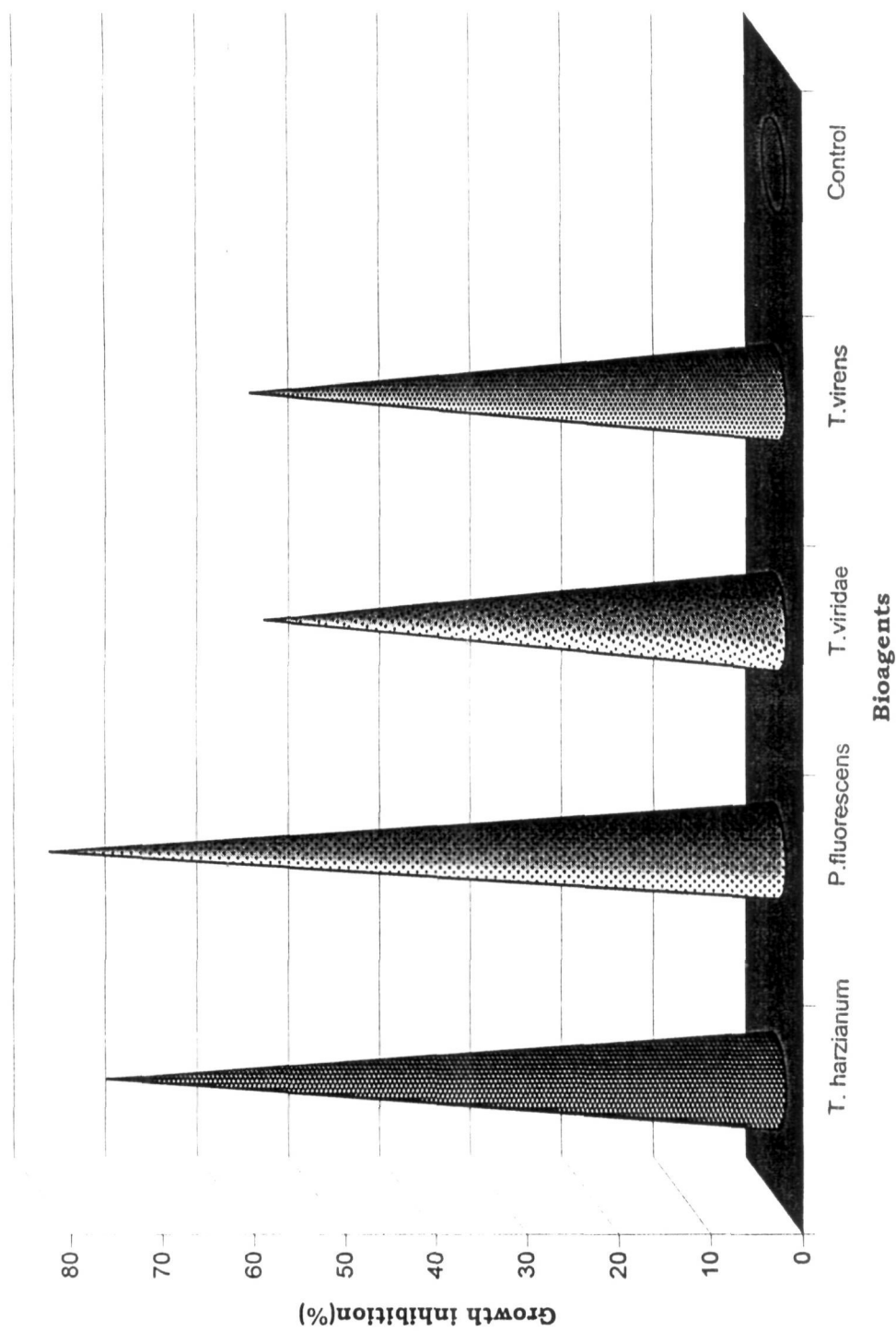


Fig. 11 .*In-vitro* inhibition of growth of *R. bataticola* by antagonists in dual culture technique

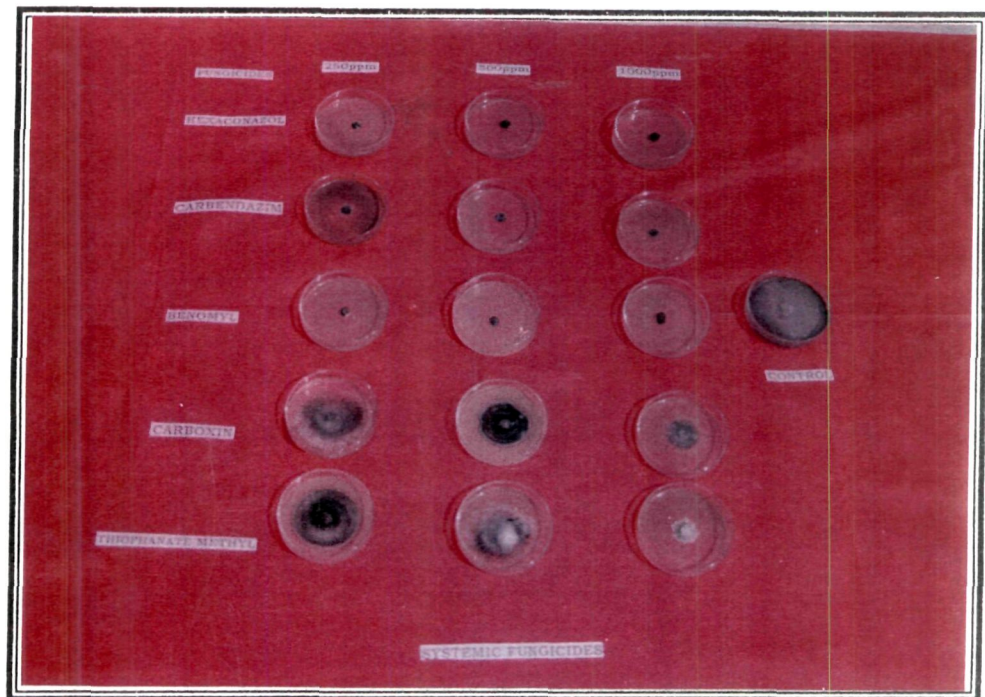


Plate 18. *In-vitro* inhibition of mycelial growth of *R.bataticola* by different systemic fungicide at different concentrations

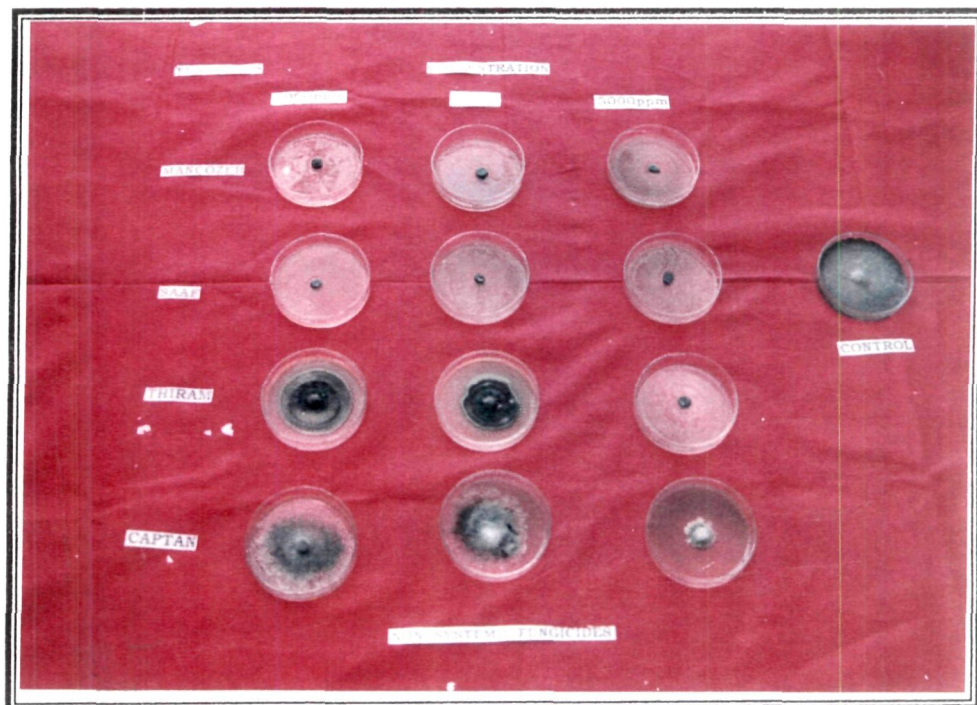


Plate 19. *In-vitro* inhibition of mycelial growth of *R.bataticola* by different non-systemic fungicides at different concentrations

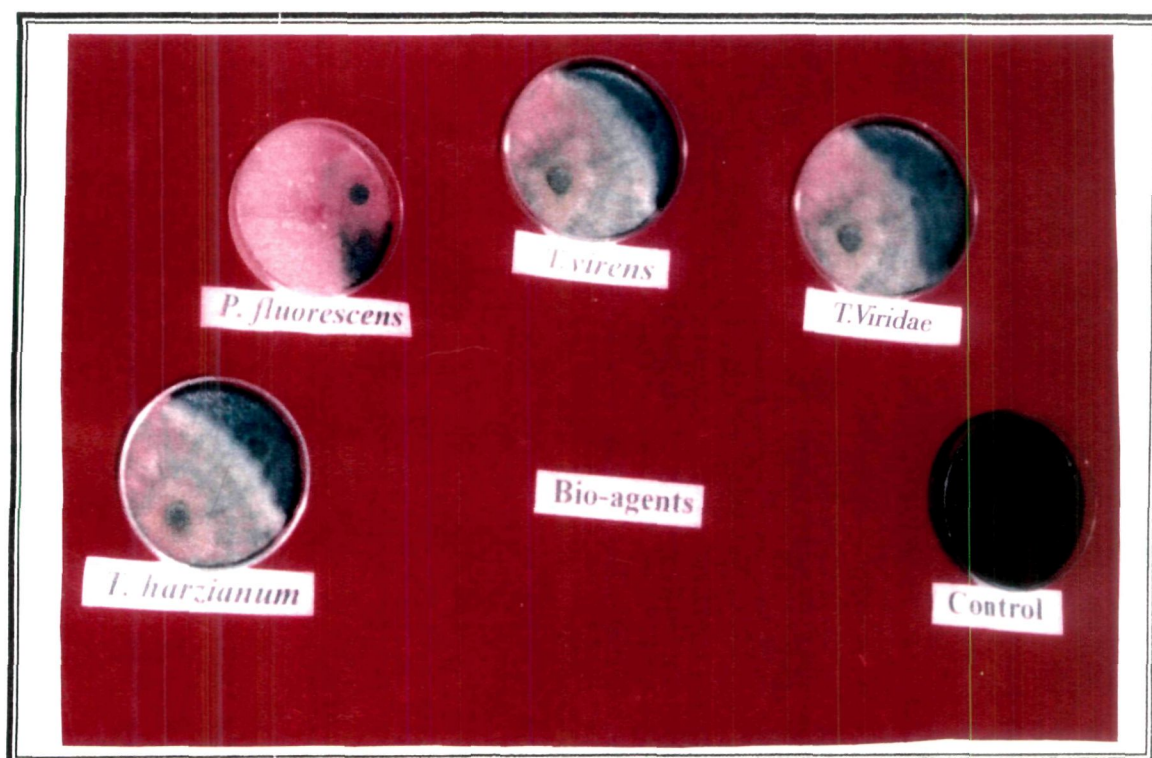


Plate 20. Effect of bio-agents on growth inhibition of *R. bataticola*

harzianum (72.9%), followed by *Trichoderma Virens* (57.3%) and the least inhibition of mycelial growth was recorded in *Trichoderma viridae* (55.8%).

4.11 SCREENING FOR RESISTANCE

4.11.1 Screening of chickpea genotypes against dry root rot disease.

In order to identify the resistant sources, twenty-five chickpea genotypes were screened against *R. bataticola* both in sick pot and blotter paper technique as described in material and methods.

4.11.1.1 Sick pot technique

The effect of *R. bataticola* on seed germination and reaction to dry root rot disease was tested under sick pot technique. The results (Table 18 and Table 19) indicated that lowest percent germination was observed in ICCV-98904 (73.0%) and ICCV-98933 (77.0%) respectively. Whereas rest of the genotypes recorded more than 80.0 per cent germination.

Among the 25 chickpea genotypes evaluated against dry root rot disease, none of them showed immune reaction. However, the genotype ICCV-94916-8 recorded least dry root rot incidence of 7.6 per cent followed by GPC-12 (10.0%) and both of them showed resistant reaction. The genotypes ICCV-98906 (14.2%) and ICCV-98904 (21.2%) recorded moderately resistant reaction. However, the maximum incidence was recorded in GPC-23 (64.5%) followed by ICCV-98907 (61.5%), AVT-5

(61.1%) and GPC-20 (58.9%) showed highly susceptible reaction. The remaining genotypes showed susceptible reaction.

4.9.1.2 Blotter paper technique.

Reaction of chickpea genotypes to blotter paper techniques are presented in Table 18 and Table 19. Among the 25 genotypes tested, ICCV-94916-8 with 8.1 per cent disease index and GPC-12 (8.3 PDI) and ICCV-98906 (10.0 PDI) were least affected genotypes. ICCV-98901 (87.7 PDI), GPC-20 (81.0 PDI) and ICCV-98905 (80.0 PDI) showed maximum dry root rot incidence compared to rest of the genotypes.

Table 18. Evaluation of chickpea genotypes to dry root rot caused by *R.bataticola* in sick pot and Blotter paper technique.

Genotypes	Sick Pot Technique		Blotter Paper Technique
	Percent germination	Percent dry root rot incidence (45 DAI)	Percent disease index
ICCV-98936	82.0	54.0	58.8
GPC-3	84.0	55.2	55.1
BG-362	88.0	40.0	46.1
PBG-68	80.0	55.1	57.7
ICCV- 98933	77.0	42.8	44.4
GPC-26	91.0	53.6	66.6
ICCV-98905	88.0	52.5	80.0
ICCV-4958	91.0	56.0	45.5
Annigeri-1	91.0	43.9	46.1
ICCV-94916-8	86.0	7.6	8.1
ICCV-98901	86.0	56.4	87.7
GPC-21	88.0	55.0	55.5
GPC-32	91.0	56.0	56.6
GPC-42	86.0	51.2	16.2
GPC-13	86.0	53.8	57.7
GPC-20	86.0	58.9	81.0
MLT-D-14	84.0	47.3	44.4
GPC-23	80.0	64.5	78.8
AVT-5	80.0	61.1	65.5
GPC-14	88.0	55.6	78.8
ICCV-98904	73.0	21.2	14.2
GPC-12	88.0	10.0	8.3
ICCV-94924-3	84.0	52.6	44.4
ICCV-98907	86.0	61.5	65.5
ICCV-98906	93.0	14.2	10.0

DAI – Days after inoculation

Table 19. Categorization of chickpea genotypes screened against dry root rot by sick pot and Blotter paper technique

Reaction	Genotypes		Genotypes	
	Rating	Sick pot	Scale	Blotter paper technique
Immune	0	NIL	1-2	NIL
Resistant	1 – 10	ICCV-94916-8 GPC-12,	3-4	ICCV-94916-8, GPC-12, ICCV-98906
Moderately resistant	11 – 25	ICCV-98906, ICCV-98904	5-6	ICCV-98904, GPC-42
Susceptible	26 – 50	BG-362, ICCV-98933, Annigeri-1, MLT-D-14	7-8	Annigeri-1, BG-362, MLT-D-14, ICCV-94924-3, ICCV-98933, ICCV-4958
Highly susceptible	51 – 100	ICCV-98936, GPC-3, PBG-68, ICCV-98905, ICCV-4958, GPC-21, ICCV-98901, GPC-32, GPC-14, GPC-13, GPC-20, GPC-23, AVT-5, GPC-14, ICCV-94924-3, GPC-26, ICCV-98907	9	GPC-3, ICCV-98936, GPC-26, PBG-68, ICCV-98905, GPC-21, ICCV-98901, GPC-32, GPC-13, GPC-23, AVT-5, GPC-20, GPC-14, ICCV-98907,

Discussion

V. DISCUSSION

Chickpea is one of the major grain legume crop of the tropics and subtropics grown predominantly under rainfed conditions. In India it is grown in an area of 6.09 million ha with a production of 5.27 million tonnes with an average productivity of 865 kg/ha in the year 2001-2002 (Mishra, 2003).

Chickpea is cultivated largely in northern parts of Karnataka. Although the black cotton soil in these parts are very ideal for cultivation of chickpea, but the crop suffers from some important diseases such as Fusarium wilt, Ascochyta blight and dry root rot in addition to viral problems. Of late, dry root rot of chickpea is a serious constraint in recent years in chickpea cultivation.

Rhizoctonia bataticola, a soil inhabiting pathogen is an important root pathogen and causes root rot, stalk rot of over 400 plant species including chickpea. The pathogen is a well known polyphagous, ubiquitous and non-target in its infectivity and is amongst the most universally distributed soilborne fungi.

The present investigations were carried out with reference to the dry root rot caused by *R. bataticola* in north eastern Karnataka, where soils are known to be problematic to soil borne pathogens. The study was initiated to know the incidence of dry root rot of chickpea in this area. In

addition, studies on cultural and physiological characters of the fungus, survivability of the pathogen in soil and plant debris, optimum inoculum level for disease development, effect of disease on chickpea growth parameter were also undertaken. For better management of the disease, some of the important antagonistic microorganisms were tested against *R. bataticola* along with commonly available fungicides. Chickpea genotypes were screened in sick pot and blotter paper to identify the resistant sources. The results obtained in the present investigations are discussed here under.

The characteristic symptoms of chickpea dry root rot observed were drooping and drying of leaves, complete destruction of lateral root system, leaves and stems of affected plants were usually straw coloured. The dark black sclerotia were observed on and with in host tissues and the disease normally appeared after flowering such plants could be easily pulled out of the soil.. The symptoms of chickpea dry root rot in our studies were similar to that of the earlier workers confirming the results on symptoms of dry root rot of chickpea (Haware *et al.*, 1990, Singh *et al.*, 1990).

The roving survey on incidence of dry root rot of chickpea was carried out in three districts of north-eastern Karnataka *viz.*, Gulbarga, Bidar and Raichur representing rainfed situations. The incidence was maximum in Gulbarga district with 9.97 per cent, followed by Raichur

(5.16%) and Bidar district with 4.28 per cent incidence. However, the highest incidence (12.21%) was noticed in Chittapur of Gulbarga district and the least incidence of 3.21 per cent was observed in Hudugi of Bidar district and Regional Agricultural Research Station farm, Raichur. The possible reason for the incidence of dry root rot in the fields surveyed may be that variation in soil type, soil temperature and moisture conditions which might be favorable for disease development during the cropping season with temperature ranging from was 20.6° C to 35.7° C with deficit rainfall occurred in Gulbarga district.

Das and Srivastava (1992) reported that variation in soil temperature and moisture stress during the cropping period leads to development of dry root rot disease. Several workers have also reported that the dry root rot disease intensity was positively correlated with soil temperature. Further Tariq Husain *et al.* (1995) observed that the colonization of *R. bataticola* on chickpea roots was higher when plants were subjected to moisture stress compared to unstressed plants.

The samples of infected plants were collected from chickpea field in Agricultural Research Station, Gulbarga. The pathogen isolated from the infected plants was identified as *R. bataticola* (Taub.) Butler, based on their morphological, cultural characters and pathogenicity in accordance with the description given by Nene *et al.* (1981) and their pathogenicity was proved by adopting blotter paper technique on chickpea. The

inoculated plants showed root rotting and drying of chickpea plants. Sclerotial production was also observed on root and basal stem portions of chickpea plants.

Cultural studies of the pathogen such as growth of the pathogen (*R. bataticola*) *in vitro* on different solid and liquid media, sclerotial production, mycelium colour, size and shape of the Sclerotia were recorded. Potato dextrose agar and Czapeck's agar were proved to be the best for growth of the fungus, followed by malt extract agar and Rose bengl agar. Good growth in PDA may be owing to provision of some additional nutrients in the media, as reported by different workers (Sahi *et al.*, 1992; Devi and Singh, (1998). The least growth of the pathogen was observed in oatmeal agar, which may be attributed to poor quality of sugar present in them.

With reference to colony character and sclerotia production on different media, potato dextrose agar showed mycelium having black colour with flat growth with uniform margin except with slightly raised growth with uneven margin in czapek's agar and with the light black colour mycelium on Rose bengal agar. However, good sclerotia production and maximum sclerotial size was noticed in PDA (68.2) whereas medium sclerotial production was observed in rest of the media tested. The present findings are in agreement with the earlier descriptions of the pathogen *R.bataticola* by Monga and Sharma (1994); Devi and Singh (1998) who

also reported good growth and sporulation of *R. bataticola* on PDA medium.

The optimum growth period for maximum mean dry mycelial weight of *R. bataticola* was obtained on the 16th day of incubation on potato dextrose broth thereafter the growth declined gradually with increase in number of days of incubation. This may be due to autolysis of the mycelium and exhaustion of nutrients in the medium after incubating for optimum number of days. However, these results are not in agreement with Satischandra (1977) who noticed maximum growth at 11 days after incubation in bean isolates of *R. bataticola*.. This variation of long incubation period for maximum dry mycelial weight may be due to variability in isolates pathogen and the host it infects in addition to exhaustion of nutrients and autolysis of mycelium in the medium.

In the present investigation, the maximum mycelial growth of *R. bataticola* was obtained at 35° C followed by 30° C. The present results are in agreement with the investigations carried out by many workers (Vishwa Dhar and Sarboy 1989; Vishwa Dhar *et al.*, 1991; Amrit Sandhu *et al.*, 1999) where they observed that, optimum temperature for germination and germ tube elongation of sclerotia of *R. bataticola* on soybean was 20-30° c and the maximum and minimum temperature were 40°c and 5° c respectively. Light is another important factor to be considered on growth and production of sclerotia by the fungus. In the

present study, significantly higher mycelial growth and sclerotia production of *R. bataticola* was noticed when culture was exposed to alternate cycles of 12hrs light and 12hrs darkness followed by continuous light. Enhanced sporulation of root rot pathogen on soybean was obtained when culture was exposed to 12 hours of alternate light and darkness (Vishwa Dhar, *et al.* 1991). On the other hand, an increased darkness period had a detrimental effect on physiology of the pathogen culture. Continuous light or light alternated with darkness favoured good growth of *M. phaseolina* on czapeck's Dox liquid medium, as reported by Muthukrishnan *et al.* (1995 b).

Relative humidity plays an important role in growth, sporulation and perpetuation of the pathogen either in soil or plant debris subsequently in causing disease epidemics of *R. bataticola*. In the present investigations revealed that maximum growth and sporulation of *R. bataticola* was observed at 75 per cent humidity level followed by 85 per cent.

Survival of *R. bataticola* mainly depends on length of the storage period, soil moisture, soil temperature and other environmental conditions. The results of present investigations also revealed that, the fungus survived to a maximum period of 210 days in plant debris as well as in soil and survived for least period of 90 days when preserved in refrigerator. Perhaps in refrigerator, the fungus *R. bataticola* must have

entered in to sclerotia stage which was not recovered on the plate. The sclerotial production of *R. bataticola* initially increased slightly up to three to four months in both soil and plant debris. The probable reason might be due to the time lag required for the acclimatization and establishment of the pathogen in the soil amongst the competitors and other microbiological complexity in the soil. However, the results are in agreement with investigations of Suriachandraselvan and Seetharaman (2000) where they reported, that the pathogen *M. phaseolina* survived for 13 and 10 months in sterilized and unsterilized soil respectively and 31 months in the plant debris when stored in ambient conditions. Mustafee and Chattopadhyay (1983) reported that maximum survival in sterilized soil might be explained due to reduction in inhibition of *M. phaseolina* resulting from the absence of the antagonistic organisms in the soil or by the absence of competition for space and nutrients at the onset of the sclerotial germination. Lewis (1979) was of the opinion that a single factor probably cannot account for the population dynamics of a pathogen in a complex natural soil. Chemical and biological properties of soil are more important than soil texture in influencing the survival of *R. solani* in various soils. It may be possible that, clays may influence microbial activity that in turn affects *R. solani*.

Since the *R. bataticola* is a soil inhabitant, its level or intensity of inoculum density in the soil is bound to influence the incidence and

severity of the disease on chickpea. The findings of the present investigations showed that, a minimum of 4% inoculum in soil (40g/kg soil) was found to be essential to incite early disease (35 days) and maximum disease intensity (76.6%) as compared to 1, 2, and 3% inoculum where the disease appeared at 58, 47, and 40 days with disease intensity of 28.5, 43.7 and 64.2 per cent respectively. The higher inoculum level of 4 % in soil influenced early symptom expression, higher disease severity and also affected all the growth parameters (plant height, shoot and root biomass at flowering and at pod formation stages) drastically as compared to control (Table 14).

Similar results were also reported by Byadgi (1979) where the infection in bean seedlings with four per cent inoculum of *R. bataticola* increased with increase in the inoculum levels. Ramamurthy (1982) also reported, the maximum infection of *R. bataticola* at higher inoculum levels in cotton seedlings. Susceptibility to *R. bataticola* [*Macrophomina phaseolina*] increased with age at inoculation. From 30 d onwards all 5 cultivars showed equal susceptibility, though they had been selected for differences in susceptibility when inoculated at 5d. Screening for resistance to this disease should therefore be carried out with plants that are at least 30 d old (Singh, *et- al.*, 1990)

Chemical control is an important tool and ultimate choice for managing any disease including soil borne pathogens, particularly when

the disease is already prevalent in the field in epidemic form. Hence, identification of effective fungicides would enable to include effectiveness as one of the components of integrated disease management strategy.

In-vitro screening of fungicides provided preliminary information regarding their efficacy against *R.bataticola*. Among the systemic fungicides, hexaconazole was found to be highly effective in inhibiting the growth of *R. bataticola* in poison food technique at all the three concentrations tested, followed by benomyl and carbendazim. However, the least inhibition of growth of *R. bataticola* was observed in carboxin at all the three concentrations. Among non-systemic fungicides, Saaf was found to be equally effective in inhibition of growth of *R. bataticola* at lowest concentration (1000ppm) under *in vitro* followed by mancozeb and thiram. These observations were in agreement with the research findings of Devi and Singh (1997) and they reported that the fungicides such as carbendazim, thiophanate methyl and mancozeb were found effective in inhibition of growth of *M. phaseolina* @ 0.2 per cent under *in-vitro* conditions. Ebenzar and Wesely (2000) reported that among the six fungicides studied under *in-vitro* condition for their inhibitory effect on the growth of *M.phaseolina*, carbendazim completely inhibited the growth of *M.phaseolina* at 0.1%. Benomyl was found to be most effective against *M.phaseolina* followed by carbendazim and thiophanate methyl in pigeonpea (Lokesha, 2002).

Biological control is a potential non-chemical means for plant disease control by reducing the harmful effects of pathogen through the use of other living entities, since, *R.bataticola* is soil borne and poses a great problem in managing the disease. Soil borne diseases are difficult to control and even seed treatment with fungicides does not protect the crop for long periods or soil drenching with fungicides are not economical. Moreover, they may establish imbalances in the microbial community unfavourable for activities of beneficial organisms. In addition, continuous use of some fungicides for the same pathogen results in the development of resistant strains of the pathogen besides polluting the environment.

Sanford (1926) was the first to realize that soil microorganisms themselves exert a natural biological control on root diseases. Then onwards there has been a dramatic increase in research efforts and several books (Baker and Cook 1974; Burges, 1981; Cook and Baker, 1983) and review articles (Naik and Sen, 1985; Jayashree *et al.*, 2000) have come up, covering the use of specific microbes, their mechanisms and commercial evaluation. It is now widely recognized that biological control of plant pathogens using antagonistic fungi and bacteria is a distinct possibility for the future and can be successfully utilized especially within the framework of integrated disease management system (Muthamilan and Jeyarajan, 1996). Use of antagonistic organisms against *Macrophomina* root rot has been well documented (Mukhopadhyay, 1987; Raguchander *et al.*, 1995) in several crops.

The use of bioagents to manage soil-borne diseases involves two types of interaction *viz.*, intermingling with test pathogen and reducing the further growth of the pathogen. Several possibilities of existence of microbial interactions have been evidenced of which, stimulating, inhibiting, non-influencing, mutual intermingling; over growth on the test pathogen and inhibition at a distance have been enumerated (Benhamou and Chet, 1993).

In the present investigation under *in-vitro* screening, among the, three fungal bioagents (*T.harzianum*, *T. Viridae* and *T. virens*) tested, *T.harzianum* recorded 72.90 percent inhibition of linear growth of *R. bataticola* under dual culture technique over other bio-agent. The biocontrol mechanisms might be due to competition by antagonist over the growth inhibition of *R.bataticola*. Similarly, Selvarajan and Jeyarajan (1996) reported that maximum inhibition zone of 5.7 mm and reducing the size of sclerotia, germination and germ tube growth of *M.phaseolina* in chickpea by *T.harzianum* and was significantly superior over other fungal antagonists. Jacob (1989) also observed wide inhibition zone of *M. phaseolina* by *T.harzianum* and *T. viridae*. The effects of antagonists on inhibition of mycelial growth in view of the fact that the reserve food material in mature sclerotia is to be supplied by vegetative hyphae. When the growth of the vegetative hyphae itself is inhibited and it is not able to support the developing sclerotia. Elad *et al* (1986) reported the inhibition

of growth and sclerotial production of *M.phaseolina* by *T.harzianum*. This may be attributed to the secretion of some toxic metabolites, growth inhibition, coagulation of cytoplasm, hyphal discoloration and dissolution of the hyphae of *M.phaseolina*. (Dubey and Dwivedi, 1988). Such mechanisms have been described by previous workers (Cook and Baker, 1983).

The bacterial antagonist, *P. fluroscens* inhibited maximum growth of 79.20 per cent of *R.bataticola* and was found more effective in comparison with the fungal antagonists tested in the present investigations. Earlier. Jacob (1989) reported the inhibition of *M.phaseolina* by *Bacillus subtilis* and inhibition zone and antagonistic effect of *P. fluroscens* on inhibited maximum growth of *R.bataticola*. Similarly, Nagina-Parmar and Dadarwal (1997) reported that bacteria with plant growth promoting and pathogen suppressive abilities were isolated from the rhizoplane and tested antifungal activity against *R.bataticola*, *Fusarium oxysporum*, *Pythium aphanidermatum*. Bacteria with antifungal activity exhibited enhanced production of flavonoids/flavones like compounds in the seedling rots, indicating their role in induction of systemic resistance in the plants. Earlier. *M. Phaseolina* in urd bean was reported by Samiyappan (1988) and Jana *et al* (2000). Further Karunanithi *et al* (2000b) found that *P. fluroscens* produced an antibiotic compound called Pyrollnitrin, which inhibited the mycelial growth of *M. phaseolina* in dual

culture by producing an inhibition zone of 12 mm. A rifampicin-resistant mutant *P. fluorescens* NBRI1303R of *P. fluorescens* NBRI1303, used to monitor chickpea root colonization, confirmed the rapid and aggressive colonization by the bacterium, suggesting that it has potential as a biocontrol agent against chickpea disease (Nautiyal, 1997).

Hence, biological control through the use of antagonists' microorganisms is a potential non-chemical means for reducing the inoculum levels and activity of the soil borne pathogen. The use of antagonists as one of the component in integrated disease management would help in preventing the pollution and hazardous effects on eco-friendly microflora in the soil.

The use of host plant resistance to manage economically important soil-borne disease has been considered to be most economical choice in all crop improvement programmes and is essential to recommend directly for cultivation in endemic area or to use as donor of resistant genes in breeding programme. Though fungicides are in use as seed treatment in case of seed-borne pathogens and its judicious use as one of the component in integrated disease management approaches appear to be a wise practice. *In-vitro* screening of chickpea genotypes against dry root rot disease caused by *R.bataticola* was carried out by employing both sick pot and blotter paper evaluation techniques. These methods were found efficient in screening for resistance when

compared to natural screening in “hot spot” locations where it takes not only longer time for chickpea seedlings to express disease symptoms and but chances of escape of disease is also high due to varied inoculum levels in soil.

In case of sick pot or blotter screening techniques the reaction of genotypes to dry root rot disease showed equal susceptibility when roots of 5-10 days old seedlings were inoculated. Hence, these evaluation methods were found preferable for screening the genotypes against *R. bataticola* to avoid disease escape under natural screening. Singh *et al.* (1990) noticed that, susceptibility of chickpea plants to *R. bataticola* increased with age of the plants of more than 30 days under field conditions.

Among 25 chickpea genotypes screened for resistant against *R. bataticola*, none of the genotypes showed immune reaction in blotter and sick pot screening methods. The reaction of genotypes in sick pot method was recorded 45 days after planting. However, in both the screening methods, the genotypes ICCV-94916-8 and GPC-12 showed resistant reaction. Further, some of the genotypes viz., ICCV-98904 and ICCV-98906 showed moderately resistant (11-25% incidence) reaction. The susceptibility of chickpea genotypes to *R. bataticola* may be due to the higher activity of pectin trans-eliminase and polygalactouronate trans-eliminase and reduced activity of these enzymes in resistant genotypes as observed by Srivastava (1987). *Cicer arietinum* accessions showed

considerable variation between years and seasons. *F. oxysporum* was the predominant fungus isolated from dead plants followed by *M. phaseolina*, *F. solani*, *Sclerotium [Corticium] rolfsii* and *R. solani*. It is suggested that this is probably the first report on the identification of combined resistance to wilt and root rots in wild *Cicer* spp (Reddy, *et al.*, 1991). Similarly, Nene, *et al.*, (1989), reported few to several lines were resistant to wilt and/or root rots. Genotypes ICC- 2862, ICC- 9023, ICC- 9032, ICC -10803, ICC-11550 and ICC -11551 were resistant to wilt and root rots at 11 to 12 of the 18 locations where they were tested.

5.6 FUTURE LINE OF WORK

During the present investigations, certain areas have been identified for further investigations for better understanding of the pathogen.

- * To conduct systematic and comprehensive surveys of all the chickpea growing areas for dry root rot of chickpea disease and estimation of their economic importance in terms of monetary loss.
- * Role of cultural practices in reducing the survival of inoculum in the soil.
- * Variability and Molecular characterization of various isolates of *R. bataticola* for better understanding the genetics of pathogen is essential in different agro-climatic regions.

- * Need to study combination components of fungicides, bioagents and resistance in the integrated management of dry root rot of chickpea
- * Identification and utilization of more number of resistant sources against the dry root rot disease and their deployment in the endemic areas and to further the cause of breeding for resistance is essential.

Summary

V1. SUMMARY

Chickpea (*Cicer arietinum* L.) is one of the major grain legume crops of the tropics and subtropics including India. Among the biotic stresses, dry root rot of chickpea caused by *R. bataticola* has become an important problem in recent years in chickpea cultivation and causes considerable loss to chickpea production.

The present investigation involves survey for dry root rot (*R. bataticola*) incidence, cultural and physiological studies, survival of the organism in soil and plant debris, effect of disease on chickpea growth parameters, screening of genotypes, evaluation of bioagents and fungicides for managing root rot of chickpea. The results obtained from the investigation are summarized here under.

The dry root rot disease symptoms on chickpea generally appears at flowering and podding time as scattered dried plants and the maximum disease incidence coincides with moisture stress and high temperature. The characteristic symptoms of the diseases are the darkening of roots and extensive rotting, destruction of lateral and finer roots. In addition, roots become brittle and shredding of bark take place. The drooping of petioles and leaflets is commonly seen. When uprooted the lower portion of stem remains in soil and dark minute sclerotial bodies are also seen around roots.

Survey for incidence of dry root rot was carried out during *rabi* season of 2002 by roving survey in three districts of northern parts of Karnataka *viz.*, Gulbarga, Bidar and Raichur. The severity of the root rot incidence was 9.97 per cent in Gulbarga district followed by Raichur (5.16%) and least incidence was observed in Bidar district (4.28%). Highest incidence of 12.21 percent was recorded in Chittapur of Gulbarga district.

The dry root rot causing pathogen was isolated in pure form by hyphal tip culture method from naturally infected chickpea plants from the field. The pathogenicity of the fungus was proved by repeated inoculation to chickpea seedlings. The inoculated seedlings showed characteristic symptoms to that of naturally infected plants caused by *R.bataticola* as drooping and drying of leaves followed by loss of vigour and premature death and production of sclerotial bodies on infected roots. The pathogen was re-isolated from inoculated seedlings and based on morphological characteristics, the fungus was identified as *R.bataticola* (Taub.) Butler.

The cultural studies revealed that the maximum mycelial growth (90mm), dry mycelial weight (181.9mg) and sclerotial production (89.3) of the pathogen was observed in potato dextrose agar followed by Czapeck's agar medium as compared to other media tested. Among the different

liquid media tested, potato dextrose broth supported the maximum dry mycelial weight when compared to other media tested.

The results of the present investigation also revealed that the optimum temperature of 35° c, relative humidity of 75 per cent and exposure to alternate cycles of light and darkness favoured maximum growth and sclerotial production of dry root rot pathogen, *R.bataticola*.

The pathogen *R.bataticola* survived/perpetuated up to 150 days in plant debris and 120 days in soil and later the fungus decreased in its growth. Where as in refrigerator, the fungus survived only 90 days and the fungus could not be recovered in subsequent isolations.

Early initiation of the disease in chickpea was observed at highest inoculum level of 4.0 per cent in soil among the four levels tested. It was also found that higher inoculum level had a profound effect on various growth parameters of plant viz., plant height, number of branches, shoot and root length and total biomass of the plant both at flowering and pod formation stages. The results indicated that total reduction of biomass is 70.68, 73.16 per cent at flowering and pod formation stages, respectively.

Five each of systemic and non-systemic fungicides were tested for their efficacy against *R. bataticola* under *in- vitro* conditions. It was found that hexaconazole and benomyl among systemic fungicides inhibited maximum growth (100 and 99%) of *R. bataticola* and was found effective

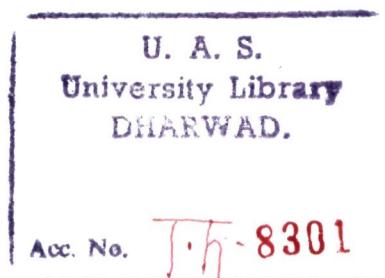
at lowest concentration of 250ppm followed by carbendazim. The least growth of inhibition of 32.5 and 34% was observed with carboxin and thiophanate methyl at 250ppm respectively. The SAAF and mancozeb among the non-systemic fungicides inhibited maximum inhibition of growth (100 and 94.7%) at 1000 ppm where as least inhibition of 36.2% was observed with captan at lowest concentration tested.

Among the four antagonists evaluated under *in- vitro* conditions, the highest percent of growth inhibition of 72.2 per cent was noticed in *P. fluorescens* followed by 72.9 and 57.3 per cent by *T. harzianum* and *T.virens* respectively. The least inhibition of 55.8 per cent was observed in *T.viride*.

Twenty-five chickpea genotypes were screened against *R. bataticola* under laboratory by employing sick pot and blotter paper technique. Among these, none of the genotypes showed immune reaction. However the genotypes ICCV-94916-8 and GPC- 12 showed resistant reaction. (1-10%), whereas ICCV-9890 and ICCV-98906 showed moderately resistant reaction (11-25%). However, GPC-23, ICCV- 98907 and AVT-5 showed highly susceptible reaction.

References

Released on — — — 1 4 MAY 2005



VII. REFERENCES

- AHMED, Y., HAMEED, A. AND ASLAM, M., 1992, Efficacy of different fungicides in controlling the maize stalk rot. *Pakistan Journal of Phytopathology*, **4**(2): 14-19.
- AMRIT SANDHU, SINGH, R.D. AND SANDHU, A., 1999, Factors influencing susceptibility of cowpea to *M. phaseolina*. *Journal of Mycology and Plant Pathology*, **29**(3): 421-424.
- ANILKUMAR AND LAKRA, B.S., 1990, Effect of heating for different duration of time on the viability of soil borne fungi in culture. *Indian Journal of Mycology and Plant Pathology*, **20**(2): 158-160.
- *ANIL KUMAR, T.B. AND SASTRY, M.N.L., 1980, Variation in *Rhizoctonia bataticola* isolates. *Zbl. Bakt. II. Abt.*, **135**: 246-251.
- ANILKUMAR, T.B. AND SASTRY, M.N.L., 1980, Nutritional and pathological variation among isolates of *R. bataticola* from sunflower. *Journal of Microbiology*, **137**: 228-232.
- ANONYMOUS, 1943, The slide germination method of evaluating protectant fungicides. The American phytopathological Society Committee on standardization of Fungicidal Test. *Phytopathology*, **33**: 627-632.

ANONYMOUS, 1997, *Hand book of Agriculture*. ICAR, New Delhi. p. 842.

ASHBY, S.F., 1927, *Macrophomina phaseolina* (Maub.) Comb. Nov. The pycnidial stage of *Rhizoctonia bataticola* (Taub.) Butler. *Phytopathology*, **17**: 141-147.

*BAKER, K.F. AND COOK, R.J., 1974, *Biological control of plant pathogens*. W.H. Freeman and Company, San Francisco. Pp. 443.

BENHAMOU, N. AND CHET, I., 1993, Hyphal interactions between *T. harzianum* and *R. solani*. *Phytopathology*, **83**: 1062-1071.

BHARGAVA, S.N., 1965, Studies on charcoal rot of potato. *Phytopathology*, **53**: 35-44.

BOOTH, C., 1971, *Methods in Microbiology*, Vol.4. Academic Press, London. P. 795.

BURGES, M.J., 1981, *Microbial control of plant diseases*. 1970-1980, London, England, Academic Press, pp. 949.

BUTLER, E.G., 1925, Identification of the fungus. *Tech. and Sci., Serv. Bull.*, **49**: 64-65.

BYADGI, A.S., 1979, Survival and variation studies on *Rhizoctonia bataticola* (Taub.), Butler, M.Sc. (Agri.) Thesis, University of Agricultural Sciences, Bangalore. P.199.

- BYADGI, A.S. AND HEGDE, R.K., 1985, Variations among the isolates of *R. bataticola* from different host plants. *Indian Phytopathology*, **38**: 297-301.
- *CHOWDHARY, S., 1944, Disease of pan (*Piper betle*) in Sylhet Assam. IV. *Rhizoctonia* root rot. *Proc. Indian Acad. Sci. Sect. B.*, **20**: 229-244.
- COOK, G.E., BOOSALIS, M.G., DUNKLE, L.D. AND ODODY, G.N., 1973, Survival of *Macrophomina phaseoli* in corn and sorghum residue. *Plant Disease Reporter*, **57**: 873-875.
- COOK, R.J. AND BAKER, K.F., 1983, The nature and practices of biological control of plant pathogens. *American Phytopathological Society, Minnesota*, pp. 539.
- DAS, N.D. AND SRIVASTAVA, N.N., 1992, Effect of environmental factors on root rot or charcoal rot disease of castor. *Indian Journal of Plant Protection*, **20**(2): 232-233.
- DEVI, T.P. AND SINGH, R.H., 1997, Screening of fungicides against seedling mortality of black gram caused by *Macrophomina phaseolina*. *Indian Journal of Plant Protection*, **25**(2): 123-127.
- DEVI, T.P. AND SINGH, R.H., 1998, Cultural variation of *Macrophomina phaseolina* isolates collected from *Vigna mungo*. *Indian Phytopathology*, **51**(3): 292-293.

- DHAR, V. AND SARBHOY, A.K., 1993, A typical isolates of *R. bataticola*. *Indian Phytopathology*, **46**: 245-246.
- DHINGRA, O.D. AND SINCLAIR, J.B., 1975, Effect of soil moisture and carbon: nitrogen ratios on survival of *Macrophomina phaseolina* in soybean stems in soil. *Plant Disease Reporter*, **58**: 1034-1036.
- *DHINGRA, C.D. AND SINCLAIR, J.B., 1978, *Biology and pathology of M. phaseolina*, Universidade Federal de Vicosa, Brazil, pp. 166.
- DINAKARAN, D. AND MARIMUTHU, T., 1988, Inhibition of *M. phaseolina* by mutants of *Trichoderma viridae*. *Journal of Biological Control*, **11**(1): 43-47.
- DUBEY, R.C, AND R.S. DWIVEDI. 1988. Antagonism between *M. phaseolina* and some rhizosphere micro fungi of soybean. *International Journal of tropical plant disease*, **6**: 89-94.
- DWIVEDI, R.S. AND DUBEY, R.C., 1987, Effect of Fungicides on survival of *M. Phoseolina* in soil. *International Journal of Tropical Plant Diseases* (India), **5**(2): 147-152.
- EBENEZAR, E.G. AND WESELY, E.G., 2000, Effect of fungicides on *Macrophomina phaseolina* causing root rot in greengram. *Pestology*, Volume XXIV. **3**: 12-14.

ELAD, Y., Y. ZVIELI AND I. CHET, 1986. Biological control of *Macrophomina phaseolina* (Tassi) Goid. by *T. harzianum*, *Crop Protection*, **5**: 288-292.

GANAGWAR, R.K., PRAJAPATI, R.K., SRIVASTAVA, S.S.L., KUMUD KUMAR AND KUMAR, K., 2002, Resistance in chickpea germplasms against the dry root rot. *Annals of Plant Protection Sciences*, **10**(2): 393-394.

*GARRETT, S.D., 1956, *Biology of root infecting fungi*. Cambridge Univ. Press, London, P. 293.

GHOSH, S.K. AND SEN, C., 1973, Comparative physiological study on four isolates of *M. phaseolina*. *Indian Phytopathology*, **26**: 615-621.

GOIDANICH, G., 1947, A revision of the genus *M. phaseolina* petreak type species: *M. phaseolina* (Tassi.) Goid, *M. Phaseolina* (Maubl.) Ashby. *Phytopathology*, **37**: 449-461.

GUPTA, 1995, Identification of chickpea genotypes with dual resistance against wilt and root rots. *International Chickpea and Pigeonpea Newsletter*, **2**: 27-28.

HAWARE, M.P., JIMENEZ-DIAZ, R.M., AMIN, K.S., PHILLIPS, J.C. AND HALILA, H., 1990, Integrated management of wilt and root rots of chickpea. *Proceedings of the Second International Workshop on Chickpea Improvement* **4-8** Dec.: 129-133.

HAWARE, M.P. AND NENE, Y.L., 1978, A root rot of chickpea seedlings caused by a sterile fungus. *Legume Research*, **1**(2): 65-68.

*HAWKSWORTH, D.L., SUTTON, B.S. AND AINSWORTH, G.C., 1983, *Ainsworth and Bisby's Dictionary of the Fungi*. Vol.III, Commonwealth Mycological Institute, Kew, Surrey, London, P.445.

HAYMAKER, H.H., 1928, Pathogenicity of two strains of the tomato wilt fungus, *Fusarium lycopersici*. *Journal of Agriculture Research*, **36**: 675-719.

*HUFFAKER, C.B. AND MESSENGER, P.S. (EDS.), 1976, Theory and Practice of Biological Control, London, U.K., *Academic Press*, pp.788.

ISRAELI, O.P. AND ALI, M.S., 1964, Effect of carbohydrates on the growth of *Rhizoctonia solani* Kuhn. *Biologia Pl.*, **6**: 84-87.

JACOB, K, 1989, Biological control of root rot of black gram caused by *M.phaseolina*, PhD thesis, T.N.A.U. Coimbatore. 128 pp.

JANA, T.K., SRIVASTAVA, A.K., CSERY, K. AND ARORA, D.K., 2000, Agglutination potential of *Pseudomonas fluorescens* against black gram and sesame root rot disease. *Journal of Biological Control*, **14**(2): 55-61.

- JAYANT, BHATT. AND BHATT, J., 1993, Reaction of chickpea cultivars to *Rhizoctonia bataticola* (Taub.) Butler. *Indian Journal of Pulses Research*, **6**(1): 118-119.
- JAYASHREE, K., SHANMUGAM, V., RAGUCHANDER, T., RAMANATHAN, A. AND SAMIYAPPAN, R., 2000, Evaluation of *Pseudomonas fluorescens* against black gram and sesame root rot disease. *Journal of Biological Control*, **14**(2): 55-61.
- *JEYARAJAN, R., RAMAKRISHNAN, G. AND SANGEETHA, P., 1991, Efficacy of *Trichoderma* as biocontrol agent for root rot disease of grain legumes. *Petria*, **1**: 143.
- KARUNANITHI, K., MUTHUSAMY, M. AND SEETHARAMAN, K., 2000, Pyrollnitrin production by *P. fluorescens* effective against *M. phaseolina*. *Crop Research*, **19**(2): 368-370.
- KEHRI, H.K. AND CHANDRA, S., 1991, Antagonism of *Trichoderma viride* to *M. phaseolina* in the control of dry root rot of mungbean. *Indian Phytopathology*, **44**: 60-63.
- LEWIS, J.A. (1979). Influence of soil texture on survival and saprophytic activity of *Rhizoctonia solani* in soils. *Canadian Journal of Microbiology*. **25**: 1310-1314.

LOKESHA, N.M., 2002, Management of dry root rot of pigeonpea (*Cajanus cajan* (L.) Millsp.) caused by *Macrophomina phaseolina* (Tassi) Goid. M.Sc. (Agri.) Thesis, *University of Agricultural Sciences*, Dharwad, p.76.

MASOOD ALI AND SHIVKUMAR, 2001, An overview of chickpea research in India. *Indian Journal of Pulses Research*, **14**(2): 81-89.

*MATANA SRIHUFFAGUM, CHARAS KITBAMROONG AND PORNPUDH PRASERTKUL, 1997, Systemic development of *M. phaseolina* from root infection. *Proceedings of National Mungbean Research Conference VII, Bangkok* (Thailand), pp. 175-185.

MEYER, W.A., SINCLAIR, J.B. AND KHARE, M.N., 1973, Biology of *Macrophomina phaseoli* in soil studied with selective media. *Phytopathology*, **63**: 613-620.

MISHRA, S.P., 2003, All India coordinated research project on chickpea, *Project Coordinators Report 2002-2003*, pp.17.

MOHAMED, B.E., SHOHLA, G.S. AND EL-ERAKI, S., 1990, Interaction of *Meloidogyne incognita* and *Macrophomina phaseolina* under different pesticide treatments. *Agricultural Research Review*, **68**(3): 581-587.

- MONGA, D. AND SHARMA, H.C., 1994. Variability in the isolates of *Macrophomina phaseolina* causing dry root rot in cotton. *Indian Phytopathology*, **47**:403-407.
- MUKHOPADHYAY, A.N., 1987, Biological control of soil borne plant pathogens by *Trichoderma* spp. *Indian Journal of Mycology and Plant Pathology*, **17**: 1-9.
- MUSTAFEE, T.P.AND S.B. CHATTOPADHYAY, 1983. Survivability of soil borne pathogens with and without host tissues. *Indian Phytopathology*, **36**: 58-61.
- MUTHUKRISHNAN, K., ARJUNAN, G. AND RAGUCHANDER, R.T., 1995, Some pathological studies on *M. phaseolina* root rot of urdbean. *Indian Journal of Pulse Research*, **8**(2): 162-165.
- MUTHUKRISHNAN, K., RAGUCHANDER, R.T. AND ARJUNAN, G., 1995b, Factors affecting survival of *M. phaseolina* in soil. *Indian Journal of Pulse Research*, **8**(2): 156-161.
- MUTHAMILAN, M.AND JEYARAJAN, R., 1996, Integrated management of *Sclerotium* root rot of groundnut involving *T. harzianum*, *Rhizobium* and Carbendazim. *Indian Journal of Mycology and Plant Pathology*, **26**: 204-209.

- NAGINA-PARMAR AND K.R. DADARWAL. (1997), Rhizobacteria from the rhizosphere and rhizoplane of chickpea. *Indian Journal of Microbiology*. **37**(4): 205-210.
- NAIK, M.K. AND SEN, B., 1985, Biocontrol of plant disease caused by *Fusarium* spp. p. 22. In: K.G. Mukherjee (Ed). Recent Developments in Biocontrol of plant diseases. *Aditya Publishing House, New Delhi, India*.
- NAUTIYAL, C.S., 1997, Selection of chickpea rhizosphere competent *P. fluorescens* NBRI-1303 antagonistic to major soil borne pathogens of chickpea. *Current Microbiology*, **35**(1): 52-58.
- NENE, Y.L., HAWRE, M.P. AND REDDY, M.V., 1981, Resistance screening techniques in chickpea disease. *Information bulletin* No. **10**. ICRI SAT.
- NENE, Y.L, HAWARE, M.P., REDDY, M.V. AND PHILLIPS, J.C., 1989, Identification of broad-based and stable resistance to wilt and root rots in chickpea. *Indian Phytopathology*, **42**(4): 499-505.
- PANDEY, G. AND SINGH, R.B., 1990, Survey of root disease of chickpea in Alahabadh region. *Current Nematology*, **1**(1): 77-78.
- PANDEY, S., SRIVASTAVA, S.N., RAMJILAL AND TRIPATHI, R.C., 1999, Role of inoculum density and incubation period an saprophytic behaviour of *Rhizoctonia solani* and *Rhizoctonia bataticola*. *Indian Phytopathology*, **52**(1): 56-58.

PATIL, S.H., 1980, Studies on charcoal rot of sorghum caused by *Macrophomina phaseoli* (Tassi) Goid. Unpublished M.Sc. (Agri.) Thesis, University of Agricultural Sciences, Bangalore, pp.78.

*PITT, D., 1964, Studies on sharp eye spot disease of cereals 1. Disease symptoms and pathogenicity of isolates of *Rhizoctonia solani* Kuhn and the influence of soil factors and temperature on disease development. *Annual Application of Biol*, **54**: 77-79.

PRADEEP KUMAR, ANUJA, KUMUD KUMAR, KUMAR, P. AND P. KUMAR, K., 2000, Biocontrol of seed borne fungal pathogens of pigeonpea. *Annals of Plant Protection Science*, **8**(1): 30-32.

RAGUCHANDER, T., RAJAPPAN, K. AND PRABHAKAR, K., 1995, Evaluation of talc based product of *T. viride* for the control of urdbean root rot. *Journal of Biological Control*, **9**: 63-64.

RAJINDER SINGH, SINDHAN, G.S. AND SINGH, R., 1998, Effect of fungicides on the incidence of dry root rot of chickpea. *Plant Disease Research*, **13**(1): 14-17.

RAMAMURTHY, R., 1982, Studies on root rot of cotton caused by *Rhizoctonia bataticola* (Taub.) Butler, M.Sc. (Agri.) Thesis, University of Agricultural Sciences, Bangalore.

- RANGESHWARAN, R. AND PRASAD, R.D., 2000, Isolation and evaluation of rhizosphere bacteria for biological control of chickpea wilt pathogens. *Journal of Biological Control*, **14**(1): 9-15.
- REDDY, M.V., RAJU, T.N AND PUNDIR, R.P.S, 1991b. Evaluation of wild Cicer accessions for resistance to wilt and root rots. *Indian Phytopathology*. 1991, **44**: 3, 388-391.
- REDDY, G.R., REDDY, A.G.R. AND RAO, K.C., 1991a, Effect of different seed dressing fungicides against certain seed borne fungi of groundnut. *Journal of Oilseeds Research*, **8**: 79-83.
- *REICHERT, I. AND HELLINGER, E., 1947, On the occurrence, Morphology and parasitism of *sclerotium bataticola*. *Palestine J. Bot.*, **6**: 107-147.
- SAHI, S.T., SHAKIR, A.S., BAJWA, M.N. AND INTIZAR UL HASSAN, M., 1992, Physiological studies on *M. phaseolina* causing dry root rot of mungbean. *Journal of Agricultural Research, Lahore*, **30**(3): 409-413.
- SANFORD, G.B., 1926, Factors affecting the pathogenicity of *actinomyces*. *Phytopathology*, **16**: 525-527.
- SAMIYAPPAN, R. 1988. Biological control of black gram root rot caused by *M. phaseolina* (Tassi) Goid. PhD thesis. *Tamil Nadu Agricultural university*, Coimbatore, India, 169 pp.

- SATISHCHANDRA, K.M., 1977, Saprophytic activity and survival of *R. bataticola* in different soils. M.Sc. (Agri.) thesis, university of agricultural sciences, Bangalore, p, 80.
- SATISH LODHA, 1998, Effect of sources of inoculum on population dynamics of *M. phaseolina* and disease intensity in cluster bean. *Indian Phytopathology*, **51**(2): 175-179.
- SAVITA PAREEK, JAIN, B.L. AND PAREEK, K.S., 1998, Efficacy of fungicides tested *in vivo* by soil drenching against *M. phaseolina*. *Journal of Phytopathological Research*, **11**(1): 83-4.
- SCHROEDER, H.D., BOOSALIS, M.G. AND MOORE, M.B., 1953, An improved inoculum of plant diseases fungi. *Phytopathology*, **43**: 401-403.
- SELVARAJAN, R.AND JEYARAJAN, 1996, Inhibition of chickpea root rot pathogens, *Fusarium solani* and *M. phaseolina* by antagonists, *Indian Journal of Mycology and Plant Pathology*, **26**(3): 248-251.
- SHANMUGAM, N. AND GOVINDASWAMY, C.V., 1973, Physiological studies on *Macrophomina phaseoli*, causing groundnut root rot. *Indian Journal of Mycology and Plant Pathology*, **3**: 1-7.
- SHARMA, B.L., GUPTA, R.N. AND GUPTA, J.S., 1983, Studies on survey of root rot incidence of *cicer arietinum* in northern region of Madhya Pradesh. *Indian Phytopathology*, **36**(1): 82-84.

- SHARMA, S.R., 1988, Evaluation of different fungicides against ashy stem blight (*M. phaseolina*) of French bean. *Indian Journal of Plant Protection*, **16**(2): 285-287.
- *SHARVELLE, E.G., 1961, *The nature and use of modern fungicides*. Burges publication Company, Minnesota, USA, pp. 308.
- SHEKAR, G., 1983, Enumeration of Micro flora from infested soil. *M.Sc. (Agri.) Thesis, University of Agricultural Sciences, Bangalore*, pp. 66.
- SINGH, P.J. AND MEHROTRA, R.S., 1980, The influence of cultivar and temperature on carbohydrate and amino acid exudation from gram seeds and on pre-emergence damping off by *R. bataticola*. *Plant and Soil*, **55**(2): 261-268.
- SINGH, P.J. and MEHROTRA, R.S., 1982, Influence of soil moisture and temperature on *Rhizoctonia bataticola* infection of gram. *Indian Phytopathology*, **35**: 327-329.
- SINGH, R.D.N. AND KAISER, S.A.K.M., 1994, Effect of different culture media and pH levels on the growth and cultural characteristics of charcoal rot pathogen (*Macrophomina phaseolina*) infecting maize. *Crop Research*, **7**(2): 282-287.
- SINGH, R.P., HARI-CHAND AND CHAND, H., 1996, Identification of multiple disease resistance in chickpea at Hisar, Haryana, India. *International Chickpea and Pigeonpea Newsletter*, **3**: 32-33.

- SINGH, S.K., NENE, Y.L. AND REDDY, M.V., 1990, Some histopathological observations of chickpea roots infected by *Rhizoctonia bataticola*. *International Chickpea Newsletter*, **23**: 24-25.
- SMITH, W.H., 1969, Comparison of mycelial and in the mortality of pine seedlings under varying soil conditions. *Phytopathology*, **59**: 379-382.
- SMITS, B.G. AND NOGUERA, R., 1988, The ontogeny and morphogenesis of sclerotia and pycnidia of *M. phaseolina*. *Agronomia Tropical*, **38**(4): 69-78.
- SRIVASTAVA, A.K., 1987, Role of pectolytic trans-eliminase during the pathogenesis of *M. phaseolina*. *Indian Journal of Plant Pathology*, **5**(10): 53-58.
- SURIACHANDRASELVAN, M. AND SEETHARAMAN, K., 2000, Survival of *Macrophomina phaseolina*, the causal agent of charcoal rot of sunflower in soil, seed and plant debris. *Journal of Mycology and Plant Pathology*, **30**(3): 402-405.
- TAUBENHAUS, J.J., 1913, The black rots of sweet potato. *Phytopathology*, **3**: 159-166.
- TRIPATHI, N.N. AND SHARMA, B.K., 1983, Incidence of chickpea dry root rot in Southern Haryana. *International chickpea Newsletter*. No. **8**: 22-23.

- TARIQ HUSAIN, ABDUL GHAFAR, HUSAIN, T. AND GHAFAR, A., 1995, Effect of soil moisture on the colonization of *R. bataticola* on roots of chickpea. *Pakistan Journal of Botany*, **27**(1): 221-225.
- VASUDEVA, R.S., 1937, Studies on the root rot disease of cotton in Punjab. III. The effect of some physical and chemical factors on sclerotia formation. *Indian Journal of Agricultural Sciences*, **7**: 259-270.
- VINCENT, J.M., 1947, Distortion of fungal hyphae in the presence of certain inhibitors. *Nature*, **159**: 850.
- VISHWA DHAR AND SARBHOY, A.K., 1989, Studies on the germination and longevity of pycnidiospores of *M. phaseolina*. *Indian Phytopathology*, **42**(1): 123-127.
- VISHWA DHAR, SARBHOY, A.K. AND DHAR, V., 1991, Induced pycnidia production by the soybean isolates of *M. phaseolina*. *Indian Phytopathology*, **44**(1): 68-72.
- WALLER, J.M., 1976, Plant disease in arid climate. *Span*, **19**: 125-126.
- WASEER, N.A., PATHAN, M.A., WONDIARM AND SOLANGI, G.R., 1990, Studies on charcoal rot of soybean caused by *M. phaseolina*. *Rajasthan Journal of Phytopathology*, **2**(1-2): 22-36.

WATANABE, T., 1973, Survivability of *Macrophomina phaseolina* in naturally infested soil and longevity of the sclerotia formed in vitro. *Annual Phytopathological Society. (Japan)*, **39**: 333-337.

WELLMAN, F.L., 1939, A technique for studying host resistance and pathogenicity in tomato Fusarium wilt. *Phytopathology*, **31**: 281-283.

*WHEELER, B.E.J., 1969, An introduction to plant disease. John Wiley and Sons Limited, London, pp. 301.

WHITE, R.P., 1927, Studies on tomato wilt caused by *Fusarium lycopersici*. *Journal of Agriculture Research*, **34**: 197-239.

* Originals are not seen

Appendix

APPENDIX - I

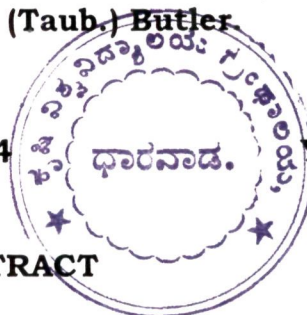
List of abbreviations used in the Thesis

PDB	: Potato dextrose broth
PDA	: Potato dextrose agar
MB	: Maltose broth
CB	: Czapecks broth
OB	: Oat meal broth
RB	: Rose bengal broth
SEM	: Scanning electron microscopy
DAI	: Days after inoculation
PDI	: Per cent disease index

**STUDIES ON DRY ROOT ROT DISEASE OF CHICKPEA (*Cicer arietinum* L.)
CAUSED BY *Rhizoctonia bataticola* (Taub.) Butler**

VEERENDRA KUMAR K.V.

2004



Y.D. NARAYANA
(Major advisor)

ABSTRACT

Chickpea (*Cicer arietinum* L.) is one of the major grain legume crops of Karnataka. Dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler is one of the constraints in chickpea production. Roving survey was conducted during the *rabi* season of 2002 revealed that the maximum severity of the root rot was noticed in Gulbarga district (9.97%) followed by Raichur (5.16%) and Bidar (4.28%). Among the culture media tested potato dextrose agar and czapeck's agar were found better for growth and sclerotia production of the fungus. A temperature of 35°C and relative humidity of 75 per cent were found favourable for growth of the pathogen. Alternate cycles of light (12 hrs) and darkness (12 hrs) favoured the good growth and sclerotia production as compared to complete darkness. The pathogen perpetuated up to 150 days in plant debris and 120 days in soil and 90 days when stored in refrigerators. Among the four levels of inoculum the early initiation of disease in chickpea was observed at highest inoculum level of 4.0 per cent. The higher inoculum level had a profound effect on various growth parameters of chickpea plant. Among fungicides evaluated, hexaconazole (0.025%) and Benomyl (0.05%) among systemic fungicides, SAAF (0.1%) and Mancozeb (0.2%) among the non systemic fungicides were effective in inhibiting the growth of the fungus under *in vitro*. Among the bioagents tested *P. fluorescens* was effective in inhibiting the growth of the pathogen followed by *Trichoderma harzianum* in dual culture technique. Among 25 chickpea genotypes screened for dry root rot resistance using sick pot and blotter paper techniques, none of the genotype were immune and ICCV-94916-8 and GPC-12 were found resistant, but ICCV-98904 and ICCV-98906 showed moderately resistant to dry root rot pathogen.